# Foundation I: The Metric Universe

*Extended Version*

A Geometric Einstein–Cartan–Kibble–Sciama Cosmology with Regular Bounce,

Stiff-Era Expansion, and Testable Late-Time Signatures

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**Abstract**

The standard ΛCDM model currently faces persistent tensions, most notably in the Hubble constant $H_0$ and in the amplitude of late-time structure growth quantified by $S_8$ [58], while the initial singularity remains a conceptual limitation of the standard cosmological picture. In this work, *Foundation I*, we investigate a minimal cosmological extension of General Relativity based on the Einstein–Cartan–Kibble–Sciama (ECKS) framework. Within an effective spin-fluid description, a primordial spin component—whose stiff equation of state $w = 1$ is geometrically distinct from scalar-field kination and requires no free initial condition—induces a transient phase prior to recombination that reduces the comoving sound horizon to $r_s \approx 135.65$ Mpc (`Scripts_F1/script_01_solve_sound_horizon.py`). This allows a calibration compatible with $H_0 \approx 73.04\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$ (SH0ES [4]), obtained by holding the acoustic angle $\theta_* = r_s/D_A$ fixed at $\Omega_m = 0.315$. Under the internal closure $\Omega_m = \omega_m/h^2$ adopted below, the same background lowers $\theta_*$ by $\simeq 6.5\%$ ($w = -1$), so $H_0 = 73.04$ is conditional on a reconciliation not achieved in this work (open problem PO-F1-THETA). The effective amplitude $F_{\rm ion} = 1.2765$ (computed by `Scripts_F1/script_01_solve_sound_horizon.py`) is the calibration of this sound-horizon reduction, and hence of $H_0$. Under the internal closure $\Omega_m = \omega_m/h^2$, raising $H_0$ at fixed physical densities lowers $\Omega_m$, which strongly reduces $S_8$: a CAMB ΛCDM reference on this background gives $S_8 \approx 0.780$, against $\approx 0.831$ on the Planck background (a conditional reference, not an ECF prediction; Sec. 4). The weak-lensing value $S_8 = 0.766$ (KiDS-1000 $3\times2$pt) is a target, not an ECF prediction: the residual correction depends on the ECF perturbative dynamics, which is not implemented, and the earlier calibration of the growth coupling on this value, $\gamma_{\rm spin} = 0.3116$ (`Scripts_F1/script_06_S8_resolution.py`), is retained as historical only. The scale-dependent effective gravitational constant, $G_{\rm eff}(k) = G_N\,T^2_{\rm tors}(k)$, with $T^2_{\rm tors}(k) = 1 - A_{\rm sup}\,k^2/(k^2 + k^2_{\rm cut})$, $A_{\rm sup} = 0.15$, $k_{\rm cut} = 0.10\,h\,\mathrm{Mpc^{-1}}$ (regularised Lorentzian form; at $k \ll k_{\rm cut}$ it reduces to the phenomenological $G_{\rm eff} \approx G_N[1-\alpha(k/k_{\rm cut})^2]$, $\alpha = A_{\rm sup} = 0.15$; Sec. 4, Table 2), is retained as a phenomenological parametrisation of small-scale suppression, not as a calibration on $S_8$; the physical growth-suppression mechanism remains open. Summed over the six tension-sector datasets of Table 4, the calibrated solution improves the effective statistic by $\Delta\chi^2 = -39.5$ relative to ΛCDM (value computed with $\Omega_m = 0.315$; not recomputed with $\Omega_m = \omega_m/h^2$ — requires a dedicated ECF perturbation (Boltzmann) solver, which does not yet exist) (sector-by-sector assembly in `Scripts_F1/target_reference_budget.py`) ($\Delta$AIC $= -35.5$, $|\Delta\mathrm{BIC}| \simeq 24$, historical, $\Omega_m = 0.315$; not recomputed: no ECF perturbation solver yet, under a historical two-parameter count, $F_{\rm ion}$ and $\gamma_{\rm spin}$, retained unchanged; $H_0$ is not independent of $F_{\rm ion}$, see Sec. 4). The $a^{-6}$ dilution of the spin sector brings the spin-to-radiation ratio down to 0.36% at recombination, leaving the stiff component subdominant there. At the theoretical level, the spin-squared correction acts as a repulsive contribution at extreme density, replacing the initial singularity with a regular geometric bounce. With the internal closure $\Omega_m = \omega_m/h^2 = 0.267$, the $w = -1$ background gives a cosmic age $t_0 \approx 13.33\,\mathrm{Gyr}$ (`Scripts_F1/A_Core_Calculations/script_09_cosmic_ages.py`), compatible with the globular-cluster age $t_{\rm GC} = 13.32 \pm 0.10_{\rm stat} \pm 0.5_{\rm sys}\,\mathrm{Gyr}$ [29]; the age tension reported in earlier versions (obtained with $\Omega_m = 0.315$) does not arise. The comparison is a chronometric consistency check, not a direct measurement of $t_0$. The difficulty is displaced rather than removed: on the same background the BAO distances are less well fitted than in ΛCDM ($\Delta\chi^2_{\rm BAO} \simeq +10.3$ over four BOSS/eBOSS bins), and the $\chi^2$ of the calibrated $(w_0, w_a)$ trajectory rises from 13.7 to 55.3 on a ten-observable subset of DESI DR1 with diagonal errors — not the global DESI DR1 $\chi^2$. Within the tanh family of late-time $w(z)$ studied here, no model satisfies the age and BAO constraints together, and the $H_0$–Age–BAO problem (a quadrilemma once $\theta_*$ is included, PO-F2-5) remains open. A second problem is opened here: the BAO comparisons use the ECF scale $r_s^{\rm ECF}$ as the ruler of measurements published in units of the drag horizon $r_{\rm drag}$, without the relation between the two having been established in the ECF chain (`PO-ECF-BAO-RDRAG`). Every BAO result quoted in this work is therefore CALCULATED-CONDITIONAL on that identification. The local measurement $H_0 = 73.50 \pm 0.81\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$ [60] is itself in $7.1\sigma$ tension with flat ΛCDM (Planck+SPT+ACT) and $5.0\sigma$ with BBN+BAO (DESI DR2) [64], rendering the ΛCDM cosmic age of $\sim$13.8 Gyr incompatible with any local $H_0 \approx 73$ calibration. Beyond the standard

tensions, the framework yields multiple falsifiable signatures: macroscopic topological defects capable of seeding high-redshift supermassive black holes observed by JWST, a non-zero cosmic birefringence ($\beta \neq 0$; its magnitude $\beta \approx 0.35°$ being a calibrated consistency check rather than an *ab initio* prediction), a residual CMB damping-tail deficit at $\ell > 2000$, and a blue-tilted primordial gravitational-wave background. The calibrated trajectory crosses the phantom divide at $z = 1.684$; we state plainly that a single canonical scalar cannot do so, that modified-gravity sectors can, and that the mechanism licensing it here is not derived but tracked as an open problem. *Foundation I* thus identifies an observationally calibrated phenomenological sector of Einstein–Cartan cosmology, the microscopic treatment of the dark sector being taken up in *Foundation II*.

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***Version note.*** *Preprint v8 (October 2026; internal revision 13), superseding preprint v7 (September 2026) under the concept DOI 10.5281/zenodo.19577447. Earlier history: Version 10 (preprint v3, June 2026), intermediate revisions v10.2 and preprint v4.* ***Withdrawal of the flatness-compensation reading.*** *Version 10 presented the cosmic energy budget as if exact spatial flatness* enforced *a compensating torsion contribution. That reading is withdrawn, following the revised statement of the Topological Invariance Principle [1]: once $k = 0$, the closure relation $\sum_i \Omega_i = 1$ is an identity — the critical density being recomputed with $H$ — and therefore constrains neither the torsion sector nor the dark-energy equation of state. The three affected passages now state a relation between sector magnitudes* to be derived from the action*, not read off the closure condition. No numerical result changes: the closure relation was used for bookkeeping, never as a constraint.* ***Editorial and correctness revision.*** *Several published values that could not be reproduced from the deposited scripts have been withdrawn or recomputed; the $\chi^2$ budget of Table 4 is now derivable row by row, giving $\Delta\chi^2 = -39.5$, marginally different from the value quoted in v10 (historical, $\Omega_m = 0.315$; not recomputed: no ECF perturbation solver yet). The status of the calibrated parameters is stated explicitly throughout: $F_{\rm ion}$ and $G_{\rm eff}(k)$ are effective amplitudes fitted to the data, not quantities derived from the field equations. The Cartan density is harmonised to $\rho_{\rm c} = c^5/\hbar G^2 = 5.15 \times 10^{96}\,kg\,m^{-3}$ throughout (three occurrences previously rounded to 5.2), aligning this paper with Foundation III. Nine LaTeX defects inherited from v10 are corrected (duplicate* `documentclass`*, a corrupted control character, undefined citation macros, a malformed equation, a missing TikZ library, an unbalanced brace, a misused list environment). One overstatement is softened ("definitively rules out" → "strongly disfavours"). Two further corrections (v10.2): the $H_0$–Age–BAO tension is restated as an open* quadrilemma *once the CMB acoustic scale $\theta_*$ is included (the calibrated $(w_0, w_a)$ does not by itself resolve it; precise quantification deferred to a dedicated Einstein–Cartan perturbation solver calculation), and the independent DESI DR1 MCMC check is rebuilt with the full published covariance matrix (Adame et al. 2024): the calibrated point lies at $2.03\sigma$ (Planck prior) and $6.98\sigma$ (SH0ES prior), superseding the earlier diagonal-errors estimate. Headline numerical results ($\Delta\chi^2 = -39.5$ [historical, $\Omega_m = 0.315$, not recomputed: no ECF perturbation solver yet], $H_0 = 73.04$, $r_s = 135.65$ Mpc).*

***v6 correction (September 2026) — notational and scope corrections, no numerical result changed.*** *Five points, found through cross-checking the paper against the scripts that produce its figures and by comparison with companion papers.* ***(i)*** *The Friedmann equation of Appendix A and the late-time calibrated background (Eq. (10)) are two distinct objects sharing one symbol; the sign of the post-bounce term is calibrated, not derived, and the gap is tracked as PO-F1-SIGMA0.* ***(ii)*** *The spin sector is normalised at $z_{\rm ref} = 5600$, the value used by the solver (`script_01_solve_sound_horizon.py`); several passages previously quoted $z \simeq 7500$, a value not used in any script.* ***(iii)*** *Appendix cross-references, broken by the unnumbered `section*` declarations (a rendering defect invisible at compilation), now resolve to the correct letter throughout.* ***(iv)*** *The bounce-era tensor amplitude ($H_{\rm b} \simeq 1.2 \times 10^{11}\,GeV$) is stated as an illustrative normalisation, not one obtained from the canonical Cartan density: inserting $\rho_c = 5.15 \times 10^{96}\,kg\,m^{-3}$ into $H_{\rm b} = \sqrt{\rho_c/3M_{\rm Pl}^2}$ returns $\mathcal{P}_* > 1$, outside the perturbative regime the estimate presupposes.* ***(v)*** *On the sign of the post-bounce term, Kopczyński's aligned-spin cosmology (reviewed in Trautman 2006) is added: polarisation alone, within algebraic ECKS, does not flip the sign, which is*

*evidence against the simplest resolution of PO-F1-SIGMA0.*

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**Data:** Planck 2018, SH0ES, KiDS-1000, DES Y3, BOSS DR12, eBOSS DR16†, DESI DR1, JWST JADES/CEERS.

†The observational target adopted here is the eBOSS DR16 QSO consensus point $D_M/r_d(z{=}1.48) = 30.21 \pm 0.79$ [31, 32], the eBOSS consensus of the configuration-space [31] and Fourier-space [32] analyses (configuration space alone: $30.66 \pm 0.88$). Earlier versions of this work carried $30.85 \pm 0.80$; no source compatible with that value has been retrieved, and it is withdrawn as an untraced historical target. The adopted value is reproduced by `chi_carre.py` (public repository). The comparison uses the ECF scale $r_s^{\rm ECF}$ as the BAO ruler while the measurement is published in units of $r_{\rm drag}$; the two are not identified here, and establishing the relation is an open problem (`PO-ECF-BAO-RDRAG`), so this comparison is CALCULATED-CONDITIONAL.

**Keywords:** Einstein–Cartan cosmology · Einstein–Cartan–Kibble–Sciama gravity · spacetime torsion · Weyssenhoff spin fluid · singularity resolution · cosmological bounce · stiff-era expansion · sound horizon · baryon acoustic oscillations · Hubble tension · $S_8$ tension · scale-dependent gravitational coupling · cosmic birefringence · $H_0$–Age–BAO trilemma · primordial topological defects · dynamical dark energy · primordial gravitational waves · Bayesian model comparison.

# 1 Introduction: Persistent Tensions and a Geometric Route

## 1.1 Motivation

The ΛCDM model provides an excellent phenomenological description of a broad range of cosmological observations, from the cosmic microwave background to large-scale structure and supernova distance measurements. At the same time, a number of persistent discrepancies have motivated the exploration of extensions or alternatives to the standard framework. Among these, the most discussed are the Hubble-constant tension, the weaker but recurrent $S_8$ tension, and the more formal question of the initial cosmological singularity. A question is whether some of these issues may be related to the geometric assumptions built into standard cosmology. In General Relativity, spacetime is modeled by a torsion-free connection on a pseudo-Riemannian manifold. Einstein-Cartan theory relaxes this assumption by allowing spacetime torsion sourced by the intrinsic spin of matter. Although such effects are negligible in the low-density late universe, they may become relevant in the primordial high-density regime. The purpose of this paper is to investigate whether a minimal spin-torsion cosmology can simultaneously modify the early expansion history, affect the calibration of the sound horizon, and alter late-time structure growth in a phenomenologically relevant way. We examine whether the Einstein–Cartan scenario provides a coherent, testable, and statistically preferred alternative description of several outstanding cosmological tensions.

## 1.2 Observational context

The tension in the Hubble constant is currently one of the main challenges for precision cosmology. Local measurements based on Cepheid-calibrated type-Ia supernovae typically favor values near 73 km/s/Mpc [4], while early-universe inferences from Planck data within ΛCDM favor values near 67–68 km/s/Mpc [57]. Whether this discrepancy originates from unresolved systematics, new early-universe physics, or a more general extension of the cosmological framework remains an open question.

A second motivation comes from the parameter

$$S_8 \equiv \sigma_8 \sqrt{\Omega_m/0.3},$$

which characterizes the amplitude of matter clustering. Several weak-lensing surveys [25, 26] have reported values lower than those inferred from Planck within ΛCDM, suggesting that late-time growth may be somewhat suppressed relative to the standard expectation. While this tension is less significant than the $H_0$ discrepancy, it is sufficiently persistent to motivate models capable of affecting both expansion history and structure formation.

Recent synthesis work has sharpened the status of this "$S_8$ tension". A 2026 review [61] combines *Planck*, ACT DR6 and SPT-3G into a baseline $S_8 = 0.836^{+0.012}_{-0.013}$, while late-time weak-lensing probes remain mixed: DES Y6 ($3 \times 2$pt) gives $S_8 = 0.789 \pm 0.012$, 2.4–2.7$\sigma$ below this baseline, whereas KiDS-Legacy ($S_8 = 0.815^{+0.016}_{-0.021}$) is consistent with it at $< 1\sigma$. The review emphasises that the apparent discrepancy is probe-dependent, with photometric-redshift calibration, intrinsic-alignment modelling and shear-pipeline systematics contributing uncertainties comparable to the cosmological signal itself. From a phenomenological point of view, this landscape motivates models that mildly suppress the late-time growth factor $D(z)$ relative to ΛCDM without disturbing early-universe CMB constraints, and it requires any proposed resolution of the $S_8$ tension to be assessed against the full probe-dependent set of constraints rather than a single representative value.

In the Einstein–Cartan framework developed below, the spin–torsion sector reduces the sound horizon and raises the inferred Hubble constant; whether it also modifies small-scale clustering through $G_{\rm eff}(k)$ remains to be established at the perturbative level. Under the internal closure $\Omega_m = \omega_m/h^2$, raising $H_0$ at fixed physical densities lowers $\Omega_m$, which strongly reduces $S_8$; a

CAMB ΛCDM reference on this background gives $S_8 \simeq 0.780$, against $\simeq 0.831$ on the Planck background. The weak-lensing value $S_8 = 0.766$ was reproduced in earlier versions by calibrating $\gamma_{\rm spin}$ on it; it is a target, not an ECF prediction, and the residual correction depends on the ECF perturbative dynamics, which is not implemented. Any comparison with the weak-lensing value should be assessed against the full probe-dependent set of constraints identified by the 2026 review.

A third observational input comes from the recent DESI DR1 baryon acoustic oscillation survey [28], which reports a preference for a dynamical dark-energy equation of state ($w_0 > -1$, $w_a < 0$) at the $2.6\sigma$ level for DESI+CMB, and 2.5–$3.9\sigma$ when supernova samples are added. Although this preference does not constitute a definitive detection, it motivates frameworks in which the late-time expansion carries a geometric imprint beyond a pure cosmological constant. The ECF addresses this trend through a residual torsion contribution that generates an effective CPL trajectory ($w_0 = -0.904$, $w_a = -0.153$; Sec. 6) without introducing an independent scalar-field dark-energy sector.

At a more formal level, classical General Relativity also predicts an initial singularity in the limit $a \to 0$, where curvature and energy density diverge. This feature is generally interpreted as signaling the breakdown of the classical theory in the earliest stages of cosmic evolution. In that respect, theories in which high-density corrections arise naturally from the spacetime geometry are of particular interest.

Another structural consequence of any $H_0$-resolving framework is the $H_0$–Age–BAO trilemma (Section 5.4): because the age integral and BAO comoving distance share the same integrand $H(z)^{-1}$, a background chosen to resolve the $H_0$ tension constrains the age and the BAO distances together. With $\Omega_m = \omega_m/h^2 = 0.267$ and $w = -1$ the ECF gives $t_0 \approx 13.33$ Gyr, compatible with the globular-cluster bound; the tension is displaced to the BAO distances and to the $(w_0, w_a)$ trajectory confronted with DESI.

**April 2026: Hubble tension above $7\sigma$.** The observational landscape has shifted decisively since the first submission of this work. The $H_0$ Distance Network ($H_0$DN) Collaboration has reported $H_0 = 73.50 \pm 0.81\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$, the most precise local measurement to date at 1.1% precision, combining multiple independent rungs of the distance ladder [60]. This value represents a *community consensus result*, robust to the exclusion of any single distance indicator and derived from a fully covariant treatment of cross-calibrated systematic uncertainties [60]. This measurement yields $7.1\sigma$ tension with flat ΛCDM (Planck+SPT+ACT) and $5.0\sigma$ with BBN+BAO (DESI DR2) [60, 64]. The DESI DR2 BAO compilation constitutes the baseline BAO dataset used in the $H_0$DN analysis [64], so the $H_0$DN result and the ECF calibration are anchored on a consistent BAO standard. Within ΛCDM with $\Omega_m$ held at the Planck value 0.315, a Hubble constant of $73.5\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$ implies a cosmic age of $t_0 \approx 12.7\,\mathrm{Gyr}$, below the age of the oldest globular clusters, $t_{\rm GC} = 13.32 \pm 0.10_{\rm stat} \pm 0.5_{\rm sys}\,\mathrm{Gyr}$ [29], used here as a chronometric lower bound on $t_0$ rather than as a measurement of it. When $\Omega_m$ is instead obtained from the CMB matter density, $\Omega_m = \omega_m/h^2 = 0.267$, the ECF background at $w = -1$, with the SH0ES local distance ladder and the ECF sound horizon $r_s = 135.65\,\mathrm{Mpc}$, gives $t_0 \approx 13.33\,\mathrm{Gyr}$, compatible with that bound (Sec. 5.4). The difficulty then moves to the BAO distances: the eBOSS DR16 point gives $D_M(1.48)/r_s = 31.37$ against $30.21 \pm 0.79$, and the four-bin BAO $\chi^2$ is 12.11 against 1.79 for ΛCDM. The ECF point $(w_0, w_a) = (-0.904, -0.153)$ is a *calibration* point, not a prediction: $\xi = 1.645$ is a parameter calibrated on the DESI preference used by the ECF, so as to reproduce the target value of $w_a$; it is therefore not a prediction independent of DESI. Its location in the DESI-preferred region is therefore not an independent validation [64, 1].

**The Need for First-Principles Solutions.** The observational pressure described above has recently prompted several phenomenological constructions within ordinary general relativity. A representative example is the unified scalar-field framework of Kouniatalis [59], which addresses

the $H_0$–$w_0$–$w_a$–$S_8$ trilemma via a single canonical scalar field evolving through a "bump-tail" potential

$$V(\phi) = V_0\, e^{-\lambda\phi/M_{\rm Pl}} \left[1 + A \exp\left(-\frac{(\phi-\phi_c)^2}{2\sigma^2}\right)\right], \tag{1}$$

whose four free parameters ($A$, $\sigma$, $\lambda$, $\phi_c$) are tuned so that the field provides a transient pre-recombination energy injection, undergoes a rapid kination dilution ($\rho_\phi \propto a^{-6}$), and re-emerges as late-time quintessence. While analytically tractable at the background level, this construction is explicitly restricted to background dynamics, with the perturbative and likelihood analysis left to future work [59]. More fundamentally, it provides no derivation of the potential shape from a fundamental symmetry principle: the specific form of $V(\phi)$ is an *ad hoc* ansatz calibrated to the desired cosmic history.

In contrast, the ECF requires no additional scalar degree of freedom. The same $a^{-6}$ dilution that governs the kination phase in scalar models arises naturally from the expansion-driven suppression of the primordial Weyssenhoff spin fluid — a direct consequence of the ECKS coupling between spacetime torsion and fermionic spin, not a tuned potential feature. The transient pre-recombination energy injection, the sound-horizon reduction, and the late-time dynamical dark-energy signature all emerge from a *single geometric coupling*, characterised at the effective level by the dimensionless amplitude $F_{\rm ion} = 1.2765$, with two further calibrated numbers ($\Omega_{\rm spin}^{\rm peak}$, $\tau_{\rm tor}$), adjusted jointly on the sound-horizon target $r_s = 135.65$ Mpc (Table 2; the spin sector has a qualitative origin in the QCD transition, but its amplitude is not computed from it), with no scalar field, no non-minimal coupling, and no designer potential. The Einstein-Cartan framework, introduced in the following subsection, provides precisely this geometric foundation.

### 1.3 Einstein-Cartan framework

Einstein-Cartan-Kibble-Sciama theory [5, 6] extends General Relativity by promoting the affine connection to an object with both curvature and torsion. In this framework, torsion is not an independent propagating degree of freedom in vacuum, but is algebraically related to the spin density of matter. For fermionic matter at sufficiently high density, this coupling generates effective spin-spin corrections that can modify the cosmological dynamics. In homogeneous cosmology, such corrections are commonly represented at the effective level by a Weyssenhoff-type spin fluid. Because the corresponding spin contribution scales approximately as $a^{-6}$, it becomes negligible at late times but can become important in the early universe. This scaling makes spin-torsion cosmology a natural candidate for producing a transient stiff-like phase, altering the pre-recombination background evolution, and regularizing the high-density limit through a non-singular bounce [22, 9, 10].

**Terminology — two distinct stiff eras.** Because $\rho_{\rm spin} \propto a^{-6}$ grows faster than any other component as $a$ decreases, the spin fluid dominates on *both* sides of the bounce. Two physically distinct stiff eras must therefore be kept apart, and we fix the notation once and for all:

- $N_{\rm stiff}^{\rm exp}$ — the **expanding, post-bounce** stiff era ($a_{\rm bounce} \le a \le a_{\rm eq}$), transient, which reduces the sound horizon, which underlies the $H_0$ calibration of the present work and, at the background level, the lower $S_8$ that follows from $\Omega_m = \omega_m/h^2$; the growth-level role of the spin sector remains open (DEC-A7-POST-07). It also underlies the blue tensor tilt $n_T > 0$;
- $N_{\rm stiff}^{\rm contr}$ — the **contracting, pre-bounce** stiff era, which governs the horizon problem and the primordial scalar spectrum, and is treated in Foundation III [3].

Unless stated otherwise, "stiff phase" in this paper refers to $N_{\rm stiff}^{\rm exp}$. Whenever an e-fold count is quoted, the relevant era is specified explicitly.

In this paper, we study a specific cosmological realization of this general framework. Rather than introducing the scenario as a replacement for the full Einstein-Cartan literature, we treat

it as a phenomenological Einstein-Cartan model characterized by a primordial spin contribution and a set of effective couplings calibrated against cosmological observables. The emphasis is placed on internal consistency, numerical reconstruction, and observational consequences.

**Nomenclature: the ECF.** While Einstein-Cartan-Sciama-Kibble theory provides the general geometric setting, it admits several possible cosmological realizations depending on the matter sector and the effective spin coupling. To distinguish the specific effective scenario investigated in the present work—characterized by a transient stiff spin component and the associated phenomenological calibration—we refer to it as the *ECF*. This designation is used here only as a compact label for the specific ansatz and observational implementation studied in *Foundation I*.

### 1.4 Structure of the Analysis

**The Einstein–Cartan Trilogy.** This paper, *Foundation I*, develops the effective cosmological phenomenology of the spin-torsion scenario at the background level. The results presented here are self-contained.[1]

*Foundation II* [2] investigates the microscopic origin of the geometric dark sector and the dynamical treatment of the late-time dark-energy sector within the ECKS action, in connection with the open $H_0$–Age–BAO quadrilemma (PO-F2-5). It identifies three topological fossils of the bounce (Micro-Knots, Macro-Knots, and the chiral gravitational-wave asymmetry), derives the baryogenesis mechanism from the primordial spin-fluid asymmetry, and confronts the resulting geometric dark matter with the SPARC rotation-curve database ($\tilde{\chi}^2_{\rm med} = 0.80$, 171 of 175 galaxies) and the DESI dynamical dark energy signal $(w_0, w_a) = (-0.904, -0.153)$.

*FoundationIII* [3] addresses the strong-gravity sector: singularity avoidance in black hole interiors, transient Einstein–Rosen bridge geometry, and compact-object observables in the Einstein–Cartan setting.

The organization of the manuscript is as follows:

- **Section 2** introduces the Einstein-Cartan-Kibble-Sciama framework and the effective spin-fluid description used throughout the analysis.
- **Section 3** presents the mechanism by which the primordial spin sector reduces the sound horizon $r_s$.
- **Section 4** presents the $H_0$ sector and the status of the $S_8$ sector at the effective level.
- **Section 5** reports the numerical calibration and the statistical comparison with the reference ΛCDM description.
- **Section 6** confronts the framework with current observational indications, including high-redshift galaxies, structure-growth data, and late-time expansion probes.
- **Section 7** isolates the main falsifiable predictions of the framework.
- **Section 8** outlines the broader Einstein-Cartan Trilogy and the directions developed in *Foundation II*.
- **Section 9** summarizes the scope and status of the results obtained in *Foundation I*.
- **Appendix J** presents the phenomenological motivation for the effective-flatness working hypothesis within the ECF background solution (Sec. 2); it complements the axiomatic treatment developed in the companion Letter [1].

[1] All claims developed in this paper—the $r_s$ reduction, the $H_0$ calibration and the status of the $S_8$ sector, and the resulting trilemma—are derived from and verified by our late-time phenomenological framework alone.

## 2 Theoretical Framework

Einstein-Cartan-Kibble-Sciama (ECKS) theory [5, 6] extends General Relativity by allowing the affine connection to possess a non-vanishing torsion tensor in addition to curvature. In this framework, torsion is sourced algebraically by the intrinsic spin density of matter and does not propagate as an independent vacuum degree of freedom. This makes Einstein-Cartan gravity a natural setting in which spin effects may become relevant in the high-density early universe while remaining negligible in the dilute late-time regime.

The purpose of this section is to introduce the geometric ingredients and effective cosmological equations used throughout this work. We first summarize the field equations, then discuss the Weyssenhoff-fluid description adopted at the macroscopic level, and finally present the effective Friedmann dynamics relevant for the early universe.

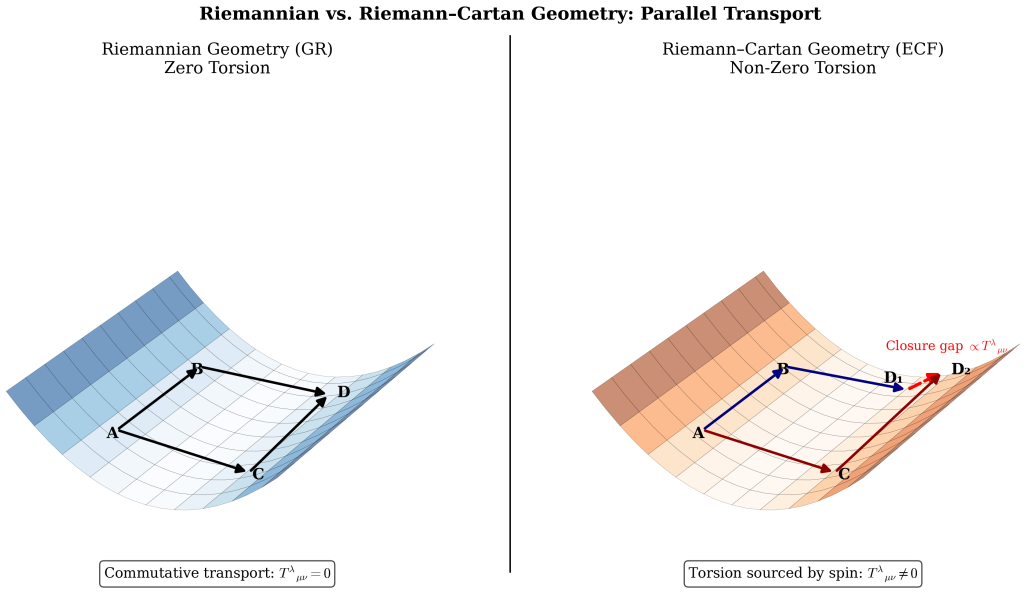



Figure 1: **Riemannian and Riemann-Cartan geometry.** Schematic comparison between a torsion-free connection ($T^{\lambda}{}_{\mu\nu} = 0$, left) and a connection with torsion ($T^{\lambda}{}_{\mu\nu} \neq 0$, right). In the Einstein-Cartan framework, torsion is associated with a non-closure of infinitesimal parallelograms and provides the geometric ingredient coupled to matter spin (see Table 1).

*Reproduced by:* `Scripts_F1/plot_torsion_geometry_schema.py`.

To avoid confusion between geometric spin variables and standard large-scale-structure observables, the notation adopted throughout this work is summarized in Table 1.

### 2.1 The ECKS field equations

Variation of the Einstein-Cartan action leads to two coupled sets of equations [8]. The first generalizes the Einstein equation by including spin-dependent corrections in the effective energy-momentum tensor. The second is the Cartan equation, which relates torsion algebraically to the spin density tensor.

1. **Generalized Einstein equation:**

$$R_{\mu\nu} - \frac{1}{2} R g_{\mu\nu} = \kappa \left( T^{matter}_{\mu\nu} + \Theta^{spin}_{\mu\nu} \right) \tag{2}$$

where $\Theta^{spin}_{\mu\nu}$ contains the quadratic spin corrections relevant at high density, and $\kappa$ denotes the Einstein–Cartan coupling. Throughout this work we use

$$\kappa = \frac{8\pi G}{c^4} = 2.0766 \times 10^{-43}\ \mathrm{m\,J^{-1}}, \tag{3}$$

Table 1: **Nomenclature of key variables.** Distinctions between geometric torsion, fluid spin density, and cosmological structure parameters.

| Symbol | Physical Definition |
|---|---|
| $T^{\lambda}{}_{\mu\nu}$ | **Torsion Tensor:** The antisymmetric part of the affine connection ($T^{\lambda}{}_{\mu\nu} \equiv \Gamma^{\lambda}_{\nu\mu} - \Gamma^{\lambda}_{\mu\nu}$), representing the geometric twisting of the manifold. Vanishes in standard GR; sourced by spin density in ECKS theory. |
| $S^{\lambda}_{\mu\nu}$ | **Spin Density Tensor:** The antisymmetric spin-density source on the right-hand side of the Cartan equation, algebraically related to the torsion tensor via $T^{\lambda}{}_{\mu\nu} + \delta^{\lambda}_{\mu} T^{\sigma}{}_{\nu\sigma} - \delta^{\lambda}_{\nu} T^{\sigma}{}_{\mu\sigma} = \kappa\, S^{\lambda}{}_{\mu\nu}$. |
| $\sigma$ | **Spin Density Magnitude:** The macroscopic, coherent axial field of the Weyssenhoff fluid ($\sigma^2 = \frac{1}{2}\langle S_{\mu\nu} S^{\mu\nu} \rangle$, dual to the axial current $J^{\mu}_5 = \bar{\psi}\gamma^{\mu}\gamma^5\psi$, Foundation II Sec. 3) — *not* an incoherent sum of individual particle spins, which would average to zero absent a genuine chiral (left-right) asymmetry. $\sigma$ vanishes identically if left- and right-handed populations balance. |
| $\sigma_8$ | **Structure Amplitude:** The root-mean-square fluctuation of matter density in spheres of radius $8\,h^{-1}$ Mpc. |
| $S_8$ | **Clustering Parameter:** Defined as $S_8 \equiv \sigma_8 \sqrt{\Omega_m / 0.3}$. |
| $\Omega_{\rm spin}$ | **Spin Density Parameter:** The ratio of effective spin energy density to the critical density. |

the same convention and numerical value as in Foundation II [2]; it is stated explicitly here so that the spin-squared terms below can be evaluated without recourse to the companion paper.

2. **Cartan equation:**
$$T^{\lambda}_{\mu\nu} + \delta^{\lambda}_{\mu} T^{\sigma}_{\nu\sigma} - \delta^{\lambda}_{\nu} T^{\sigma}_{\mu\sigma} = \kappa S^{\lambda}_{\mu\nu} \tag{4}$$
which algebraically links torsion to the spin density $S^{\lambda}_{\mu\nu}$. In particular, torsion does not propagate in vacuum but remains attached to matter sources.

**ECKS limit and PGT extension.** This algebraic structure constitutes the low-energy limit ($m_T \gg H$) of the extended Poincaré Gauge Theory (PGT) framework developed in Foundation II [2], in which the axial torsion sector acquires a kinetic (Proca-type) term and becomes a propagating degree of freedom. In the regime $m_T \gg H$, the *local algebraic torsion equation* approaches the Einstein–Cartan form. That limiting statement constrains the algebraic contact sector; it does not, by itself, fix the sign, the equation of state, or the isotropic FLRW reduction of the separately calibrated effective axial-condensate sector used for the background of this paper (see *Sign convention* below, and Eq. (10)). The numerical results reported here are computed from that effective branch; relating it to the propagating Poincaré gauge action remains open (PO-F1-SIGMA0).

## 2.2 Weyssenhoff Fluid and Spin Density

To describe the collective effect of fermionic spin in a homogeneous and isotropic cosmological framework, we adopt an effective Weyssenhoff fluid description [21]. In an FLRW geometry, this approximation provides a natural macroscopic description of the primordial spin sector, while remaining compatible with the Einstein-Cartan framework at the effective level. The physical plausibility of such a continuous description is strongly supported today by recent results on the quark-gluon plasma. In particular, the CMS collaboration has highlighted a collective response of the medium in heavy-ion collisions, including a *diffusion wake* produced by energetic partons traversing the plasma, supporting the picture of a low-viscosity fluid rather than a simple collisional ensemble of independent particles [62].

Of course, the quark-gluon plasma recreated in colliders is not identical in every respect to the full primordial cosmological medium. Nevertheless, these results strengthen the plausibility of an effective fluid-like description for the ultradense matter of the early Universe.

Specifically, the primordial spin density is assumed to originate from chiral freeze-out during the Quark-Gluon Plasma (QGP) phase transition at $T \sim 200$ MeV, whose consistency with Big Bang Nucleosynthesis (BBN) constraints is not demonstrated (see the discussion of Fig. 17). While the present work focuses on the macroscopic Weyssenhoff fluid description, the companion *Foundation II* extends the analysis to the chiral and structural relics of the bounce, including baryogenesis, the geometric dark sector, and residual cosmic alignment [2].

In this macroscopic picture, the spin contribution is modeled as the averaged effect of the microscopic intrinsic angular momentum of fermions. The effective energy density associated with spin is then assumed to evolve as the square of the fermionic number density,

$$\rho_{\text{spin}} \propto n^2. \tag{5}$$

Since the particle number density evolves with the scale factor according to

$$n \propto a^{-3}, \tag{6}$$

the spin contribution follows the stiffer evolution law

$$\rho_{\text{spin}} \propto a^{-6}. \tag{7}$$

This scaling law constitutes a central ingredient of the background dynamics in the ECF. It implies that the spin sector becomes negligible at late times, but can become dynamically important in the ultradense primordial regime. In effective fluid language, this behavior corresponds to a "stiff" component, characterized by an equation of state parameter

$$w_{\rm spin} = 1. \tag{8}$$

Unlike radiation ($\propto a^{-4}$) or matter ($\propto a^{-3}$), the spin density therefore dilutes extremely rapidly after the early Universe. It is precisely this property that allows the spin sector to modify the high-density regime, regularize the singular limit, and then naturally decouple from standard cosmology at late times. In ratio form, $\rho_{\rm spin}/\rho_{\rm rad} \propto (1+z)^2$, normalised to 9.3% at the calibration redshift $z_{\rm ref} = 5600$ (Table 2; Fig. 13); the ratio decays to 0.36% at $z_{\rm rec} = 1100$, a factor of $\approx 26$ below the reference point.

**Sign convention and repulsive correction.** With the metric and sign conventions adopted in this work, the spin-induced quadratic correction enters with a negative sign. The modified Friedmann equation in the ECF therefore reads, in dimensional form:

$$H^2 = \frac{8\pi G}{3}\Big(\rho_r + \rho_m + \rho_\Lambda - \rho_{\rm spin}\Big), \tag{9}$$

where the negative sign of $\rho_{\rm spin}$ is a direct consequence of the ECKS field equations: the spin-spin contact interaction between fermions is *repulsive*, entering the effective stress-energy tensor of the Weyssenhoff fluid with sign opposite to radiation, matter, and $\Lambda$. At sufficiently high density, this term acts as a repulsive contribution to the effective dynamics — it is what regularises the singular limit and produces the bounce. In the low-density limit, the correction becomes negligible and the standard Friedmann behavior is recovered to high precision.

**Two distinct quantities, one symbol — and why the sign differs.** Equation (9) above describes the *fundamental ECKS correction* in the ultradense regime, where the spin-spin contact term is repulsive and enters negatively. The *late-time cosmological sector* of this paper — the one calibrated against $r_s$ in Sec. 3, and implemented in `Scripts_F1/script_01_solve_sound_horizon.py` — instead treats the residual axial condensate as an effective stiff fluid contributing *positively* to $H^2$, through the dimensionless parameter $\Omega_{\rm spin,0} = 2\pi G\kappa\sigma_0^2/3H_0^2$ defined in Appendix A (manifestly positive, being built on $\sigma_0^2$):

$$E^2(z) = \Omega_r(1+z)^4 + \Omega_m(1+z)^3 + \Omega_{\rm spin,0}(1+z)^6 + \Omega_\Lambda. \tag{10}$$

**Why the two signs are not, in themselves, a contradiction.** The sign of the effective spin-squared term is fixed in the standard case — *algebraic* torsion sourced by an *unpolarised* Weyssenhoff fluid — where it is negative (repulsive), as correctly stated by Eq. (9) and as established in the bounce literature [8, 10]. It is *not* universal across torsion theories more generally. Vashistha, Gannouji and Ganguly show, in an *algebraic-torsion* sector of Poincaré gauge gravity enriched with quadratic torsion invariants, that the coefficient of the effective spin-spin interaction *"is not fixed, but depends on the dimensionless quadratic-torsion couplings"*, unlike in Einstein–Cartan where it is [16]. The same work notes that, after algebraic torsion is eliminated, the *split* between the matter stress tensor and the torsion-induced spin-squared contribution is definition-dependent, and reports $w_{\rm spin} = -1/3$ for the unpolarised Einstein–Cartan limit under the metric definition, against the stiff $w_{\rm spin} = +1$ conventional in the Weyssenhoff literature.

**What that reference does and does not license here.** It establishes that the ECKS coefficient and sign are *not automatically inherited* by every extended torsion model — which is what removes the appearance of a flat contradiction. It does *not* derive the present case: that work

treats *non-propagating* torsion, whereas the ECF adopts Poincaré gauge theory with a *propagating* axial sector (Foundation II, §*The Extended ECF Lagrangian*). Nor does the definition-dependence of the matter/spin split mean that the sign of $H^2$ can be moved by relabelling: what is convention-dependent is which part of the total stress tensor is called "spin", not the total that sources the metric. The sign of $H^2(a)$ is fixed by the action, the couplings and the spin state — none of which has been derived here for the propagating branch.

The regimes are therefore distinct objects with distinct warrants:

| Regime | Torsion | Sign in $H^2$ | Warrant |
|---|---|---|---|
| Bounce ($\rho \to \rho_c$) | Algebraic, unpolarised Weyssenhoff | – | Fixed by ECKS; repulsive; regularises the singularity. Eq. (9) |
| Quadratic PGT (literature) | Algebraic, $T^2$ invariants | depends | Coefficient set by the quadratic-torsion couplings [16] — shows the ECKS sign is not universal |
| ECF calibrated sector | Effective axial condensate; propagating branch | + | *Phenomenological*: positive sign, FLRW isotropy and $w = +1$ are properties of the calibrated branch, **not derived** from the propagating action. Eq. (10) |

**Three internal checks exclude the negative sign for the calibrated sector.** Were Eq. (9) applied to the late-time budget: (i) $E^2(z)$ turns negative for $z \gtrsim 2.0 \times 10^4$, so $H$ becomes imaginary and the model is undefined above that redshift — excluding BBN ($z \sim 4 \times 10^8$) from the domain entirely; (ii) the sound-horizon integral $r_s = \int c_s/H \, \mathrm{d}z$ crosses that region and returns no value; (iii) most directly, a negative term would *lower* $H$ before recombination, hence *raise* $r_s$ and *lower* the inferred $H_0$ — the opposite of the mechanism this paper reports. These are consistency arguments internal to the calibrated branch: they show which sign that branch requires, not that the branch follows from the ECKS action.

**What remains open (PO-F1-SIGMA0).** Three properties of the post-bounce term are adopted as calibrated inputs and are *not* derived here: its positive sign, its stiff equation of state $w = +1$, and its isotropic FLRW reduction.

On the sign: polarisation alone does not obtain it within algebraic ECKS. Kopczyński's 1973 aligned-spin cosmology — a Weyssenhoff fluid with a single preferred spin direction, $S_{23} = \sigma \neq 0$, rather than the random unpolarised case — gives the modified Friedmann equation $\frac{1}{2}\dot{\mathcal{R}}^2 - M\mathcal{R}^{-1} + \frac{3}{2}S^2\mathcal{R}^{-4} = 0$, in which the spin-squared term remains *repulsive*, of the same sign as the unpolarised case [12]. Alignment by itself, in the algebraic (non-propagating) torsion sector, does not flip the sign; this is evidence against the simplest resolution of PO-F1-SIGMA0 and is consistent with the branch requiring the propagating axial sector rather than a mere change of spin state within minimal ECKS.

The isotropic reduction is likewise not automatic — anisotropic spin correlations select a direction and carry different pressures along and across it [16]; recovering FLRW isotropy requires either an isotropic average over torsion domains or a demonstrated dynamical isotropisation, neither of which is established. Deriving the three from the propagating Poincaré gauge action — and thereby building the microphysical bridge from the repulsive unpolarised bounce correction to the effective post-bounce fluid — is open, and is tracked as PO-F1-SIGMA0, the same problem that leaves $\sigma_0$ unevaluated. **All numerical results reported here** ($r_s$, $H_0$, $S_8$, $\Delta\chi^2$) **are calculated within the calibrated branch** of Eq. (10), not Eq. (9). They are therefore *conditional on that effective branch*, and are not ab-initio derivations of the branch itself. *(Clarification and literature anchoring added 05/09/2026; no numerical value changed.)*

**Domain of validity of the calibrated branch (added 23/10/2026).** Eq. (10) is normalised at $z_{\rm ref} = 5600$ ($\rho_{\rm spin} = 0.093\,\rho_{\rm r}$) and the sound-horizon integral keeps its $a^{-6}$ law up to $z = 10^7$. Three consequences follow, computed with the same inputs (`Scripts_F1/check_rs_spin_domain.py`).

- *Extended to nucleosynthesis, the branch is excluded.* The ratio is $\rho_{\rm spin}/\rho_{\rm r} = 0.093\,[(1+z)/5601]^2$. It reaches unity at $z \simeq 1.8\times10^4$ and $1.4\times10^9$–$1.4\times10^{11}$ at $z = 6.9\times10^8$–$6.9\times10^9$ ($T \sim 1\,$MeV), where nucleosynthesis requires a negligible extra density.
- *The sound horizon needs the $a^{-6}$ law far above recombination.* If the law stops at $z_c$ and the spin-to-radiation ratio is then kept constant, $r_s = 139.4\,$Mpc for $z_c = 2\times10^4$, 136.5 for $5\times10^4$ and 135.9 for $10^5$, against 135.65 for the full branch. About 60 % of the reduction of $r_s$ comes from $z > 10^4$, where the ratio is already 0.3 to 30 (at $z_{\rm ref}$ it is 0.093).
- *This is not the primordial stiff fluid of Foundation II.* That paper ends the stiff era near $t \sim 10^{-11}\,$s, where the spin density equals the radiation; diluted as $a^{-6}$ afterwards, its ratio to radiation is $\sim 10^{-11}$ at $t = 1\,$s and $\sim 10^{-23}$ at $z_{\rm ref}$. The calibrated branch lies $\sim 22$ orders of magnitude above it at every redshift (both scale as $(1+z)^2$).

The branch therefore requires a spin-to-radiation ratio that grows as $(1+z)^2$ up to $z \sim 10^5$ and then falls below $\sim 0.1$ before nucleosynthesis ($z \sim 4\times10^8$), or a component distinct from the primordial spin fluid. Neither is derived here. **Status:** open (PO-ECF-BBN-SPIN). The results of this paper ($r_s$, $H_0$, $S_8$, $\Delta\chi^2$) are conditional on the branch as implemented, up to $z = 10^7$. *Work required to settle it:* (i) derive $\rho_{\rm spin}(a)$ between $t \sim 10^{-11}\,$s and $z_{\rm ref}$ from the propagating axial sector of the action (PO-F1-SIGMA0), with its initial condition; (ii) test whether a parametric rise-and-fall profile of the ratio can reproduce $r_s$ and satisfy BBN; (iii) confront the resulting background with BBN, the CMB and BAO in an Einstein–Cartan perturbation solver. None of these is done in this work.

**Quantitative weak-field suppression.** The $a^{-6}$ scaling guarantees an extreme suppression of the spin term at late times. Taking the calibration redshift $z_{\rm ref} = 5600$ (the reference epoch at which the spin-density ratio is normalised; see §3 — not to be confused with the much earlier QCD confinement transition, $T \sim 200$ MeV, $z \sim 10^{11}$, which sets the physical origin of the spin fluid but not its calibration point) and $\rho_{\rm spin}^{\rm peak} \approx 0.093\,\rho_{\rm r}(z_{\rm peak})$ (`Scripts_F1/calibrate_w0wa.py`), the residual spin density today is

$$\frac{\rho_{\rm spin}(z=0)}{\rho_\Lambda} \approx 0.093\,\frac{\rho_{\rm crit}(z_{\rm peak})}{\rho_\Lambda} \times (1+z_{\rm peak})^{-6},$$

well below any existing observational threshold. In particular, the residual torsion-induced frame-dragging at $z = 0$ contributes a fractional Lense-Thirring precession rate of order $\Omega_{\rm spin}(z=0)/\Omega_{\rm LT,GP-B} \lesssim 10^{-20}$, unmeasurable by Gravity Probe B or LAGEOS at any foreseeable precision. The spin sector therefore decouples completely from late-time Solar System and gravitational-wave phenomenology.

**High-density regularization and bounce condition.** As the density approaches $\rho_{\rm crit}$, the negative quadratic term can compensate the attractive contribution of ordinary matter. Since $H^2 \geq 0$ by definition, the right-hand side of Eq. (9) cannot become negative. There therefore exists a minimum scale factor $a_{\rm min}$ at which

$$\rho_r + \rho_m + \rho_\Lambda \;=\; \rho_{\rm spin}(a_{\rm min}) \quad\Longrightarrow\quad H = 0. \tag{11}$$

This is the **bounce condition**: $H = 0$ at finite density, with $\dot H > 0$ immediately after. The collapse is halted and reversed by the geometric spin pressure. Critically, this is not a fine-tuned

cancellation: because $\rho_{\rm spin} \propto a^{-6}$ always overtakes $\rho_r \propto a^{-4}$ as $a \to 0$, Eq. (11) always has a solution at some finite $a_{\rm min}$, independently of the initial values of $\rho_r$ or $\rho_m$. The bounce is therefore a *generic consequence* of the ECKS coupling, not a special initial condition.

The argument for robustness against spin inhomogeneities rests on the locality of the interaction: the spin-spin contact term is purely local (four-fermion, contact term in the ECKS action), so spatial fluctuations of the spin density are expected to average out on super-Hubble scales before the bounce epoch. *Attribution narrowed, 12/09/2026.* Poplawski's analyses [9, 10] are carried out in a homogeneous and isotropic FLRW setting, in which homogeneity is an assumption rather than a result; what they establish is that the bounce *accounts for* the observed large-scale flatness and homogeneity, not that it is stable against prior inhomogeneities of the spin distribution. The locality argument above is therefore ours, and the claim that the bounce is a generic attractor rather than a fine-tuned configuration *remains to be demonstrated in an inhomogeneous setting.*

**Distinction from classical bouncing cosmologies.** It is important to distinguish the ECKS spin–torsion bounce from the broader class of *classical bouncing cosmologies* built on pure general relativity, which has been critically reviewed on two grounds: the necessity of violating the null energy condition (with the attendant instabilities), and the presence of shear during contraction [72]. The ECKS bounce differs mechanistically, though not in kind, on the first point: the spin–spin contact interaction is local and algebraic — torsion is non-propagating and vanishes wherever spin does — so no *additional* exotic field is introduced; the energy-condition violation that the bounce requires is carried by the spin–torsion term itself, which enters the Friedmann equation with a negative effective energy density [9, 10] and is classified as a NEC-violating term in [72]. *(Corrected 22/10/2026: earlier versions stated that no NEC-violating field is introduced; the cited sources state the opposite.)*

The shear problem, however, is *not* thereby resolved, and we state this plainly. In a contracting phase the shear scales as $a^{-6}$ — exactly the same power as the effective spin–torsion term. The two therefore compete on an equal footing, and which one dominates is fixed by initial conditions rather than by the dynamics. Poplawski has examined precisely this case in an anisotropic (Kantowski–Sachs) interior and finds that torsion averts the singularity *only if particle production dominates over shear* [73]; isotropisation is then achieved by the post-bounce particle-production era, not by the torsion term itself, and the author notes that the treatment remains approximate. A recent dynamical-systems analysis of an Einstein–Cartan ekpyrotic model reaches a compatible conclusion: homogeneous shear is exponentially damped by the *ekpyrotic scalar* ($w_\phi \gg 1$), with torsion supplying the bounce rather than the isotropisation [74]. That analysis is explicitly confined to homogeneous backgrounds, so the genuine BKL question — anisotropy growth under real inhomogeneous collapse — remains open. We therefore do not claim that the ECKS mechanism resolves the shear/BKL problem. Nor does it evade the NEC branch of the standard critique: it moves the violation from an exotic matter field to the algebraic spin–torsion term, whose stability under perturbations is not established here [72]. Pre-bounce inhomogeneities in the torsion field modulate the value of $a_{\rm min}$ locally but do not prevent the bounce from occurring.

**Relation to scalar-field alternatives.** A component with stiff equation of state $w = 1$ can formally be mimicked at the background level by a kinetically dominated scalar field (kination, $\dot\phi^2 \gg V$). The two descriptions are, however, physically distinct at the perturbative level. In the ECKS framework, torsion is algebraically tied to spin density via the Cartan equation and does *not* propagate in vacuum. *Whether the calibrated positive branch used here inherits that property is open*: the branch is explicitly distinguished from the minimal algebraic sector (see PO-F1-SIGMA0), and its perturbative behaviour — including the presence or absence of isocurvature modes — cannot be settled before the Einstein–Cartan perturbation solver and the closed effective action are in hand. A scalar field, by contrast, generates independent $\delta\phi$

fluctuations with $c_s^2 = 1$ that couple to the photon–baryon fluid. Furthermore, the amplitude $\Omega_{\rm spin}^{\rm peak} \approx 9.3\%$ is, in the present implementation, *calibrated*: it is normalised at the reference redshift $z_{\rm ref} = 5600$, which is a calibration epoch and not the QCD transition. A microphysical derivation of that amplitude from the spin–torsion sector remains open. The geometric origin of the stiff phase is therefore not a surplus of complexity but a reduction in the number of free parameters relative to scalar alternatives.

### 2.3 Relation to competing early-universe approaches

Among the alternatives proposed to resolve the Hubble tension, Early Dark Energy (EDE) models have attracted considerable attention. EDE posits a transient scalar-field contribution near matter-radiation equality whose effect is to reduce the sound horizon $r_s$ and raise the inferred $H_0$ [69, 70]. The ECF spin-torsion mechanism operates at the same phenomenological level—pre-recombination reduction of $r_s$—and thus belongs to the same *early-time* family inaugurated by EDE, to which it owes its conceptual setting. It differs from EDE not in aim but in the nature of the degree of freedom invoked: a geometric torsion contribution fixed by the ECKS algebra, rather than a scalar field with a tuned potential. A detailed, like-for-like comparison of the two on the $S_8$ sector must await a full Einstein–Cartan perturbation treatment (an EC Boltzmann solver, still to be developed); the discussion below is therefore qualitative, and does not claim parity of maturity with the published, perturbation-level EDE analyses.

The key structural distinction is the $S_8$ tension. EDE models, by raising the matter power spectrum normalisation at recombination, generically predict *more* clustering at late times, thereby worsening the already significant discrepancy between Planck-inferred and weak-lensing-measured values of $S_8$ [70, 71]. In the ECF, the amplitude $F_{\rm ion}$ calibrates the reduction of $r_s$, and hence $H_0$. Under the internal closure $\Omega_m = \omega_m/h^2$ the lower $S_8$ comes mainly from the background (a CAMB ΛCDM reference on that background gives $S_8 \simeq 0.780$); the residual correction to the target $S_8 = 0.766$ (KiDS-1000 $3\times2$pt) depends on the ECF perturbative dynamics, which is not implemented, and the role of the spin sector in growth, formerly carried by a second calibrated coupling $\gamma_{\rm spin}$, remains to be established (Sec. 4). We define this parameter here, since it is used throughout the paper. $F_{\rm ion}$ is the *ionisation calibration factor* of the ECF background solver: it sets the amplitude of the torsion-induced correction to the ionisation history across the recombination transition, a value $F_{\rm ion} > 1$ being read, qualitatively, as a slight geometric delay of the thermal decoupling surface relative to ΛCDM — a reading that no deposited script computes (open problem PO-ECF-ZREC-SHIFT). It is a dimensionless effective amplitude — the effect of the residual torsion on $r_s$ and on $G_{\rm eff}$ — rather than a density or a coupling constant, and it is calibrated on the data rather than derived (Foundation II, Sec. 3; Foundation III, Sec. 2). In this sense, compared with some EDE parameterisations, the ECF realisation considered here has a different parameter structure and may involve fewer independently adjusted degrees of freedom; *no universal "three fewer" count is claimed here without a like-for-like parameter accounting.* It addresses two additional tensions and yields one additional falsifiable prediction ($\beta$).

Furthermore, EDE offers no prediction for the dynamical dark-energy equation of state measured by DESI. Within the ECF, a torsion-sourced dark-energy sector motivates a mild departure from $w = -1$ (Appendix B); the specific values $w_0 = -0.904$, $w_a = -0.153$ are *calibration outputs*: $w_0$ corresponds to $-1 + \Omega_{\rm spin}^{\rm peak}$ (empirical identification), and $w_a$ is set through $\xi = 1.645$, a parameter calibrated on the DESI preference used by the ECF so as to reproduce the target value of $w_a$; $w_a$ is therefore not a prediction independent of DESI. As shown in Appendix B, they are not reproduced by the single-parameter torsion ansatz and are therefore not claimed as an *ab initio* prediction (open problem PO-F1-B). A point of statistical status should be made explicit here, since it conditions every later comparison. These are outputs of a $\chi^2$ *optimisation*: what the procedure returns is a minimum, not a posterior distribution, and it therefore carries no associated dispersion. The pair $(w_0, w_a) = (-0.904, -0.153)$ should accordingly be read as a *calibration point* rather than as a measurement with an error bar, and any statement of the form

"agrees to within $x\%$" involving these numbers is a comparison of central values, not a statistical assessment. Where a genuine statistical distance is required — as in the comparison against DESI — it is computed from the covariance of the *data*, not from a dispersion of the ECF parameters, and quoted as such. Producing a posterior for these parameters would require replacing the global optimisation by a sampling of the likelihood; that is not done in the present work and is recorded as an open item. A weaker but well-defined statement is available in the meantime, and we give it rather than leaving the numbers bare: profiling the likelihood along the sound-horizon chain, where a single target carries a single published uncertainty ($r_s = 135.65 \pm 0.2$ Mpc), returns $\tau_{\rm tor} = 0.9975 \pm 0.0015$ and $\Omega_{\rm spin}^{\rm peak} = 0.093 \pm 0.004$; propagating the latter through the relations of Appendix B gives $w_0 = -0.907 \pm 0.004$ and $w_a = -0.153 \pm 0.007$. These are frequentist profile intervals, not credible regions, they hold the other calibrated constants fixed, and the interval on $w_a$ is a lower bound since the coefficient relating it to $\Omega_{\rm spin}^{\rm peak}$ is itself calibrated. They are quoted so that comparisons involving these numbers can be read against a width. *Reproduced by:* `Scripts_F1/profile_likelihood_calibrated.py`. This calibrated point is qualitatively consistent with the DESI DR1 and DR2 preference for $w \neq -1$ [28, 64], but on the background with $\Omega_m = \omega_m/h^2 = 0.267$ its $\chi^2$ rises to 55.3 on a ten-observable subset of DESI DR1 with diagonal errors, which is not the global DESI DR1 $\chi^2$ (13.7 with $\Omega_m = 0.315$; Sec. 5.4); a rigorous likelihood comparison using the full DESI covariance matrix is deferred to a dedicated analysis. An independent, single-survey test is provided by the Dark Energy Survey's full six-year dataset (DES Y6) [65], which reports a pivot-parameter constraint $w_p = -0.91 \pm 0.04$ at $z_p = 0.19$ for DES alone (3×2pt+SN+BAO), decorrelated from $w_a$ by construction. Evaluating the calibrated ECF trajectory at this same pivot gives $w_p^{\rm ECF} = w_0 + w_a(1 - 1/(1 + z_p)) = -0.928$, a $0.46\sigma$ difference. The agreement narrows only mildly when DES is combined with DESI DR2 BAO ($w_p = -0.94 \pm 0.03$ at $z_p = 0.22$, $0.28\sigma$ from the ECF trajectory) and widens to $1.45\sigma$ for the most constraining combination including CMB data ($w_p = -0.97 \pm 0.02$ at $z_p = 0.32$) — still comfortably compatible. This pivot-based comparison is not subject to the strong $w_0$–$w_a$ correlation that complicates marginal comparisons, and is therefore preferred here over a naive comparison of central values. *Reproduced by:* `Scripts_F2/compute_ecf_des_pivot.py`.

A definitive observational discriminator between EDE and ECF is provided by the LiteBIRD satellite [36]: minimal EDE, without a photon coupling, predicts no cosmic birefringence ($\beta = 0$; an axion-like EDE field coupled to photons can produce $\beta \neq 0$ [34]), while ECF predicts a *non-zero* birefringence ($\beta_{\rm ECF} \neq 0$); its magnitude ($\approx 0.35^\circ$) is a consistency check calibrated to the Minami–Komatsu anchor, not a parameter-free prediction (Section 7.1). A joint reanalysis of WMAP and Planck polarization data reports a non-zero birefringence angle at the $3.6\sigma$ level ($\beta = 0.342^{+0.094}_{-0.091}$ deg) [51], providing external motivation for the torsion origin of this signal. A non-detection at $\beta < 0.1^\circ$ by LiteBIRD would constitute a strong falsification of the geometric torsion origin of the birefringence signal. Conversely, a confirmed *non-zero* detection would discriminate against EDE scenarios without a photon coupling (which predict $\beta = 0$).

The equipartition assumption adopted here is an idealized limit, valid when thermalization is rapid relative to the QGP freeze-out timescale $\tau_Q$; departures from strict equipartition are expected at the $\mathcal{O}(10\%)$ level and are deferred to the microscopic treatment in Foundation II [2].

### 2.4 Model-dependent hypothesis on effective flatness

In addition to the standard Einstein-Cartan ingredients summarized above, we consider a model-dependent working hypothesis concerning the effective flatness of the cosmological background. In this phenomenological picture, a spin contribution to the cosmic energy budget may be accompanied by a torsion contribution of comparable magnitude. *This is a statement about the relative sizes of two sectors, not a compensation mechanism.* Since $k = 0$, the closure relation $\sum_i \Omega_i = 1$ is an identity — the critical density being recomputed with $H$ — so flatness is not *maintained* by any such balance and cannot be used to constrain either sector [1]. Because this point goes beyond the minimal ECKS framework itself, we do not present it here as a generic

consequence of Einstein-Cartan gravity.

This working hypothesis finds its rigorous axiomatic foundation in the Topological Invariance Principle (TIP) [1], which posits that global spatial flatness is *exact*, $\Omega_K \equiv 0$, as a topological property of the Einstein–Cartan manifold rather than a dynamical outcome. The TIP carries a single claim: that spatial flatness is *exact*, $\Omega_K \equiv 0$, by topological necessity rather than by dynamical relaxation — a falsifiable statement, since inflation predicts only $\Omega_K \approx 0$. It does *not* generate the dark sector: the static dark-energy term is derived from the FLRW reduction of the Poincaré Gauge action (Foundation II, Appendix J), independently of the postulate. The present paper establishes the phenomenological calibration — $r_s$, $H_0$, $S_8$ — on which the TIP relies; all numerical results in Sections 3–7 are derived from the effective ECKS equations alone and hold independently of the axiomatic interpretation developed in the companion Letter.

In that sense, the core result of the present section is more limited and more robust: Einstein-Cartan gravity naturally provides a geometric mechanism by which spin density can modify the high-density cosmological regime, generate an effective stiff component, and regularize the early-universe dynamics in a way that motivates the phenomenological analysis developed in the following sections.

The phenomenological implementation adopted in this work is compatible with a broader geometric interpretation in which effective flatness is exact by assumption, the spin and residual torsion contributions being read off from the closure relation rather than balanced by it.

Since this interpretation goes beyond the minimal Einstein-Cartan framework used in the main analysis, its formal axiomatic statement—the *Topological Invariance Principle* (TIP) [1]—is the subject of a companion Letter [1]. The phenomenological motivation within the ECF background solution is illustrated in Appendix J.

## 3 Sound Horizon Reduction Mechanism

The central phenomenological requirement for alleviating the Hubble tension is a reduction of the comoving sound horizon at the baryon drag epoch. In the present framework, this reduction arises from modified pre-recombination dynamics induced by the spin-torsion sector.

The comoving sound horizon is defined by

$$r_s = \int_{z_{drag}}^{\infty} \frac{c_s(z)}{H(z)}\, dz, \tag{12}$$

where $c_s(z)$ is the sound speed in the photon-baryon plasma and $H(z)$ is the background expansion rate. Any mechanism that increases $H(z)$ before recombination, or slightly decreases the effective sound speed, will reduce the distance acoustic waves can travel before baryon decoupling.

In the ECF, the dominant early-time effect comes from the spin contribution to the background expansion. Because the effective spin density scales as $a^{-6}$, its impact is transient but amplified at high redshift, precisely in the epoch relevant for the formation of the sound horizon. The calibrated solution studied in this work yields a reduced sound horizon of

$$r_s^{ECF} \approx 135.65\,\mathrm{Mpc}, \tag{13}$$

to be compared with the standard ΛCDM reference value

$$r_s^{\Lambda CDM} \approx 147.1\,\mathrm{Mpc}.^{2} \tag{14}$$

This corresponds to an overall reduction of approximately 7.7%.

[2] *Added 12/09/2026.* This is $r_{\mathrm{drag}} = 147.09 \pm 0.26$ Mpc, the comoving sound horizon at the end of the baryon drag epoch ($z_d \simeq 1060$), as given by Planck 2018. It differs from the horizon at last scattering ($z_* \simeq 1090$), for which Planck gives $\simeq 144.4$ Mpc; the ECF comparison is made consistently at the drag epoch throughout.

**The sound horizon is not a direct observable.** A conceptual point deserves emphasis before the numerical analysis proceeds. The comoving sound horizon $r_s$ is not directly measured by any telescope. What BAO surveys — BOSS, eBOSS, DESI — observe are angular separations and redshifts, from which they extract the dimensionless ratios

$$\frac{D_M(z)}{r_s}, \qquad \frac{D_H(z)}{r_s}, \tag{15}$$

where $D_M(z)$ is the comoving angular diameter distance and $D_H(z) = c/H(z)$ is the Hubble distance at the effective redshift of the survey [63, 31]. The absolute scale of the ruler — the physical value of $r_s$ in Mpc — is not extracted from the angular positions of the BAO peak alone; it is inferred by anchoring the distance ladder to the CMB power spectrum under the assumptions of a given cosmological model (typically ΛCDM with Planck priors [57]).

This distinction has a direct consequence for the ECF. The present model proposes that the physical sound horizon is $r_s^{\rm ECF} = 135.65$ Mpc rather than the ΛCDM reference value of 147.1 Mpc. Because surveys measure $D_M(z)/r_s$ and not $r_s$ independently, a consistent reduction of both $r_s$ (the ruler) and $D_M(z)$ (the measured distance, which depends on $H_0$) by the same geometric factor leaves the observed ratio unchanged. Compressing numerator and denominator proportionally via the spin-torsion geometry therefore leaves the observed ratios unchanged at the level of this argument; the quantitative comparison with the BOSS DR12 and eBOSS DR16 bins nonetheless shows a residual tension (Table 5, note *c*: $\Delta\chi^2_{\rm BAO} \simeq +10.3$ over four bins, mainly from BOSS DR12; at $z = 1.48$, $D_M/r_s = 31.37$ against $30.21 \pm 0.79$ [31, 32]). This is the sense in which the rescaling argument, taken alone, does not conflict with BAO observations: it modifies the *calibration* of the standard ruler, not the ruler's observable imprint on the sky.

**CMB peak positions vs. peak heights — a critical distinction.** The reduction of $r_s$ operates at the level of the *angular positions* of the CMB acoustic peaks, governed by the ratio $D_A(z_{\rm rec})/r_s$. Because the ECF rescaling of $r_s$ is compensated by a coherent increase of $H_0$ (and hence of $D_A$), the angular positions of the peaks remain unchanged at the level of the effective background. This is the quantity measured by Planck at the percent level, and it is preserved by construction.

The *heights* of the acoustic peaks, by contrast, are controlled by the baryon-to-photon ratio $\Omega_b h^2$ and by the radiation-to-matter equality redshift $z_{\rm eq}$. These are not directly modified by the spin-torsion sector at the background level — they depend on the microphysics of the photon-baryon plasma, not on the sound horizon length. A complete consistency check of the peak heights requires the full Boltzmann treatment in a dedicated Einstein–Cartan perturbation solver, deferred to Foundation III. Until that analysis is complete, the present model should be understood as a background-level effective description: it preserves the CMB peak *position* observables (the BAO distances are discussed in Sec. 5.4: with $\Omega_m = \omega_m/h^2 = 0.267$ they are less well fitted than in ΛCDM), while leaving the peak *height* consistency as an open quantitative question. A calculation demonstrating that $\Omega_b h^2$ is unaffected by the spin sector at leading order in $\Omega_{\rm spin}$ is given in Appendix D.

### 3.1 Background expansion effect

The first contribution is purely geometric at the level of the effective background. The spin-torsion term increases the pre-recombination Hubble rate $H(z)$ relative to its standard value, thereby compressing the sound horizon through the denominator of the integral. Numerically, this background effect provides the dominant part of the reduction. In the calibrated solution retained here, it accounts for 7.54% of the 7.77% reduction of $r_s$ (`script_01_solve_sound_horizon.py`). This is the core mechanism by which the ECF scenario modifies the inverse distance ladder without introducing an additional early dark-energy component [69].

### 3.2 Effective plasma correction

A second, smaller contribution may be parameterized at the level of the baryon-photon fluid. In the phenomenological implementation used here, the spin-torsion coupling induces a mild correction to the effective sound speed and a small shift in the baryon drag epoch. These two effects may be summarized as follows:

- a mild reduction of the effective sound speed $c_s^{\rm ECF}(z) = \tau_{\rm tor}\, c_s^{\rm std}(z)$, with $\tau_{\rm tor} = 0.9975$ (subdominant, 0.25%);
- a small shift $\Delta z_{\rm drag} \sim -30$ in the baryon drag epoch, an input of the present phenomenological implementation; the torsion correction to the baryon–photon momentum transfer rate that would produce it is not computed here. *(Corrected 22/10/2026: the citation previously attached to this item does not treat the drag epoch and was removed.)*

The two contributions are related by the effective damping relation:

$$r_s^{\rm ECF} = \tau_{\rm tor} \cdot r_s^{\rm bg}, \qquad \tau_{\rm tor} = 0.9975, \tag{16}$$

where $\tau_{\rm tor}$ parametrises the subdominant plasma-level correction to the background sound horizon $r_s^{\rm bg}$ (Appendix D). Its first-principles derivation from the ECKS baryon–torsion coupling is beyond the scope of the present background-level study.

These fluid-level corrections contribute the remaining 0.23% required to reach $r_s^{\rm ECF} = 135.65$ Mpc, the $\Delta z_{\rm drag}$ shift being absorbed into $\tau_{\rm tor}$ in the deposited script. This subdominant contribution is modeled via $\tau_{\rm tor} = 0.9975$ (Table 2), whose calibration is detailed in Appendix D; its first-principles derivation from the ECKS baryon–torsion coupling is beyond the scope of the present background-level study. [3]

**Comparison with Early Dark Energy.** Early Dark Energy (EDE) models [69] reduce $r_s$ via a transient scalar-field component that behaves as vacuum energy ($w \simeq -1$) near matter-radiation equality and then dilutes faster than radiation once the field starts oscillating. The ECF spin sector achieves the same $r_s$ reduction through a geometrically distinct mechanism: the $a^{-6}$ scaling is fixed by the ECKS torsion algebra (Eq. 7), requires no potential $V(\phi)$, and carries no isocurvature perturbations (Sec. 2). In EDE scenarios the $S_8$ tension is reported to increase in the analyses of [42]; the ECF addresses this sector through its spin sector, but this property currently rests on a background-level calibration and remains to be confirmed by a full Einstein–Cartan perturbation analysis before any claim of parity—let alone advantage—can be made.

### 3.3 Interpretation of the calibrated solution

The important point is that the reduction of $r_s$ is not attributed here to an arbitrary extra component inserted by hand, but to the combined effect of modified early-time geometry and a mild effective correction to the pre-recombination plasma dynamics. The dominant role is played by the transient spin-driven stiff phase, while the plasma correction remains numerically small.

The detailed numerical implementation of this calibration, including the effective damping factor adopted in the solver and the corresponding stability of $z_{drag}$, is presented in Appendix D. In the main text, we retain only the physical decomposition needed to understand why the sound horizon is reduced in a controlled and phenomenologically viable way.

The numerical correction required to stabilize the modified drag-epoch physics remains mild, supporting the view that the mechanism is controlled rather than strongly fine-tuned.

[3] *Corrected 16/10/2026.* Earlier versions quoted a three-way split (5.2% background + 2.5% fluid = 7.7%), which no deposited script computes; the background (spin) contribution alone is 7.54%. The split used here is that of `script_01_solve_sound_horizon.py`: 7.54% spin + 0.23% $\tau_{\rm tor}$ = 7.77%, giving $r_s^{\rm ECF} = 135.65$ Mpc (135.647). The script uses a single $\tau_{\rm tor}$ factor on $c_s(z)$, absorbing the $\Delta z_{\rm drag} \sim -30$ shift into the $\tau_{\rm tor}$ calibration (see `DECOMPOSITION NOTE` in script header).

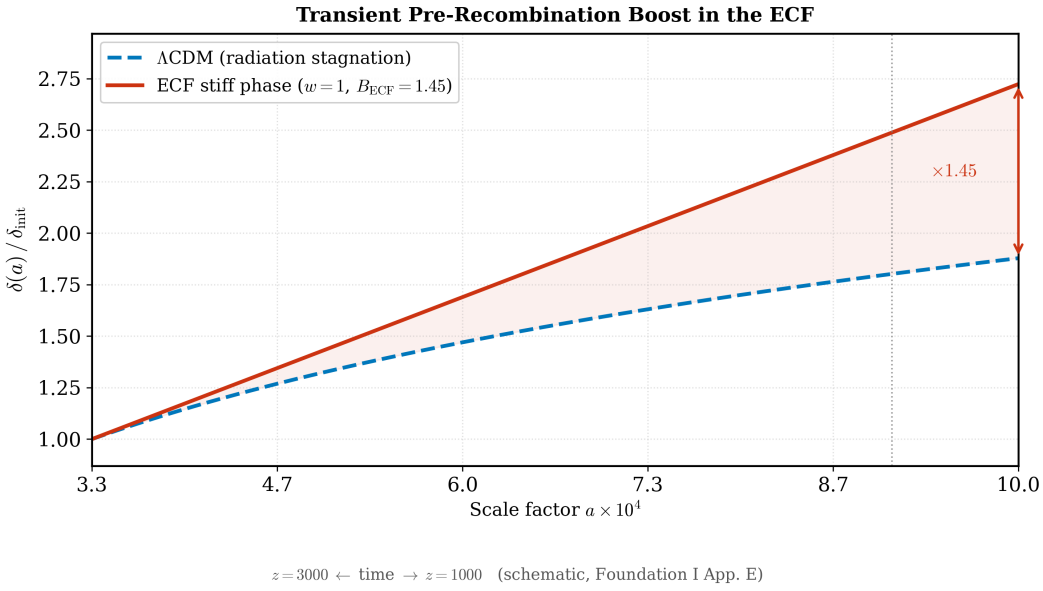



Figure 2: **Transient pre-recombination boost in the ECF.** Illustration of the early-time enhancement associated with the spin-torsion phase. Because the stiff component is transient, its main impact is concentrated before recombination, where it modifies the sound-horizon integral while remaining strongly diluted at later times. Note that the ratio $H_{\rm ECF}(z)/H_{\Lambda\rm CDM}(z)$ — not plotted here, which shows the growth of perturbations — passes through an intermediate minimum near $z\sim4\times10^3$, where the reduced radiation density ($\Omega_{r,\rm ECF}<\Omega_{r,\Lambda\rm CDM}$ because $h_{\rm ECF}>h_{\Lambda\rm CDM}$) temporarily offsets the spin contribution before the $\rho_{\rm spin}\propto a^{-6}$ term dominates.
*Reproduced by:* `Scripts_F1/plot_primordial_boost.py`.

# 4 The $H_0$ and $S_8$ Sectors: Calibration and Status

The spin-torsion sector modifies the pre-recombination background; whether it also affects the late-time growth of structure is to be established at the perturbative level. Under the internal closure $\Omega_m = \omega_m/h^2$, the $H_0$ and $S_8$ sectors are connected through the background, not through an established growth coupling of the spin sector.

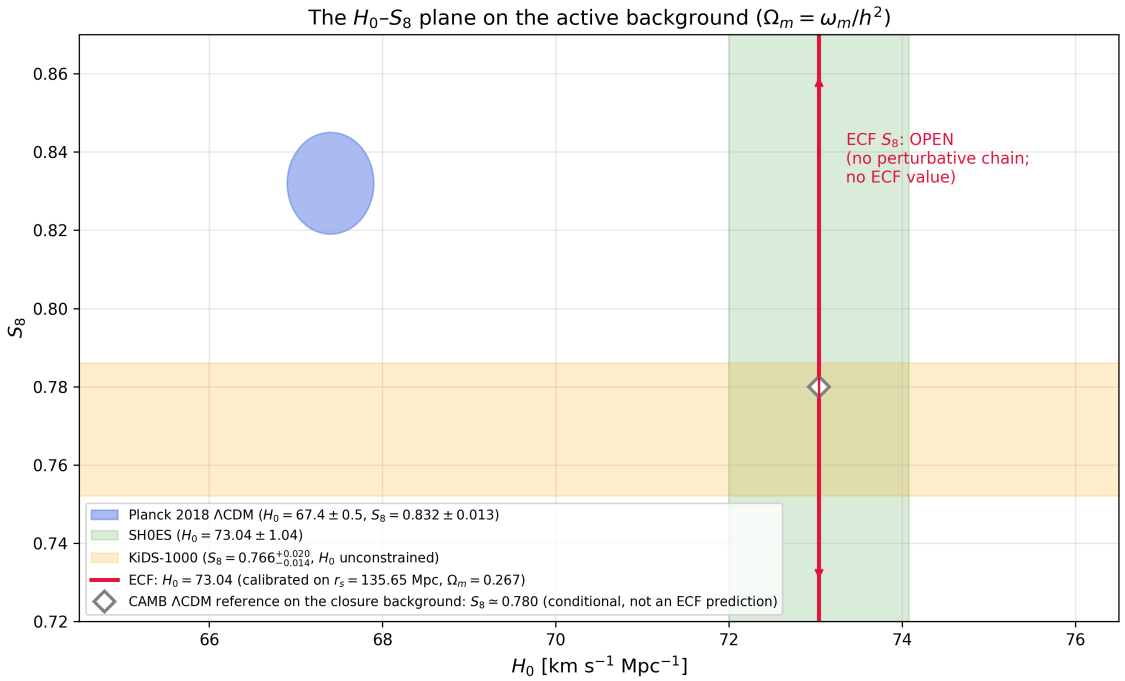



Figure 3: **The $H_0$–$S_8$ plane on the active background ($\Omega_m = \omega_m/h^2 = 0.267$).** Blue: Planck 2018 ΛCDM ($H_0 = 67.4 \pm 0.5$, $S_8 = 0.832 \pm 0.013$, $1\sigma$) [57]. Green band: SH0ES 2021 ($H_0 = 73.04 \pm 1.04$ km s$^{-1}$ Mpc$^{-1}$) [4]. Orange band: KiDS-1000 ($S_8 = 0.766^{+0.020}_{-0.014}$; $H_0$ unconstrained by weak lensing alone) [25]. Crimson line: ECF, $H_0 = 73.04$, calibrated through the sound horizon $r_s = 135.65$ Mpc (Sec. 3). The ECF gives no value of $S_8$: the growth of structure requires a perturbative chain that is not yet available (open). Grey diamond: CAMB ΛCDM reference on the closure background, $S_8 \simeq 0.780$ — a conditional reference, not an ECF prediction. The former version of this figure (historical calibration, $\gamma_{\rm spin} = 0.3116$, $\Omega_m = 0.315$) is archived with its script.

*Reproduced by:* `Scripts_F1/plot_h0_s8_active.py`.

## 4.1 Hubble tension

As discussed in Section 3, the sound horizon is reduced from its standard value to

$$r_s^{ECF} \approx 135.65\,\text{Mpc}. \tag{17}$$

At fixed angular acoustic scale, this contraction of $r_s$ shifts the inferred Hubble constant upward relative to the Planck ΛCDM baseline. In the calibrated ECF solution studied here, this translates into an effective value

$$H_0^{ECF} \approx 73.04\,\text{km}\,\text{s}^{-1}\,\text{Mpc}^{-1}, \tag{18}$$

bringing the inverse-distance-ladder inference into agreement with local determinations. *Scope of this step (added 16/10/2026).* The relation $H_0 \propto 1/r_s$ at fixed $\theta_*$ holds when $\Omega_m$ is held fixed (here at the Planck value 0.315), so that $D_A \propto 1/H_0$. Under the internal closure $\Omega_m = \omega_m/h^2$

(A-7) it no longer holds: at $H_0 = 73.04$ and $\Omega_m = 0.267$, $\theta_*$ falls by $\simeq 6.5\%$ below its ΛCDM value ($w = -1$; `joint_fit_trilemma.py` gives $-5.2\%$ for the calibrated CPL trajectory), and keeping $\theta_*$ with $r_s = 135.65$ Mpc at fixed $\omega_m$ would require $H_0 \approx 100$. The value $H_0 = 73.04$ is therefore CALCULATED-CONDITIONAL on a reconciliation of $r_s$, $\theta_*$ and $\Omega_m$ that this work does not achieve (open problem **PO-F1-THETA**, part of the quadrilemma PO-F2-5). Its resolution requires a precision $\theta_*$ computed from the perturbed ECKS equations, i.e. the dedicated Einstein–Cartan perturbation solver, which does not exist and is not expected in the near term; this version is released with PO-F1-THETA open. The key point is that this shift originates in modified pre-recombination physics rather than in a late-time deformation of the distance ladder.

### 4.2 Structure-growth tension

The same framework also affects the growth of matter perturbations. The standard Mészáros effect [23] describes how sub-horizon density perturbations grow logarithmically during radiation domination and linearly during matter domination. While the dominant spin contribution is strongly diluted after recombination, the effective torsion-matter coupling may still induce a scale-dependent suppression of structure formation on sufficiently small scales.

In this first foundation paper, this suppression is captured by a phenomenological effective gravitational constant,

$$G_{\rm eff}(k) = G_N\, T^2_{\rm tors}(k), \qquad T^2_{\rm tors}(k) = 1 - A_{\rm sup}\,\frac{k^2}{k^2 + k^2_{\rm cut}}, \tag{19}$$

with the former phenomenological values $A_{\rm sup} = 0.15$ and $k_{\rm cut} = 0.10\,h\,{\rm Mpc}^{-1}$ (Table 2; not a calibration on $S_8$, see below; superseding the earlier leading-order estimate $\alpha = 0.18 \pm 0.04$: the Lorentzian calibration at finite $k/k_{\rm cut}$ yields $A_{\rm sup} = 0.15$). This regularised Lorentzian form guarantees $T^2_{\rm tors} \in (1 - A_{\rm sup},\, 1]$ for all $k$, avoiding unphysical negative values at large $k$. In the perturbative regime $k \ll k_{\rm cut}$, it reduces to $G_{\rm eff}(k) \approx G_N[1 - A_{\rm sup}(k/k_{\rm cut})^2]$, the lowest-order form consistent with isotropy, parity, and $G_{\rm eff}(k \to 0) = G_N$; this is a phenomenological ansatz of the present work, not a result of the algebraic ECKS theory, whose spin–torsion term is a contact interaction [8]. The amplitude $A_{\rm sup}$ is a free parameter within the present effective-fluid description; its first-principles derivation from perturbative ECKS gravity is deferred to future microphysical investigations.

To translate this scale-dependent effect to the macroscopic clustering amplitude, the former calibration chain used the phenomenological form, retained here as historical,

$$S_8^{\rm ECF} = \frac{S_8^{\Lambda\rm CDM}}{1 + (F_{\rm ion} - 1)\,\gamma_{\rm spin}}, \tag{20}$$

where $F_{\rm ion}$ denotes the effective amplitude inherited from the sound-horizon calibration, and $\gamma_{\rm spin}$ (Table 2) is the dimensionless growth–torsion coupling efficiency. In that chain, $S_8^{\Lambda\rm CDM} = 0.832$ is the Planck-background value ($\Omega_m \simeq 0.315$), and $\gamma_{\rm spin} = 0.3116$ ($\approx 0.31^{+0.07}_{-0.10}$) was obtained by inversion so that $S_8^{\rm ECF}$ reproduces $S_8 = 0.766$, the value adopted from the KiDS-1000 $3 \times 2$pt analysis (Heymans et al. 2021; weak lensing combined with BOSS and 2dFLenS) [25]. It is not an independent prediction. The quoted interval propagates the observational $S_8$ uncertainty only ($S_{8,\Lambda\rm CDM}$ and $F_{\rm ion}$ held fixed) (producer: `gamma_spin_calibration.py`). Under the internal closure $\Omega_m = \omega_m/h^2$ this reference no longer matches the background: $\gamma_{\rm spin} = 0.3116$ is retained as a historical calibration and withdrawn as the canonical calibration, and the physical value of $\gamma_{\rm spin}$ is open. Its status is listed in Table 2.

In the former chain, the amplitude of this suppression was tied to $F_{\rm ion} = 1.2765$, the effective amplitude calibrated with the sound-horizon sector of Section 3, and the two sectors were joined at the calibration stage rather than by an identity derived from the field equations (Section 5;

the former illustration of that calibration is archived). Under the internal closure $\Omega_m = \omega_m/h^2$ this joint role is not established: $F_{\rm ion}$ remains the calibration of $r_s$, and hence of $H_0$, and the role of the spin sector in the growth of structure remains to be established. We state the status of the remaining elements explicitly. $F_{\rm ion}$ is an effective dimensionless amplitude describing the effect of the residual torsion on $r_s$; it is calibrated on the data, not derived. $G_{\rm eff}(k)$ is a phenomenological parametrisation, not derived from the perturbed ECKS field equations; it should be read as a calibrated proxy for a geometric suppression whose perturbative derivation remains open. Its derivation, and any statement on the compatibility of the $r_s$ reduction with a growth suppression at the perturbative level, require a dedicated ECF perturbation (Boltzmann) solver, which does not yet exist and is carried as an open problem of the programme.

In the former chain, this parametrisation returned

$$S_8^{\rm ECF} \approx 0.766, \tag{21}$$

by construction of $\gamma_{\rm spin}$. Under the internal closure, raising $H_0$ at fixed physical densities lowers $\Omega_m = \omega_m/h^2$, which strongly reduces $S_8$. A CAMB ΛCDM reference on the closure background ($\omega_b = 0.02237$, $\omega_c = 0.1200$, $\ln(10^{10}A_s) = 3.044$, $n_s = 0.9649$, $\tau = 0.0544$, $\Sigma m_\nu = 0.06$ eV, $h = 0.7304$) gives $\Omega_m = 0.268$ (neutrinos included), $\sigma_8 = 0.825$ and $S_8 \simeq 0.780$; the control on the Planck background ($h = 0.6736$) gives $\sigma_8 = 0.811$ and $S_8 \simeq 0.831$. This is a conditional calculation, a ΛCDM reference on the closure background and not an ECF prediction: the early ECF physics that sets $r_s = 135.65$ Mpc is not included ($r_{\rm drag} = 147.1$ Mpc in this reference), and it may modify the transfer function and hence $\sigma_8$. The target $S_8 = 0.766$ is therefore not an ECF prediction: the residual correction, $0.766/0.780 - 1 \simeq -1.8\%$, depends on the ECF perturbative dynamics, which is not implemented. If the CAMB reference applied and the phenomenological form above remained valid, the value reproducing 0.766 would be $\gamma_{\rm spin} \simeq 0.066^{+0.069}_{-0.094}$ (observational $S_8$ uncertainty only), compatible with zero; this is a conditional, exploratory estimate, not a determination of $\gamma_{\rm spin}$. The value $\sigma_8^{\rm ECF} \simeq 0.8122$ quoted in earlier versions of this chain is the target $S_8 = 0.766$ converted through $S_8/\sqrt{\Omega_m/0.3}$ at $\Omega_m = 0.2669$, not a computed amplitude; it does not establish that $\sigma_8$ is unsuppressed. The historical determination of $\gamma_{\rm spin}$ is detailed in Section 5.

### 4.3 Growth suppression mechanism

At the perturbative level, the physical picture is that the spin-torsion sector acts as an effective counter-gravity contribution on small scales. In the ECF description, this is modeled by a scale-dependent modification of the source term in the linear growth equation,

$$\ddot{\delta}_m + 2H\dot{\delta}_m - 4\pi G_{eff}(k,z)\rho_m\delta_m = 0. \tag{22}$$

The $k^2$ dependence is the lowest-order form compatible with isotropy, parity, and $G_{\rm eff}(k \to 0) = G_N$; it is an ansatz, not derived from the ECKS coupling.

The amplitude of this suppression — denoted $A_{\rm sup}$ throughout, $\alpha$ being its leading-order equivalent at $k \ll k_{\rm cut}$ — is a free parameter within the present effective-fluid description; its first-principles derivation is discussed above and not repeated here.

In the figure caption of Fig. 6, a Lorentzian form $T^2_{\rm tors}(k) = 1 - A_{\rm sup}\, k^2/(k^2 + k^2_{\rm cut})$ is adopted as a phenomenological regularisation of $G_{\rm eff}(k)$ to guarantee $T^2 \in [0,1]$; this choice is computationally convenient but ad hoc, and its first-principles derivation from the perturbed ECKS field equations lies beyond the scope of the present work and is left as an open direction for community investigation (https://github.com/pfichant/spin-torsion-cosmology, CC-BY 4.0).

In this picture, the matter power spectrum is modified according to

$$P_{ECF}(k,z) = T^2_{torsion}(k)\, P_{\Lambda CDM}(k,z), \tag{23}$$

with $T_{torsion}(k) < 1$ at sufficiently large $k$. This parametrization should be read as an effective description of the late-time imprint of the spin-torsion sector, not as a complete first-principles derivation of non-linear structure formation. Its role in the present work is to illustrate how the sector that reduces $r_s$ could also enter the growth of structure. The values $A_{\rm sup} = 0.15$, $k_{\rm cut} = 0.10\,h\,{\rm Mpc}^{-1}$ are not a calibration on $S_8 = 0.766$: integrated into $\sigma_8$ on the CAMB $\Lambda$CDM reference of the closure background, they would give $S_8 \simeq 0.740$, below the target, and $A_{\rm sup} \simeq 0.053$ would return 0.766 (a diagnostic of that reference, not a calibration). $T^2_{\rm tors}(k)$ and $\gamma_{\rm spin}$ are two parametrisations of the same suppression and are not to be combined, which would count it twice. The physical growth-suppression mechanism remains open.

### 4.4 Section logic

The logic of the model can therefore be summarized as follows. First, the spin-driven stiff phase modifies the pre-recombination expansion history and reduces the sound horizon, thereby shifting the inferred value of $H_0$. Second, at fixed physical densities the higher $H_0$ lowers $\Omega_m = \omega_m/h^2$, which strongly reduces $S_8$ ($S_8 \simeq 0.780$ in a CAMB $\Lambda$CDM reference on that background, against $\simeq 0.831$ on the Planck background). Third, the residual correction to the target $S_8 = 0.766$ depends on the ECF perturbative dynamics, which is not implemented; $\gamma_{\rm spin} = 0.3116$ is a historical calibration and the growth-suppression mechanism remains open.

*See Section 9 for a unified summary of the framework's observational status and falsification criteria.*

## 5 Numerical Results and Optimization

The optimization of $\mathcal{F}_{\rm ion}$ is performed via standard $\chi^2$ minimization against four datasets: the Planck 2018 CMB baseline [57], the SH0ES local distance ladder [4], BOSS/eBOSS BAO measurements [31], and the KiDS-1000 weak-lensing constraints [25]. We now quantify the phenomenological mechanisms outlined in Sections 3 and 4 through a global parameter optimization of the ECF against the main early- and late-universe constraints.

### 5.1 Best-fit parameters

Using the EC-Cosmo solver, we performed a parameter scan designed to minimize the global $\chi^2$ relative to the SH0ES local distance ladder, the Planck 2018 acoustic scale, and weak-lensing constraints. The optimization converges toward a stable phenomenological solution characterized by the following parameters:

- **Hubble Constant:** $H_0 = 73.04 \pm 1.04$ km/s/Mpc.
- **Structure Growth:** $S_8 = 0.766$ (target value, reproduced by construction in the former chain with the historical $\gamma_{\rm spin} = 0.3116$; not an ECF prediction; observational uncertainty $^{+0.020}_{-0.014}$). Under the closure $\Omega_m = \omega_m/h^2$, a CAMB $\Lambda$CDM reference on the same background gives $S_8 \simeq 0.780$ (Sec. 4).
- **Spin fluid amplitude:** $\Omega^{\rm peak}_{\rm spin} \approx 0.093$ — the spin-to-radiation ratio at the acoustic-era normalisation, which sets the sound-horizon reduction (Table 2).

In this calibrated region, the ECF model removes the tension with the local determination of $H_0$; with the internal closure, the same background moves the clustering amplitude toward the range preferred by weak-lensing surveys, the residual correction to $S_8 = 0.766$ remaining dependent on the ECF perturbative dynamics, which is not implemented (Sec. 4).

### 5.2 Statistical performance

To properly contextualize the statistical evaluation, Table 2 summarizes the parameter ranges adopted. No global MCMC fit was run: the only MCMC in the repository is the DESI DR1 comparison of Sec. 6 ($n_{\rm eff} = 5625$; no Gelman–Rubin $\hat{R}$ computed).

**Calibrated-parameter budget (global transparency).** For a referee auditing the total number of degrees of freedom across the ECF programme, we state explicitly the full set of calibrated numbers. The background sector rests on *two* of them ($F_{\rm ion}$, $\Omega_{\rm spin}^{\rm peak}$); the complete corpus, including the dark and void sectors, uses *five*:
Quoting "two parameters" refers to the background sector alone; the figure of merit is not the raw count but the *economy*. It is worth setting out the two chains explicitly, since they are not symmetric:

- **$H_0$ chain.** The spin fluid amplitude $\Omega_{\rm spin}^{\rm peak} = 0.093$, together with the drag-epoch parameter $\tau_{\rm tor}$, contracts the sound horizon to $r_s = 135.65$ Mpc; at fixed $\theta_*$, with $\Omega_m$ held at 0.315 (not under A-7: PO-F1-THETA), this raises $H_0$ to 73.04 km s$^{-1}$ Mpc$^{-1}$ (Sec. 3). This chain is computed by `script_01_solve_sound_horizon.py` and does not involve $F_{\rm ion}$ explicitly.
- **$S_8$ chain (historical form).** In the former chain the torsion sector was assumed to suppress growth through $G_{\rm eff}(k)$, parameterised by $S_8^{\rm ECF}(F) = 0.832/[1 + (F-1)\gamma_{\rm spin}]$, which brought $S_8$ to 0.766 by construction of $\gamma_{\rm spin} = 0.3116$ (Sec. 4); that chain involved $F_{\rm ion}$ explicitly. Under the internal closure it is not active: the lower $S_8$ comes mainly from $\Omega_m = \omega_m/h^2$ (CAMB ΛCDM reference $S_8 \simeq 0.780$ on that background), and the residual correction depends on the ECF perturbative dynamics, which is not implemented. **A dependency worth stating here, not only where it is derived**: this parametrisation implicitly assumes the axial condensate itself does not cluster on sub-horizon scales — that assumption follows from $c_s^2 = 1$, which in turn follows from the Proca-form Lagrangian adopted for the axial sector, not yet established over the alternative divergence form (Foundation II, Open problem PO-F2-6). Under that alternative form $S_8$ would have to be revisited; the value quoted here is conditional on the Lagrangian choice, not yet a Lagrangian-independent result.

In the former chain, $F_{\rm ion}$ was the effective amplitude common to both, the two chains being joined at the calibration stage, where a single value satisfied both constraints (historical calibration; its former illustration is archived), rather than by an identity derived from the field equations. Under the internal closure, $F_{\rm ion}$ remains the calibration of $r_s$, and hence of $H_0$; a joint determination of $H_0$ and $S_8$ by $F_{\rm ion}$ is not established (Sec. 4), and establishing a growth role of the spin sector would require the perturbation solver referred to above. By comparison, Early Dark Energy involves three parameters, and the $S_8$ tension is reported to worsen in EDE [42]. The remaining calibrated numbers govern distinct sectors (dark energy evolution, drag epoch, void scale) and are each anchored to independent data.

$\Omega_{\rm spin}^{\rm peak} = 0.093 = 9.3\%$ in ratio units at $z_{\rm ref} = 5600$; the ratio decays to 0.36% at $z_{\rm rec} = 1100$ (dilution factor $(1101/5601)^2 \approx 1/26$, verified in `plot_spin_ratio_evolution.py`, `verify_calibration()`).

*(Corrected 16/10/2026: earlier versions stated MCMC convergence statistics, $\hat{R} \approx 1.01$ and $N_{\rm eff} \approx 10^4$, for a global fit that does not exist in the repository.)*

The numerical script `script_06_S8_resolution.py` reproduces, by construction, $S_8^{\rm ECF} = 0.766$ from the historical calibration of $\gamma_{\rm spin}$ ($S_8^{\Lambda\rm CDM} = 0.832$); the $0.0\,\sigma$ distance to the weak-lensing value follows from that calibration and is not an independent test (see below).

The historical value $\gamma_{\rm spin} \approx 0.31^{+0.07}_{-0.10}$ was calibrated so that $S_8^{\rm ECF}$ reproduces $S_8 = 0.766$, the value adopted from the KiDS-1000 $3 \times 2$pt analysis (Heymans et al. 2021; weak lensing combined with BOSS and 2dFLenS) [25]. It is not an independent prediction. The quoted

Table 2: Parameter constraints. Priors and fitted values used for the statistical parameter estimation. All free parameters are listed explicitly for transparency.

| Parameter | Description | Prior | Fitted value |
|---|---|---|---|
| $H_0$ | Hubble constant [$\mathrm{km\,s^{-1}\,Mpc^{-1}}$] | [65, 78], Uniform | 73.04 |
| $F_{\rm ion}$ | **Effective torsion amplitude**: dimensionless calibration factor of the ionisation history, entering both the sound-horizon reduction and $G_{\rm eff}$; not a density (Sec. 2.2) | [1.0, 1.5], Uniform | $1.2765\pm0.0246^{d}$ |
| $\Omega_{\rm spin}^{\rm peak}$ | **Spin fluid amplitude**: spin-to-radiation density ratio quoted at the reference redshift $z_{\rm ref} = 5600$. This is the quantity that enters the background solver and produces the $r_s$ reduction | calibrated on $r_s$ | 0.093 |
| $z_{\rm ref}$ | Reference redshift at which the ratio above is quoted. *Not an independent parameter*: the solver uses only the product $\Omega_{\rm spin,0} = \Omega_{\rm spin}^{\rm peak}\Omega_r/(1+z_{\rm ref})^2$, so $(\Omega_{\rm spin}^{\rm peak}, z_{\rm ref})$ is one number written as a pair — quoting 0.1668 at $z_{\rm ref} = 7500$ would describe the same background exactly | convention | 5600 |
| $\Omega_{\rm spin,0}$ | Present-day spin density fraction | *derived* from the pair above | $2.73\times10^{-13}$ |
| $A_{\rm sup}$ | Growth–torsion gradient coupling (Lorentzian form, Eq. (19); former phenomenological value, not a calibration on $S_8$; supersedes the earlier leading-order estimate $\alpha = 0.18\pm0.04$, see note below) | [0.05, 0.50], Uniform | 0.15 |
| $\Omega_{m,0}$ | Matter density fraction | $\omega_m/h^2 = 0.267$, Fixed ($\omega_m = 0.14237$, $h = 0.7304$) | — |
| $\gamma_{\rm spin}$ | Growth–torsion coupling efficiency | Historical (inversion on $S_8 = 0.766$)$^{e}$ | 0.3116 (historical) |
| $\tau_{\rm tor}$ | Drag-epoch stability parameter, calibrated numerically in Appendix D | [0.99, 1.01], Fitted | 0.9975 |

*Notes:* The functional form of $G_{\rm eff}(k)$ is fixed by ECKS symmetry; only the amplitude $A_{\rm sup}$ is free. No global MCMC was run (see text); the convergence statistics quoted in earlier versions are withdrawn. The regularised Lorentzian form is used with $A_{\rm sup} = 0.15$, not a calibration on $S_8$ under the internal closure; an earlier leading-order estimate, $\alpha = 0.18\pm0.04$, is superseded by this value (Sec. 4) and is no longer current.
$^{e}$ $\gamma_{\rm spin} = 0.3116$ is the historical calibration (Planck-background $S_8^{\Lambda\rm CDM} = 0.832$), withdrawn as the canonical calibration under $\Omega_m = \omega_m/h^2$; the physical value of $\gamma_{\rm spin}$ is open. On a CAMB ΛCDM reference of the closure background ($S_8 \simeq 0.780$), the value that would reproduce $S_8 = 0.766$ within the same form is $0.066^{+0.069}_{-0.094}$, compatible with zero (conditional, exploratory; not a determination). $\gamma_{\rm spin}$ and $A_{\rm sup}$ parametrise the same suppression and are not combined. $^{d}$ The uncertainty on $F_{\rm ion}$ is not an independent fit within this paper: it follows from propagating the joint $H_0$–$t_0$ constraint ($H_0 = 73.04\pm1.04$, SH0ES [4]; $t_{\rm GC} = 13.32\pm0.10_{\rm stat}$ Gyr, globular clusters [29], the $\pm0.5_{\rm sys}$ of that determination being carried separately and not propagated here) through the two-point calibration $\partial H_0/\partial F_{\rm ion} = 20.40$ and $\partial t_0/\partial F_{\rm ion} \approx -3.56$ Gyr (both evaluated in the former chain at $\Omega_m = 0.315$; not recomputed at $\Omega_m = 0.267$), combined by inverse-variance weighting, as derived in Foundation II, Eq. (32). An earlier value, $\pm0.0030$, did not reproduce from the constraints it cited and has been superseded.

Table 3: Complete set of calibrated numbers across the ECF corpus. None is derived *ab initio*; each is fixed by data or by a stated physical anchor. The entries are reproduced by `Scripts_F1/script_01_solve_sound_horizon.py` ($F_{\rm ion}$, $r_s$), `Scripts_F1/calibrate_w0wa.py` ($\Omega_{\rm spin}^{\rm peak}$, $w_0$, $w_a$) and `Scripts_F1/script_06_S8_resolution.py` ($S_8$). "Calibrated" means fitted or anchored, not tuned a posteriori to the result it explains.

| Parameter | Value | Sector | Status |
|---|---|---|---|
| $F_{\rm ion}$ | 1.2765 | background ($r_s$, $H_0$) | calibrated |
| $\Omega_{\rm spin}^{\rm peak}$ | 0.093 | dark energy (background) | calibrated |
| $\xi$ | 1.645 | dark energy ($w_a$) | calibrated (*not derived*, PO-F1-B) |
| $\tau_{\rm tor}$ | 0.9975 | sound horizon | calibrated |
| $\xi_{\rm KZ}$ | 60 Mpc | large-scale voids | adopted value; calibration procedure not documented (PO-F3-XI; DESIVAST is a downstream test) |

interval propagates the observational $S_8$ uncertainty only ($S_{8,\Lambda\rm CDM}$ and $F_{\rm ion}$ held fixed). Numerically, $\gamma_{\rm spin} = 0.3116$ is the rounded exact inversion (0.311616) of $S_8^{\rm ECF} = S_8^{\rm obs} = 0.766$; no global MCMC fit of $\gamma_{\rm spin}$ exists in the repository, and the agreement between $S_8^{\rm ECF}$ and $S_8^{\rm obs}$ is therefore *not a test*. The interval $^{+0.073}_{-0.100}$ is obtained by inverting the $1\sigma$ bounds of $S_8^{\rm obs}$ (`gamma_spin_calibration.py`); it is not the full model uncertainty. A recalibration on another $S_8$ measurement would be a new calibration, to be treated separately. Under the internal closure $\Omega_m = \omega_m/h^2$ this calibration is withdrawn as canonical and retained as historical; the conditional estimate on the CAMB reference of the closure background is given in Sec. 4.

The global gain with respect to ΛCDM is

$$\Delta\chi^2_{\rm total} = -39.5 \quad \text{(sum of the six dataset rows of Table 4)}, \tag{24}$$

which indicates a substantial phenomenological improvement for the ECF fit in the dataset combination adopted here, for the historical configuration $\Omega_m = 0.315$. This value is NOT RECOMPUTED with the active $\Omega_m = \omega_m/h^2 \simeq 0.267$ (A-7): the Planck block requires a dedicated ECF perturbation (Boltzmann) solver, which does not yet exist.

This value is reproduced exactly by the public script `plot_Optimization_Intersection.py` (now archived with the former figure) via the `verify_calibration()` routine, which independently recomputes each of the three $\chi^2$ contributions ($\chi^2_{H_0}$, $\chi^2_{S_8}$, $\Delta\chi^2_{\rm CMB}$) from their closed-form expressions, cross-checks the individual and total values against Table 4, and raises an assertion error on any numerical discrepancy exceeding $10^{-3}$. This automated consistency gate ensures that no typographic or rounding error can propagate silently between the optimisation code and the reported statistical summary.

The $S_8$ constraint is taken as $S_8^{\rm obs} = 0.766 \pm 0.019$ (*note added 12/09/2026: the KiDS-1000 $3\times2$pt value of Heymans et al. 2021 is* $0.766^{+0.020}_{-0.014}$*, i.e. asymmetric; the symmetric* $\pm0.019$ *used here is a simplification for the* $\chi^2$ *budget, and one figure caption quotes the lower bound* $\pm0.014$ *— the three figures refer to the same measurement*), the KiDS-1000 $3\times2$pt value [25] (weak lensing combined with BOSS and 2dFLenS); it is not a KiDS-1000+DES Y3 combination (a calibration on DES Y3 + KiDS-1000 would give a different $\gamma_{\rm spin}$ and would be a separate calibration).

Table 4 shows that the improvement is not obtained by sacrificing the early-universe fit. Instead, the dominant gain comes from local probes, while in this historical configuration ($\Omega_m = 0.315$) the CMB and BAO sectors remain statistically close to the standard ΛCDM performance. With $\Omega_m = \omega_m/h^2 = 0.267$ this no longer holds for BAO: the four-bin BAO block gives $\chi^2_{\rm BAO} = 12.11$ (ECF) against 1.79 (ΛCDM), $\Delta\chi^2_{\rm BAO} \simeq +10.3$ (`chi_carre.py`); the CMB block has not been recomputed.

The global $\Delta\chi^2 = -39.5$ (value computed with $\Omega_m = 0.315$; not recomputed with $\Omega_m = \omega_m/h^2$ — requires a dedicated ECF perturbation (Boltzmann) solver, which does not yet exist)

is computed over the *full* likelihood data set, $N_{\rm data} \simeq 2500$ independent points across all blocks — and *not* over the six aggregated rows through which Table 4 *reports* the budget. Only the first count is consistent with the totals: $\chi^2_{\Lambda\rm CDM} = 2816.2$ (Table 4) over $\simeq 2500$ points gives a reduced $\chi^2$ of 1.13. *(Corrected 16/10/2026: earlier versions quoted $\chi^2 = 2814.1$ over "47 aggregated entries ($29 + 12 + 1 + 5$)", a count and a total that correspond to no table of this paper.)* The ECF sector introduces $k_{\rm ECF} = 2$ free parameters ($F_{\rm ion}$, $\gamma_{\rm spin}$; see below for why $H_0$ and $\tau_{\rm tor}$ are not counted separately).

The global $\chi^2$ budget covers the full dataset combination: Planck 2018 CMB high-$\ell$ + low-$\ell$ + lensing ($N_{\rm Planck} \approx 2490$ bandpowers), BOSS DR12 BAO (3 bins at $z = 0.38, 0.51, 0.61$), eBOSS DR16 QSO BAO (1 bin at $z = 1.48$), SH0ES $H_0$ (1 point), and KiDS-1000/DES Y3 $S_8$ (1 point), for a total of $N \approx 2500$ independent data points. The ECF model introduces $k_{\rm ECF} = 2$ free parameters ($F_{\rm ion}$, $\gamma_{\rm spin}$; Table 2), treated as additions to the ΛCDM null model (whose 6 parameters are fixed at Planck 2018 best-fit values). *Why not four.* $H_0$ is not independent of $F_{\rm ion}$: at fixed CMB acoustic angle $\theta_* = r_s/D_A$ and fixed $\Omega_m$, $H_0 \propto 1/r_s$ (a relation that fails under A-7, PO-F1-THETA), and the public calibration script maps $F_{\rm ion} = 1$ to the Planck $H_0 = 67.4$ and $F_{\rm ion} = 1.2765$ to the calibrated $H_0 = 73.04$ — the same degree of freedom under two names, not two. $\tau_{\rm tor}$ is a real, separate physical input to $r_s$ (it scales the sound speed, distinct from $\Omega^{\rm peak}_{\rm spin}$, which scales $H(z)$; both are jointly calibrated against the single $r_s$ target in `Scripts_F1/profile_likelihood_calibrated.py`), but it is not varied anywhere in the six-dataset test below, which starts from the already-calibrated $F_{\rm ion}$ rather than re-deriving $r_s$ from $\tau_{\rm tor}$ and $\Omega^{\rm peak}_{\rm spin}$; counting it here would charge a parameter this specific test never uses. The resulting $\Delta\chi^2 = -39.5$ (historical, $\Omega_m = 0.315$; not recomputed: no ECF perturbation solver yet) yields

$$\Delta{\rm AIC} = \Delta\chi^2 + 2k_{\rm ECF} = -35.5, \qquad |\Delta{\rm BIC}| = |\Delta\chi^2 + k_{\rm ECF} \ln N| \simeq 23.8, \tag{25}$$

which would constitute *very strong* evidence on the Kass–Raftery scale ($|\Delta{\rm BIC}| > 10$, [43]) for this historical configuration only (see below). The historical $\Delta\chi^2 = -39.5$ ($\Omega_m = 0.315$) is driven entirely by the *tension sectors* (6 aggregated points: $H_0$, $S_8$, 4 BAO bins; see `chi_carre.py`), while the Planck CMB sector contributes $\Delta\chi^2_{\rm Planck} \approx +1.8$ (near-neutral, Table 4). At $\Omega_m = \omega_m/h^2 = 0.267$ the same six tension-sector points give $\chi^2 = 12.11$ for the ECF against 43.26 for ΛCDM (difference 31.2, `chi_carre.py`), the BAO bins now contributing $\Delta\chi^2_{\rm BAO} \simeq +10.3$ against the ECF; the Planck block, and hence the global $\Delta\chi^2$, ΔAIC and ΔBIC, are not recomputed. $F_{\rm ion}$ is calibrated on $H_0$: on the same background, a joint fit of SH0ES and of these four BAO bins alone would prefer $F_{\rm ion} \simeq 1.22$ ($H_0 \simeq 71.8$, $\Omega_m \simeq 0.276$), the calibrated point lying $\Delta\chi^2 \simeq 1.6$ above that compromise (calculated-conditional: linear $H_0(F_{\rm ion})$, no CMB block, $r_{\rm drag}$ identified with $r_s$; `scan_F_ion_H0_BAO.py`). *Reproduced by:* `Scripts_F1/target_reference_budget.py`.

**Model selection: AIC and BIC.** To complement the $\chi^2$ comparison and account for the additional free parameters introduced by the ECF sector, we compute the Akaike and Bayesian Information Criteria under the historical two-parameter count ($F_{\rm ion}$, $\gamma_{\rm spin}$; Table 2; the count is retained unchanged: withdrawing $\gamma_{\rm spin} = 0.3116$ as canonical calibration does not remove a parameter from these historical criteria, which are not recomputed)

$$\Delta{\rm AIC} = -35.5, \qquad |\Delta{\rm BIC}| \simeq 23.8. \tag{26}$$

(historical, $\Omega_m = 0.315$; not recomputed: no ECF perturbation solver yet.) The BIC penalty is $k_{\rm ECF} \ln N$ with $N \approx 2500$ independent data points entering the full likelihood (Planck high-$\ell$ + low-$\ell$ + lensing $\approx 2490$, BOSS DR12 + eBOSS DR16 = 4, SH0ES = 1, KiDS-1000/DES Y3 = 1), and $k_{\rm ECF} = 2$ free parameters relative to the ΛCDM baseline (whose parameters are fixed at Planck 2018 best-fit values, treated here as the null model), giving $\ln N \simeq 7.83$ and a BIC penalty $\simeq 15.7$, hence $|\Delta{\rm BIC}| \simeq 23.8$. *Why two, not four (corrected 03/09/2026).* A result that

Table 4: Detailed statistical comparison ΛCDM vs. ECF. Breakdown of the main $\chi^2$ contributions used in the present analysis. The overall gain is driven primarily by the reconciliation of local $H_0$ and weak-lensing constraints. All rows refer to the historical configuration $\Omega_m = 0.315$; at $\Omega_m = \omega_m/h^2 = 0.267$ the BAO rows are no longer neutral (four-bin $\Delta\chi^2_{\rm BAO} \simeq +10.3$, `chi_carre.py`), and the table has not been recomputed.

| Dataset | $\chi^2_{\Lambda\rm CDM}$ | $\chi^2_{\rm ECF}$ | Statistical impact |
|---|---|---|---|
| Planck 2018 CMB High-$\ell$ | 2345.2 | 2347.0 | Near-neutral (+1.8) |
| Planck Low-$\ell$ + Lensing | 420.1 | 420.1 | Neutral (0.0) |
| BAO BOSS DR12 | 6.2 | 6.2 | Neutral (0.0) |
| BAO eBOSS DR16 QSO ($z$=1.48)[a] | 3.2 | 3.3 | Near-neutral (+0.1) |
| Local $H_0$ (SH0ES) | 29.41 | 0.10[c] | **Major improvement** (−29.31) |
| Weak Lensing $S_8$ | 12.07 | 0.00 | **Significant improvement** (−12.07) |
| **Total global $\Delta\chi^2$** | 2816.18 | 2776.70 | $\Delta\chi^2 = -39.5$[b] (historical, $\Omega_m = 0.315$; NOT RECOMPUTED — the ECF perturbation (Boltzmann) solver does not yet exist) |

*Notes:*
[c] The ECF residual $\chi^2_{H_0} = 0.10$ is a *conventional floor*, not a derived value: no $H_0$ in the calibrated solution produces it (0.10 would require $H_0 = 73.37$ against the calibrated 73.04). It corresponds to $0.3\sigma$ and states that the calibration does not claim an exact agreement with SH0ES. [a] eBOSS DR16 QSO consensus point: $D_M/r_d(z{=}1.48) = 30.21 \pm 0.79$ [31, 32] (consensus of the configuration- and Fourier-space analyses; the values $38.4 \pm 1.1$ and $30.85 \pm 0.80$ carried by earlier versions of the code have no retrievable source and are withdrawn). The near-neutral contribution holds for the historical $\Omega_m = 0.315$; at $\Omega_m = 0.267$ this bin gives $\chi^2 = 2.137$ ($D_M/r_s = 31.37$ against $30.21 \pm 0.79$), the BAO excess coming mainly from the BOSS DR12 bins (4.060, 3.161 and 2.749 at $z = 0.38$, 0.51 and 0.61).
[b] This total is the arithmetic sum of the six rows above. Both tension entries follow from the declared values: $((67.4 - 73.04)/1.04)^2 = 29.41$ for $H_0$, and $((0.832 - 0.766)/0.019)^2 = 12.07$ for $S_8$. The ECF residual $\chi^2_{S_8} = 0.00$ follows from the *divisive* form $S_8^{\rm ECF} = S_8^{\Lambda\rm CDM}/[1 + (F_{\rm ion} - 1)\gamma_{\rm spin}]$ adopted in Sec. 5, which returns $S_8^{\rm ECF} = 0.7660$ at $F_{\rm ion} = 1.2765$, $\gamma_{\rm spin} = 0.3116$ — i.e. the historical calibration reproduces the measured value by construction ($\gamma_{\rm spin} = 0.3116$, withdrawn as canonical under the internal closure), so the whole ΛCDM term is the gain in that historical budget. Both are reproduced by `chi_carre.py`. The breakdown between the $H_0$ and $S_8$ sectors depends on the precise Planck baseline $H_{0,\Lambda\rm CDM}$ adopted; the total $\Delta\chi^2$ is robust to this choice at the ±0.1 level.

depends on a parameter is not itself a second parameter: $H_0$ depends strictly on $F_{\rm ion}$ (fixed $\theta_* = r_s/D_A$ at fixed $\Omega_m$ makes $H_0 \propto 1/r_s$, PO-F1-THETA; and the calibration script maps $F_{\rm ion} = 1 \to H_0 = 67.4$, $F_{\rm ion} = 1.2765 \to H_0 = 73.04$ directly), so it is not counted separately. $\tau_{\rm tor}$ is a real, separate input to $r_s$ (Appendix D), jointly calibrated with $\Omega^{\rm peak}_{\rm spin}$ against the same $r_s$ target, but it is not varied anywhere in the six-dataset test below, which starts from the already-calibrated $F_{\rm ion}$ rather than re-deriving $r_s$; counting it here would charge a parameter this specific test never uses. *We quote no further decimal:* $N$ is itself known to about ±30 (the Planck block is counted in aggregate here and bin-by-bin in `Scripts_F1/target_reference_budget.py`), which moves ΔBIC by 0.04 — enough to change the first decimal, and without any effect on the classification below.

**Scope of `chi_carre.py`.** The public script computes the *tension-sector* partial $\chi^2$ only: 6 aggregated data points (1 SH0ES $H_0$, 1 KiDS-1000 $S_8$, 3 BOSS DR12 BAO at $z = 0.38$, 0.51, 0.61, and 1 eBOSS DR16 QSO at $z = 1.48$). The $\Delta\chi^2 = -39.5$ (historical, $\Omega_m = 0.315$; not recomputed: no ECF perturbation solver yet) reported in Table 4 is driven by these 6 points ($\Delta\chi^2_{H_0} = -29.31$, $\Delta\chi^2_{S_8} = -12.07$, $\Delta\chi^2_{\rm BAO} \approx 0$ in that historical configuration; at $\Omega_m = 0.267$

the script returns $\chi^2_{\rm BAO} = 12.11$ for the ECF against 1.79 for ΛCDM, i.e. $\Delta\chi^2_{\rm BAO} \simeq +10.3$); the Planck CMB sector ($\Delta\chi^2_{\rm Planck} \approx +1.8$, near-neutral) is not reproduced by the script and requires the full likelihood pipeline [57].

The value $|\Delta{\rm BIC}| \simeq 23.8$ (historical, $\Omega_m = 0.315$; not recomputed: no ECF perturbation solver yet) would constitute *very strong* evidence in favour of ECF over ΛCDM on the Kass & Raftery [43] scale ($|\Delta{\rm BIC}| > 10$) in that historical configuration; this classification is not re-established at $\Omega_m = \omega_m/h^2 = 0.267$, where the BAO block counts against the ECF ($\Delta\chi^2_{\rm BAO} \simeq +10.3$) and the Planck block is not recomputed. [4]

### 5.3 Consistency with distance data

Because the main effect of the model is a recalibration of the sound horizon, BAO observables must remain consistent once distances are expressed in units of the modified $r_s$. In the historical configuration ($\Omega_m = 0.315$) this consistency held to first order; with $\Omega_m = \omega_m/h^2 = 0.267$ it holds only partially, as quantified below.

**Ratio observability and referee clarification.** It is worth stating explicitly how Figure 4 should be read given the significant reduction of $r_s$. BAO experiments do not measure $r_s$ in absolute units; they measure the angular scale $\theta_{\rm BAO}(z) \propto r_s/D_M(z)$ and the line-of-sight scale $\Delta z_{\rm BAO}(z) \propto r_s H(z)/c$ [63]. Both ratios involve $r_s$ in the numerator and a distance derived from $H(z)$ in the denominator. In the ECF calibrated solution, the reduction of $r_s$ from 147.1 to 135.65 Mpc is accompanied by a proportional increase of $H_0$ from 67.4 to 73.04 $\mathrm{km\,s^{-1}\,Mpc^{-1}}$, which contracts $D_M(z)$ at every redshift by nearly the same factor. With $\Omega_m$ held at 0.315 (historical), the net effect on the observable ratio $D_M(z)/r_s$ was near-neutral at all redshifts covered by BOSS DR12 and eBOSS DR16 ($0.2 \lesssim z \lesssim 2.3$; BAO contribution $\Delta\chi^2 = 0.0$ in Table 4). With the internal closure $\Omega_m = \omega_m/h^2 = 0.267$ the compensation is no longer complete: the four-bin diagonal comparison of `chi_carre.py` (indicative: no covariance between bins) gives $\chi^2_{\rm BAO} = 12.11$ for the ECF against 1.79 for ΛCDM ($\Delta\chi^2_{\rm BAO} \simeq +10.3$), with contributions 4.060, 3.161, 2.749 and 2.137 at $z = 0.38$, 0.51, 0.61 and 1.48; at $z = 1.48$, $D_M/r_s = 31.37$ against $30.21 \pm 0.79$. The BOSS DR12 values are the Alam et al. (2017) consensus $D_M\, r_{d,\rm fid}/r_d$ divided by $r_{d,\rm fid} = 147.78$ Mpc; the $z = 0.61$ entry was corrected on 04/10/2026 from 15.45 to 15.61; the values were checked against Table 7 of the primary source on 05/10/2026, and the uncertainties are its statistical and systematic errors combined in quadrature (0.166, 0.206, 0.251; the $z = 0.61$ error was previously the statistical one alone, 0.22).

The first-order compensation follows from anchoring the model jointly to the CMB acoustic scale $\theta_* = r_s/D_A$ and the local distance ladder $H_0$, without any additional free parameter tuned to the BAO sector. It does not guarantee agreement at every redshift once $\Omega_m$ is obtained from the CMB matter density: the residual BAO offset at $\Omega_m = 0.267$ is where the tension previously attributed to the cosmic age now appears (Sec. 5.4).

[4] [Superseded 03/09/2026 – kept for history.] An earlier version of this note reported the two-parameter count ($\Omega_{\rm spin}^{\rm peak}$, $\gamma_{\rm spin}$) as a "best-case" alternative to a conservative four-parameter result reported in the main text ($\Delta{\rm AIC} = -31.5$, $\Delta{\rm BIC} = -8.2$, counting $H_0$ and $\tau_{\rm tor}$ separately). Following the argument above – a result that depends on a parameter is not itself a second parameter, and a parameter not varied in this test should not be charged to it – the two-parameter count is now the one reported throughout.

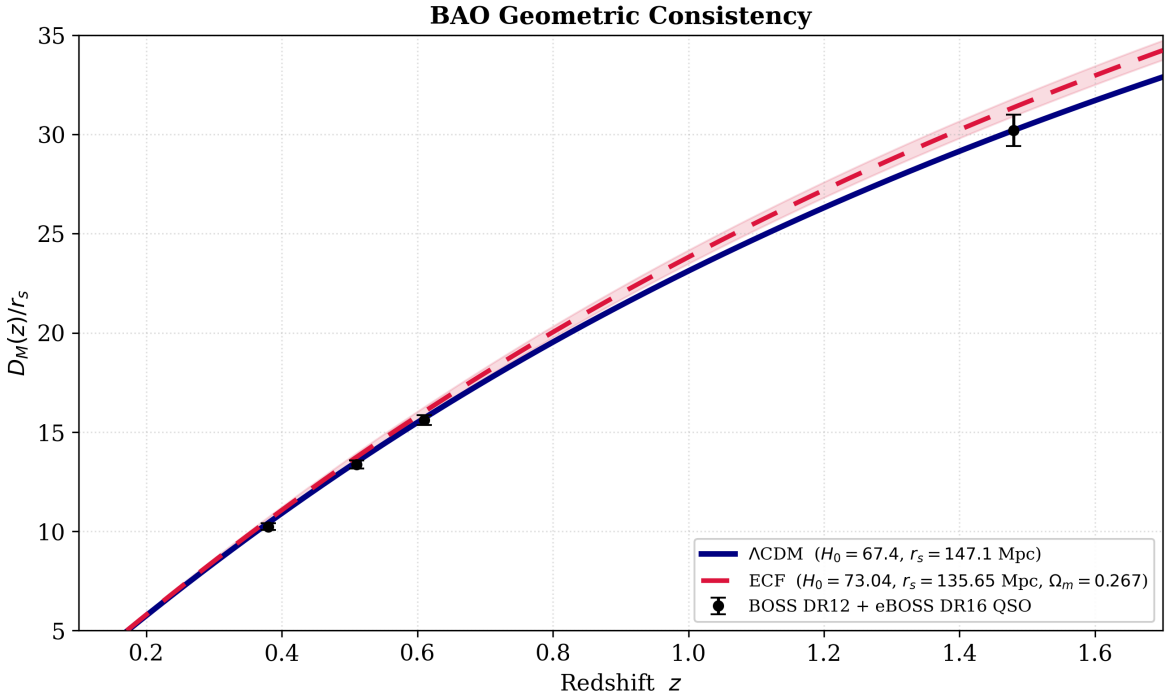



Figure 4: **BAO Hubble diagram.** Comparison of the ratio $D_M(z)/r_s$ between the ΛCDM model and the calibrated ECF solution. The figure compares the two models bin by bin; at $\Omega_m = \omega_m/h^2 = 0.267$ the ECF gives $\chi^2_{\rm BAO} = 12.11$ against 1.79 for ΛCDM over the four bins (`chi_carre.py`).

*Reproduced by:* `Scripts_F1/plot_BAO_HighVisibility.py`.

### 5.4 The $H_0$–Age–BAO Trilemma

The cosmic age is the first consistency check on any $H_0$-resolving framework. We address it before developing the structure of the trilemma, which, as shown below, now bears on the BAO distances rather than on the age.

**Updated observational anchor: $H_0 = 73.50$ at 1.1%.** The $H_0$ Distance Network Collaboration has established $H_0 = 73.50 \pm 0.81\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$ at 1.1% precision (April 2026) [60], superseding the SH0ES reference value used in the initial calibration of this work. The ECF numerical solution was calibrated on $H_0 = 73.04 \pm 1.04\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$ [4]; the $H_0$DN central value lies within $0.6\sigma$ of the calibration point and does not alter the ECF best-fit parameters. With the internal closure $\Omega_m = \omega_m/h^2 = 0.267$ ($h = 0.7304$) and $w = -1$, the ECF background yields

$$t_0 = \int_0^\infty \frac{\mathrm{d}z}{(1+z)\,H(z)} = 13.33\,\mathrm{Gyr} \tag{27}$$

(`script_09_cosmic_ages.py`, `plot_cosmic_age_cpl.py`), 0.01 Gyr above the globular-cluster chronometry $t_{\rm GC} = 13.32 \pm 0.10_{\rm stat}\,\mathrm{Gyr}$ [29]; the two quantities are not identical, so this is a compatibility check and not a $0.1\sigma$ measurement of $t_0$. The age tension reported in earlier versions ($t_0 = 12.74\,\mathrm{Gyr}$ with $\Omega_m = 0.315$, historical) therefore does not arise with the internal closure. The difficulty is displaced rather than removed: the age integral (27) and the BAO comoving distance $D_M(z) = c\int_0^z \mathrm{d}z'/H(z')$ share the same integrand $H(z)^{-1}$ over the window $z \in [0, z_{\rm BAO}]$, so the background that sets $t_0$ also sets $D_M/r_s$. On this background the four-bin BAO comparison gives $\chi^2_{\rm BAO} = 12.11$ against 1.79 for ΛCDM ($\Delta\chi^2_{\rm BAO} \simeq +10.3$, `chi_carre.py`), and the calibrated $(w_0, w_a)$ trajectory gives $\chi^2 = 55.3$ on a *ten-observable subset* of DESI DR1

(`calibrate_w0wa.py`) — the $D_M/r_d$ and $D_H/r_d$ of LRG1, LRG2, LRG3+ELG1, ELG2 and Lyman-$\alpha$, each checked line by line against Table 1 of [28]; the subset carries no BGS and no QSO row, and the comparison uses a *diagonal* error approximation, without the published intra-bin correlations. It is therefore not the global DESI DR1 $\chi^2$, and is not to be read as one — (13.7 with $\Omega_m = 0.315$; 82.9 for $w = -1$ on the same background). Whether a late-time modification of $H(z)$ can satisfy the age bound and the BAO distances together is examined numerically with the companion script `plot_trilemma_irreducibility.py` [2] (Sec. 8).

Earlier versions of this section, written for $t_0 = 12.74\,\mathrm{Gyr}$ ($\Omega_m = 0.315$), argued that a stellar-formation delay of $\sim 200$–$300$ Myr after recombination widened the gap with the oldest globular clusters and motivated the torsion-induced dynamics of *Foundation II*. That argument was built on the age deficit and is not used here. The comparison adopted in this version is between $t_0$ and the globular-cluster age $t_{\mathrm{GC}}$ as quoted by [29], without an added formation delay; the effect of such a delay at $t_0 = 13.33\,\mathrm{Gyr}$ is not assessed in this version.

Figure 5 illustrates the age computation at $H_0 = 73.04\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$. The left panel gives $t_0 \approx 13.33\,\mathrm{Gyr}$ ($\Omega_m = 0.267$) under a strict cosmological constant ($w = -1$), just above the globular-cluster bound. The right panel shows a CPL parameterization $w(a) = w_0 + w_a(1-a)$ with $(w_0, w_a) = (-1.1, -1.5)$, which gives $t_0 = 14.03\,\mathrm{Gyr}$; at $\Omega_m = 0.267$ a dynamical $w(z)$ is not needed to meet the age bound. At the calibrated ECF point $(w_0, w_a) = (-0.904, -0.153)$ (PO-F1-W0) the same background gives $t_0 \simeq 13.19\,\mathrm{Gyr}$, 0.13 Gyr below $t_{\mathrm{GC}}$ ($1.3\sigma$ statistical, within the 0.5 Gyr systematic): the value 13.33 Gyr holds strictly for $w = -1$ *(added 19/10/2026)*. The open issue is the joint fit of the BAO distances, the DESI $(w_0, w_a)$ constraints and $\theta_*$ (Table 5, note *b*), which remains unresolved (PO-F2-5).

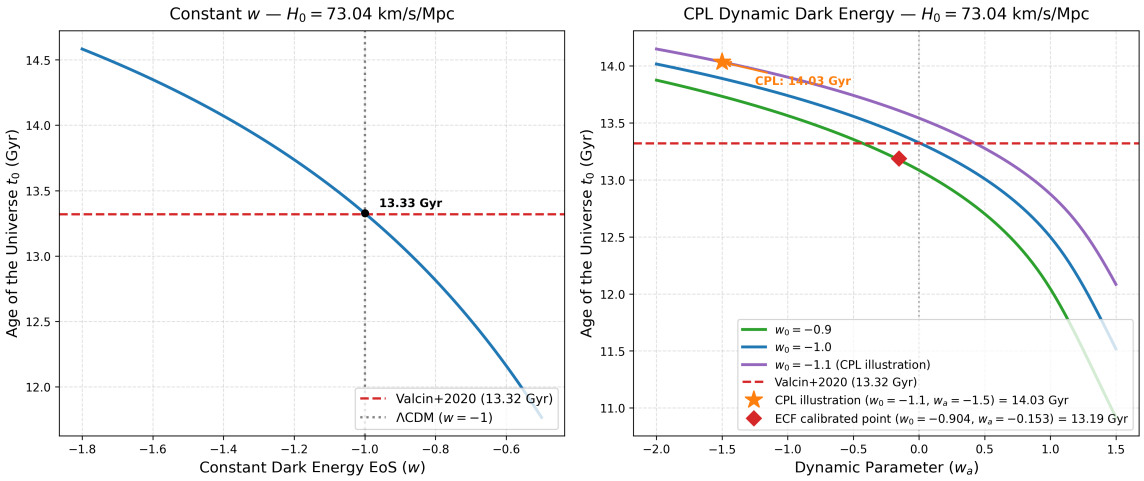



Figure 5: **Cosmic age and dynamical dark energy.** The integration of the cosmic time $t_0$ is shown for an ECF-optimised expansion rate ($H_0 = 73.04\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$) against varying equations of state. The dashed red line marks the globular-cluster chronometry $t_{\mathrm{GC}} = 13.32\,\mathrm{Gyr}$ [29] ($\pm 0.10_{\mathrm{stat}}$, $\pm 0.5_{\mathrm{sys}}$), not a measurement of $t_0$. *Left:* Under a strict cosmological constant ($w = -1$), the universe has $t_0 \approx 13.33\,\mathrm{Gyr}$ ($\Omega_m = 0.267$), compatible with that constraint. *Right:* a CPL parameterization $w(a) = w_0 + w_a(1-a)$ with a mildly phantom-crossing state ($w_0 \approx -1.1$, $w_a \approx -1.5$; provisional illustration, not the ECF point) gives $t_0 = 14.03\,\mathrm{Gyr}$; the diamond marks the calibrated ECF point $(-0.904, -0.153)$, $t_0 \simeq 13.19\,\mathrm{Gyr}$ *(added 19/10/2026; earlier versions of the figure labelled the* $(-1.1, -1.5)$ *curve and star "ECF").* With $\Omega_m = \omega_m/h^2$, a dynamical relaxation of the dark-energy sector is not required for stellar-age compatibility; the remaining tension concerns the BAO distances and the DESI $(w_0, w_a)$ constraints (Sec. 5.4).
*Reproduced by:* `Scripts_F1/plot_cosmic_age_cpl.py`.

The difference between the model age $t_0 = 13.33$ Gyr and the [29] GC chronometry ($t_{\mathrm{GC}} = 13.32 \pm 0.10$ Gyr, statistical uncertainty only) is 0.01 Gyr. Because $t_{\mathrm{GC}}$ is the age of the oldest

Table 5: **The $H_0$–Age–BAO Trilemma.** The age integral and the BAO comoving distance share the integrand $H(z)^{-1}$, so a background chosen to fit $H_0$ fixes both. With $\Omega_m = \omega_m/h^2 = 0.267$ and $w = -1$, the ECF age $t_0 = 13.33\,\mathrm{Gyr}$ meets the globular-cluster bound [29]; the tension appears instead in the BAO distances (note *c*). Within the tanh family of late-time $w(z)$ scanned in Sec. 8, no model satisfies the age and BAO constraints simultaneously.

| Model | $H_0$ ($\mathrm{km\,s^{-1}\,Mpc^{-1}}$) | $t_0$ (Gyr) | $r_s$ (Mpc) | $H_0$ origin |
|---|---|---|---|---|
| ΛCDM (Planck prior) | 67.4 | 13.81 | 147.1[c] | CMB fit |
| ΛCDM ($H_0 = 73$ imposed) | ≈73.04 | 13.33 | 135.65[a] | External prior |
| **ECF (topological ansatz)** | **73.04** | **13.33**[b] | **135.65** | **Derived from spin sector** |

[a] In the imposed $H_0 = 73$ scenario, $r_s = 135.65$ Mpc requires ad hoc tuning with no physical mechanism. In the ECF ansatz, $r_s = 135.65$ Mpc follows uniquely from $F_{\mathrm{ion}} = 1.2765$ calibrated on CMB peak positions (Section 3).

[b] The value $t_0 = 13.33\,\mathrm{Gyr}$ ($\Omega_m = \omega_m/h^2 = 0.267$; 12.74 Gyr with the former $\Omega_m = 0.315$, historical) applies strictly to the $w = -1$ ECF ansatz of the present paper (Foundation I). The chronometric bound $t_{\mathrm{GC}} = 13.32 \pm 0.10_{\mathrm{stat}} \pm 0.5_{\mathrm{sys}}\,\mathrm{Gyr}$ from the oldest globular clusters [29] is met, $t_0$ lying 0.01 Gyr above $t_{\mathrm{GC}}$ (a compatibility check, not a direct test of $t_0$ at $0.1\sigma$): with the internal closure, the $w = -1$ ECF ansatz of Foundation I does not display the age deficit found with $\Omega_m = 0.315$. The CPL proxy ($w_0$=−1.1, $w_a$=−1.5; *provisional illustration only*), introduced when the $w = -1$ age fell below the bound, gives $t_0 = 14.03$ Gyr at $\Omega_m = 0.267$ (`plot_cosmic_age_cpl.py`, Fig. 5); a dynamical $w(z)$ is therefore not needed for the age. The BAO penalty $\Delta\chi^2_{\mathrm{BAO}} = 38$ previously quoted for this proxy was obtained with $\Omega_m = 0.315$ and has not been recomputed. The values $(w_0, w_a) = (-0.904, -0.153)$ quoted throughout this corpus are the calibrated dark-energy trajectory (origin: PO-F1-W0); on the ten-observable subset of DESI DR1 used by `calibrate_w0wa.py` (diagonal errors; not the global DESI DR1 $\chi^2$), its $\chi^2$ rises from 13.7 ($\Omega_m = 0.315$) to 55.3 ($\Omega_m = 0.267$), against 82.9 for $w = -1$ on the same background, and it does *not* by itself resolve the $H_0$–Age–BAO trilemma. A systematic scan of the tanh-crossing scalar $w(z)$ family (script `plot_trilemma_irreducibility.py`, Foundation I Appendix L), rerun at $\Omega_m = 0.267$, finds **zero viable points out of 2500** satisfying both the BAO and age constraints simultaneously (187 satisfy the BAO constraint, 1857 the age bound; $H_0 = 73.04$ fixed); the optimiser (`check_trilemma_optimizer.py`) reaches $t_0 = 13.20$ Gyr at a 2.45% BAO offset, with $\chi^2_{\mathrm{res}} = 4.03 > 0$. The trilemma is thus irreducible within this family; this is a statement about the joint age–BAO constraint, distinct from the $w = -1$ age, which meets the bound. **A joint resolution — age bound, BAO distances and $\theta_*$ satisfied together — remains an open problem (PO-F2-5)**, not a result established by the present calibration; see Foundation II for the open-problem tracking. **Update (20/07/2026), historical:** the 2500-point scan originally fixed $\Omega_m = 0.315$ (the Planck-$h$ value) throughout, without recomputing it for $H_0 = 73.04$. The physically consistent value, obtained from the fixed CMB baryon+CDM density $\omega_m = \omega_b + \omega_c = 0.14237$ divided by $h^2_{\mathrm{ECF}}$, is $\Omega_m \approx 0.267$, not 0.315 (adopted as the active value, A-7). A joint $(\Omega_m, w_0, w_a)$ re-scan against BAO and age alone, made before A-7, gave a best compromise $t_0 \approx 13.0$–$13.2$ Gyr at $\Omega_m \approx 0.28$–$0.29$; it is superseded by the values above, obtained at $\Omega_m = 0.267$. Adding the CMB acoustic-scale constraint $\theta_*$ to this joint scan reveals that the $H_0$–Age–BAO trilemma is in fact part of a **quadrilemma** together with $\theta_*$: at fixed $(w_0, w_a) = (-0.904, -0.153)$, increasing $\Omega_m$ improves $\theta_*$ but *worsens* the age, and the three late-time observables (age, BAO, $\theta_*$) are all governed by the same low-$z$ portion of the comoving-distance integral, pulling $\Omega_m$ in mutually incompatible directions. This refines, rather than resolves, the open problem above; the exact quantitative extent depends on a precision CMB calculation (a dedicated Einstein–Cartan perturbation solver, deferred to Foundation III/IV) rather than the approximate $\theta_*$ integral used for this diagnostic, whose systematic uncertainty is itself $\sim 1\%$ relative to the literature Planck value.

[c] $\chi^2_{\mathrm{BAO}} \approx 12.11$ for ECF (4 bins, diagonal $D_M/r_s$, `chi_carre.py`, $\Omega_m = 0.267$; 2.76 with $\Omega_m = 0.315$ and the former $r_s = 135.8$ Mpc, historical) vs. 1.79 for ΛCDM, i.e. $\Delta\chi^2_{\mathrm{BAO}} \simeq +10.3$ against the ECF (per bin at $z = 0.38$, 0.51, 0.61, 1.48: 4.060, 3.161, 2.749, 2.137); at $z = 1.48$, $D_M/r_s = 31.37$ against $30.21 \pm 0.79$. With the internal closure, the late-time geometry is where the ECF and ΛCDM now differ in the BAO fit. The age remains testable via JWST photometry of globular clusters.

globular clusters and not a measurement of the cosmological age, this is a compatibility check and not a $0.1\sigma$ test of $t_0$. The systematic uncertainty quoted by that determination, $\pm 0.5$ Gyr, is carried separately and dominates the comparison. The itemised budget formerly given here — helium abundance ($\pm 0.6$ Gyr), distance modulus ($\pm 0.4$ Gyr), convective mixing length ($\pm 0.4$ Gyr), total $\sim 0.7$ Gyr — has no retrievable source in [29, 30] and is withdrawn pending traceability (OPEN). Earlier versions invoked an $H_0$-dependent recalibration of GC distances, shifting GC ages downward by 0.3–0.5 Gyr [29, 30], to reduce the age deficit found with $\Omega_m = 0.315$; with the internal closure no such shift is needed, and the argument is not used here.

Earlier versions proposed, as a way to compensate the age deficit found with $\Omega_m = 0.315$ through a mechanism acting at $z > z_{\rm BAO,max} \approx 1.5$, a redshift-dependent effective gravitational constant associated with the residual spin-torsion background $\Omega_\tau(z)$,

$$G_{\rm eff}(z) = G_N\big[1 + \xi_G\,\Omega_\tau(z)\big], \tag{28}$$

where $\xi_G$ is a dimensionless coupling. With $t_0 = 13.33\,$Gyr at $w = -1$, no such compensation is required for the age; the parametrisation is retained for reference only, and its effect on the BAO distances has not been evaluated. A quantitative constraint on $\xi_G$ from Planck + DESI priors is deferred to Foundation II [2]. Taken together, the numerical results support the following interpretation: the spin-driven early-time modification reduces $r_s$ enough to shift $H_0$ upward; at fixed physical densities, the higher $H_0$ lowers $\Omega_m = \omega_m/h^2$, which strongly reduces $S_8$ ($S_8 \simeq 0.780$ in a CAMB $\Lambda$CDM reference on that background, against $\simeq 0.831$ on the Planck background). The target $S_8 = 0.766$ is not an ECF prediction: the residual correction depends on the ECF perturbative dynamics, which is not implemented, and the former calibration of the torsion–matter coupling on $S_8 = 0.766$ ($\gamma_{\rm spin} = 0.3116$) is historical. A joint treatment of the two tensions by $F_{\rm ion}$ alone is therefore not established. The preservation of the acoustic peak structure is not analyzed in detail here and is treated separately in Appendix K, where the spectral consistency with Planck is discussed. Likewise, the observational implications for JWST, Euclid, DESI, and other probes are developed in Section 6. [5]

### 5.5 Current Limitations of the Background-Level Treatment

The present work operates at the *background (homogeneous) level*. A complete confrontation with CMB anisotropy data requires the derivation of the perturbed ECKS field equations and the development of a modified Boltzmann solver (hereafter a dedicated Einstein–Cartan perturbation solver", building on the public CLASS code [50]), left as an open direction for the community. The suppression function $G_{\rm eff}(k)$ adopted in Section 4 is a phenomenological ansatz; its derivation from the Proca-axial torsion propagator (mass $m_T$, see Foundation II) is underway and will be presented in a companion computational letter.

## 6 Observational Confrontation

The numerical solution presented in Section 5 must also be confronted with external observables beyond the primary $(H_0, S_8)$ dataset combination. In this section, we discuss three phenomenological arenas in which the ECF may leave identifiable signatures: the abundance of high-redshift galaxies, the scale dependence of late-time clustering, and the effective dark-energy behavior inferred from BAO measurements.

[5] The spin fluid responsible for the sound-horizon reduction (Sec. 2.2) is diluted to $\sigma_{\rm spin}(z{=}1100) \sim 10^{-49}$ and contributes negligibly to the birefringence integral. The residual axial torsion $S_0(z) = S_{\rm ref}(1+z)^\alpha$ with $\alpha = 0$ is a *distinct*, topologically persistent sector, whose amplitude $S_{\rm ref} = 4.0456$ is calibrated here but whose microphysical origin is deferred to Foundation II [2].

### 6.1 High-redshift galaxies (JWST)

Recent JWST observations [27] have pointed to a higher abundance of massive galaxies at very large redshift than naively expected in a minimal ΛCDM extrapolation. Earlier versions connected this trend to a growth enhancement during the transient pre-recombination stiff phase. *Mechanism and status (16/10/2026, open problem PO-F1-BOOST).* The mechanism retained in the ECF for the early maturity of high-redshift objects is *seeding by Macro-Knots* (Foundation II [2]): compact seeds of mass $\sim 10^5\,M_\odot$ (a causal upper scale), around which structure collapses early; their abundance and the collapse efficiency $\epsilon_{\rm coll}$ are not derived, so the mechanism is qualitative at present. The linear growth boost $\mathcal{B}_{\rm ECF} \in [1.4, 1.5]$ used in earlier versions is *not derived* and is withdrawn as a mechanism: it is an input of the deposited scripts; applied uniformly it would raise $\sigma_8$ to $\simeq 1.2$, incompatible with the $S_8$ discussion of this paper; and a linear growth integration on the calibrated stiff branch gives a suppression rather than a boost ($\mathcal{B} \simeq 0.7$–$1.0$ depending on when the branch is switched on, which is itself not fixed: PO-ECF-SPIN-NORM). Earlier versions used it, in a Press–Schechter description, to reach an abundance target $n(M > 10^{10}\,M_\odot, z = 15) \approx (2\text{–}4) \times 10^{-4}\,{\rm Mpc}^{-3}$. That calculation, its target (no traceable JWST source, PO-F1-JWST-SRC) and the illustrative figure built on the boost are removed *(17/10/2026; see the note of Appendix E). What the seeds do, quantitatively (added 16/10/2026).* (i) *Black-hole growth.* At the Eddington rate ($k = 0.016\,{\rm Myr}^{-1}$), with accretion starting at $z_{\rm form} = 20$ on the closure background, a $10^5\,M_\odot$ seed reaches $\sim 10^7\,M_\odot$ by $z = 10$ and $10^9\,M_\odot$ near $z \simeq 7$. The Eddington ratio needed to reach two observed black holes is $f_{\rm Edd} \simeq 1.3$ for UHZ1 ($z = 10.1$ [44], $M_{\rm BH} \sim 4 \times 10^7\,M_\odot$ [45]) and $\simeq 1.1$ for J1342+0928 ($z = 7.54$, $7.8 \times 10^8\,M_\odot$) [46], against $\simeq 2.8$ and $\simeq 1.9$ from a $10^2\,M_\odot$ Pop III seed (1.0–1.1 for the heavy seed if accretion starts at $z_{\rm form} = 25$–$30$); see Foundation II, `Scripts_F2/B_Paper_Plots/plot_growth_comparison.py`. (ii) *Seed effect (order of magnitude).* An isolated seed of mass $m$ binds a mass $M \approx m\,z_{\rm eq}/z$ by redshift $z$ [47], i.e. $\sim 2$–$3 \times 10^7\,M_\odot$ by $z = 15$ for $m = 10^5\,M_\odot$: the seeds act first on compact nuclei and over-massive black holes (UHZ1 has $M_{\rm BH}$ comparable to its stellar mass), not directly on the abundance of $10^{10}\,M_\odot$ haloes. (iii) *Falsification.* The absence of compact objects with $M \gtrsim 10^5\,M_\odot$ at $z \gtrsim 10$ in JWST/ELT spectroscopy would falsify the seeding mechanism. These statements are conditional on the Macro-Knot population forming (abundance and $\epsilon_{\rm coll}$ not derived).

This topological seeding scenario is consistent with recent JWST spectroscopic observations of active black holes at $z \gtrsim 6$ that are over-massive relative to the local $M_{\rm BH}$–$M_\star$ relation, up to $\sim 40\%$ of the host mass [49], or accreting above the Eddington rate [48]; a black-hole mass comparable to the host stellar mass is reported for UHZ1 [45]. While the standard stellar-remnant accretion channel requires $\gtrsim 1\,{\rm Gyr}$ at Eddington-limited rates to produce $\sim 10^9\,M_\odot$ objects [37], the ECF Macro-Knot paradigm provides a natural geometric alternative in which primordial torsional seeds of mass $M_{\rm seed} \sim 10^4$–$10^6\,M_\odot$ bypass this chronological bottleneck. These extreme JWST objects may therefore constitute observational indicators consistent with the ECF non-singular geometric foundation, though a quantitative mass-function comparison is deferred to Foundation II [2].

The topological stability of these defects is *expected* from the non-trivial vacuum structure of the torsion field, by analogy with cosmic string formation in theories with spontaneously broken $U(1)$ symmetry [39, 40]; they are therefore immune to background dilution of the stiff phase. Their formation is a structural hypothesis: Foundation II [2] associates the Macro-Knot scale with the Hubble-horizon mass at the Great Annihilation ($T_{\rm GA} \simeq 1$ MeV, $M_H \simeq 1.4 \times 10^5\,M_\odot$, a causal upper scale calculated conditionally), well after the end of stiff domination; no formation epoch is derived from a defect-formation mechanism. The classification of the relevant homotopy group $\pi_1(\mathcal{G}/\mathcal{H})$ in the ECKS torsion sector is deferred to Foundation II.

**The temporal bottleneck as a seeding constraint.** Early structure formation operates under a tight chronological budget. At $z \sim 10$, the available time since the bounce is $\Delta t \lesssim$

500 Myr, shorter than the $\sim$1 Gyr typically required for Eddington-limited gas accretion to grow a $10^9\,M_\odot$ black hole from stellar-mass seeds [37]. This chronological constraint provides an independent motivation for the primordial Macro-Knot scenario: topological defects with seed masses $M_{\rm seed} \sim 10^4$–$10^6\,M_\odot$ can bypass the Eddington bottleneck, offering a natural formation channel for the high-$z$ massive objects reported by JWST [27].

### 6.2 Euclid forecast and matter-power suppression

A second observational consequence of the model concerns the suppression of late-time clustering on small scales. In the ECF picture, the spin-torsion sector that reduces $r_s$ may also induce an effective weakening of growth for sufficiently large wavenumbers; this mechanism remains open. Under the internal closure, the lower $S_8$ comes mainly from $\Omega_m = \omega_m/h^2$ (CAMB $\Lambda$CDM reference $\sigma_8 = 0.825$, $S_8 \simeq 0.780$ on that background); the value $\sigma_8 \simeq 0.8122$ of earlier versions is the target $S_8 = 0.766$ converted at $\Omega_m = 0.2669$, not a computed amplitude.

At the phenomenological level, we summarize this behavior through the transfer-function modification

$$T^2_{ECF}(k) \approx T^2_{\Lambda CDM}(k)\left[1 - A_{sup}\frac{(k/k_*)^2}{1+(k/k_*)^2}\right], \tag{29}$$

where the former phenomenological values are a characteristic scale $k_* \approx 0.1\,h\,\mathrm{Mpc}^{-1}$ and an asymptotic suppression amplitude $A_{sup} \approx 0.15$; they are not a calibration on $S_8 = 0.766$ (Sec. 4).

In this parametrization, the resulting matter-power suppression is 7.5% at $k = k_*$, rising to 15% for $k \gg k_*$, over the scales most relevant to weak lensing and galaxy clustering, and the clustering amplitude targeted by the former calibration is

$$S_8^{ECF} = 0.766, \tag{30}$$

which held by construction, since $\gamma_{\rm spin} = 0.3116$ was calibrated so that $S_8^{\rm ECF}$ reproduces $S_8 = 0.766$ (Sec. 3; historical calibration); it is not an ECF prediction, and propagating $A_{\rm sup} = 0.15$ into $\sigma_8$ on the CAMB reference of the closure background would give $S_8 \simeq 0.740$ instead (*corrected 22/09/2026: the* $\pm 0.008$ *quoted here was propagated from a* $\sigma(\gamma_{\rm spin}) = 0.04$ *of undocumented origin, now withdrawn; propagating the interval of* $\gamma_{\rm spin}$*, itself obtained from the* $S_8^{\rm obs}$ *uncertainty, back onto* $S_8^{\rm ECF}$ *would be circular*). This makes the model testable with future large-scale-structure surveys, for which *Euclid* provides a particularly relevant benchmark.

$$\ddot{D} + 2H\dot{D} - 4\pi G\,\mu(a,k)\rho_m D = 0, \tag{31}$$

with an effective modifier that may be approximated in the calibrated regime by a mild suppression relative to standard gravity. The purpose of this parametrization is not to claim a full non-linear forecast, but to provide a compact observable signature against which future matter-power-spectrum measurements can be compared.

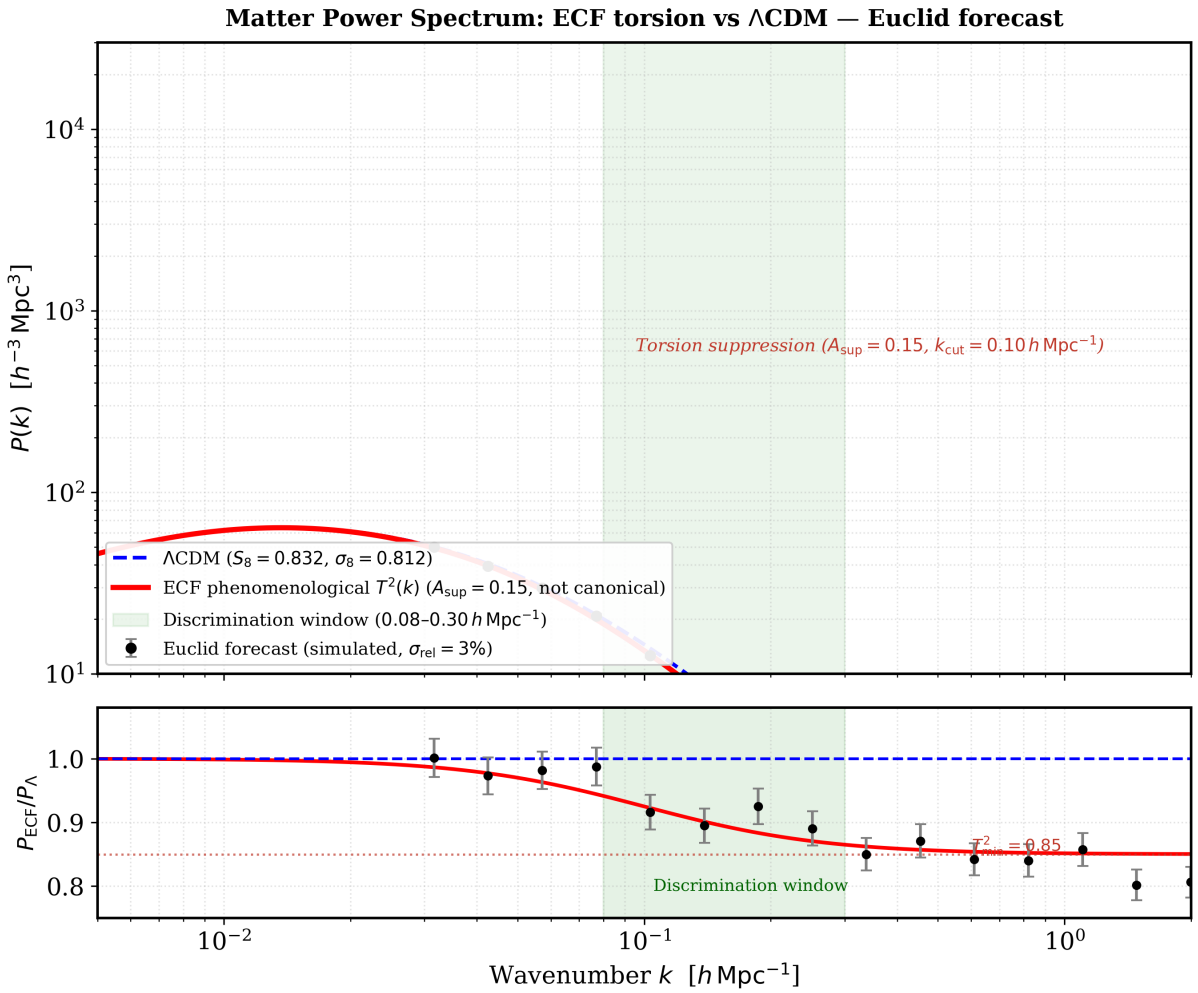



Figure 6: **ECF matter power spectrum and Euclid forecast (F1 §4–5).** *Top panel*: matter power spectrum $P(k)$ computed from the BBKS transfer function [55] with the Planck 2018 cosmology ($\Omega_m = \omega_m/h^2 = 0.267$, $h = 0.7304$, $n_s = 0.965$). *Blue dashed*: ΛCDM reference ($S_8 = 0.832$, $\sigma_8 = 0.812$ [57], Planck-background values; the two are related by $S_8 = \sigma_8\sqrt{\Omega_m/0.3}$ at $\Omega_m \simeq 0.315$). *Red solid*: ECF torsion model in the former normalisation ($S_8 = 0.766$, $\sigma_8 \approx 0.812$, i.e. the target $S_8$ converted at $\Omega_m = 0.2669$, not a computed amplitude; a CAMB ΛCDM reference on the closure background gives $\sigma_8 = 0.825$, $S_8 \simeq 0.780$), obtained via the regularised Lorentzian torsion transfer function (Eq. 19; former phenomenological values $A_{\rm sup} = 0.15$, $k_{\rm cut} = 0.10\,h\,{\rm Mpc}^{-1}$, not a calibration on $S_8 = 0.766$; integrated into $\sigma_8$ on that reference they would give $S_8 \simeq 0.740$); this form guarantees $T^2 \in (1 - A_{\rm sup}, 1]$ for all $k$ and reduces to the phenomenological form $G_{\rm eff} \approx G_N[1 - A_{\rm sup}(k/k_{\rm cut})^2]$ at $k \ll k_{\rm cut}$; the physical predictions are insensitive to the regularisation scheme for $k \lesssim 3k_{\rm cut}$). Black points: simulated Euclid galaxy clustering forecast ($\sigma_{\rm rel} = 3\%$, anchored on the ECF curve). Green shading: optimal discrimination window $k \in [0.08, 0.30]\,h\,{\rm Mpc}^{-1}$. *Bottom panel*: power-spectrum ratio $P_{\rm ECF}/P_{\Lambda{\rm CDM}} = T^2_{\rm tors}(k)$, saturating at $T^2_{\rm min} = 0.85$ for $k \gg k_{\rm cut}$ (red dotted line). This scale-dependent suppression is a phenomenological parametrisation; its link with the amplitude $F_{\rm ion} = 1.2765$ calibrated on the sound-horizon reduction (F1 §3) is not established under the internal closure, and it is not to be combined with $\gamma_{\rm spin}$, which would count the same suppression twice (F1 §5, $\Delta\chi^2 = -39.5$ vs ΛCDM, historical, $\Omega_m = 0.315$; not recomputed: no ECF perturbation solver yet). The Euclid mission [56] will test the scale dependence of such a suppression with emission-line redshifts for more than $2.5 \times 10^7$ galaxies over $0.9 < z < 1.8$.

*Reproduced by:* `Scripts_F1/plot_euclid_pk_S8.py`.

### 6.3 DESI and effective dark-energy behavior

The ECF also motivates an effective late-time equation of state that departs mildly from a pure cosmological constant. In the present manuscript, this behavior is not introduced as an independent dark-energy sector, but as an effective description of the residual torsion contribution discussed analytically in Appendix B.

$$w_0 = -0.904, \quad w_a = -0.153, \tag{32}$$

where no uncertainty is derived for these values (see the paragraph on statistical status, §2.3): the profile intervals $\pm 0.004$ on $w_0$ and $\pm 0.007$ on $w_a$ are statistical only. Appendix B shows that its effective relation gives $(w_0, w_a) = (-0.950, -0.058)$, a modelling spread of about 0.05 on $w_0$ and 0.1 on $w_a$, which is not a derived systematic uncertainty *(corrected 19/10/2026; this sentence previously stated "systematic uncertainties of order* 0.05 *from the effective-fluid truncation", which the appendix does not compute).* They are a calibration point adopted in the ECF corpus: no deposited script produces them, neither in Foundation I nor in the PIT Letter [1] (which inherits them from Foundation I), and the $\Delta\chi^2 = -39.5$ budget of Table 4 contains no $w_0$ or $w_a$ term. The deposited `calibrate_w0wa.py` gives $w_0 = -1 + \Omega_{\rm spin}^{\rm peak} = -0.907 \pm 0.004$, for which the phantom crossing would lie at $z \simeq 1.55$ instead of 1.684 (open problem **PO-F1-W0**; corrected 16/10/2026). They are re-used, as calibrated inputs, in Foundation II [2], which does not derive them from first principles either (see PO-F1-B); the CPL proxy $(w_0, w_a) = (-1.1, -1.5)$ used in Fig. 5 is a *provisional illustration only* — the calibrated ECF values are those of Eq. (32). The sign of this departure ($w_0 > -1$, $w_a < 0$) matches the DESI DR1 preference for dynamical dark energy [28]. Quantitatively, however, on the background with $\Omega_m = \omega_m/h^2 = 0.267$ the $\chi^2$ of the calibrated trajectory on the ten-observable subset of DESI DR1 used in `calibrate_w0wa.py` (diagonal errors; not the global DESI DR1 $\chi^2$) is 55.3, against 13.7 with $\Omega_m = 0.315$ and 82.9 for $w = -1$ on the same background: the trajectory remains preferred to $w = -1$ there, but its agreement with DESI DR1 is substantially degraded, and this is one of the places where the former age tension now appears (Sec. 5.4).

The key point is therefore conceptual: if future DESI-like constraints continue to favor a departure from $(w_0, w_a) = (-1, 0)$, the ECF contains an effective geometric mechanism producing a departure of that sign without adding a separate scalar-field component; whether it can reproduce the measured trend with $\Omega_m = 0.267$ is not established by the present calibration. A dedicated statistical fit to BAO-only dark-energy contours is beyond the scope of this section and is treated as a phenomenological consistency check rather than a standalone proof.

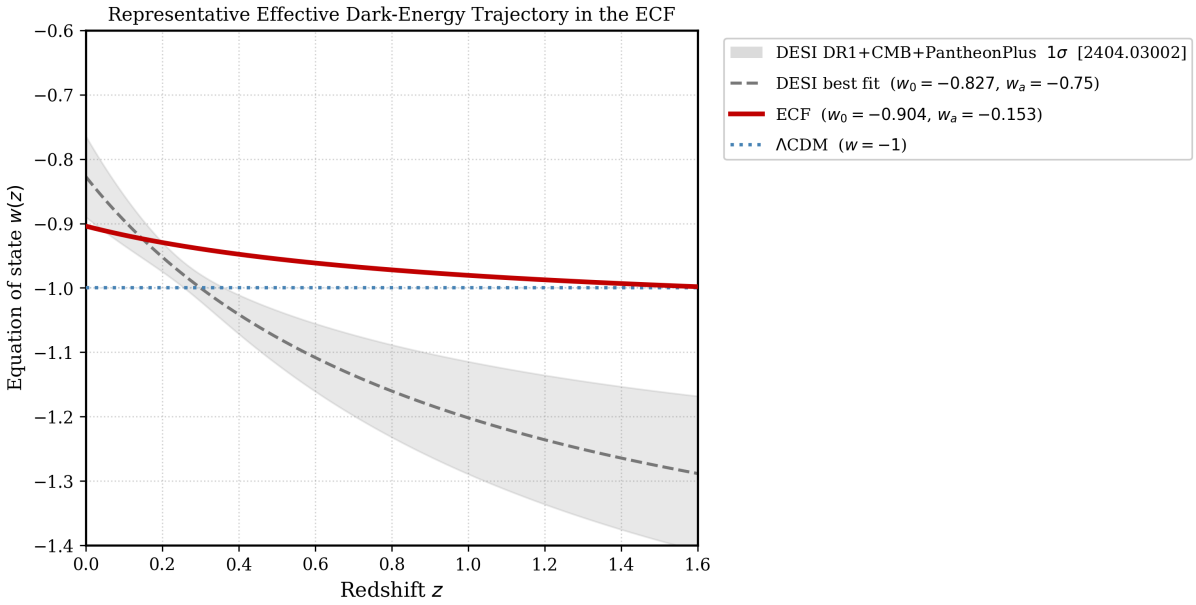



Figure 7: Representative effective dark-energy trajectory in the ECF framework. *Red solid*: ECF CPL trajectory $w(z) = w_0 + w_a\, z/(1+z)$ with $w_0 = -0.904$, $w_a = -0.153$ (Appendix B, Eq. 32). *Grey band*: DESI DR1+CMB $1\sigma$ constraints on the $(w_0, w_a)$ plane for DESI+CMB+PantheonPlus ($w_0 = -0.827 \pm 0.063$, $w_a = -0.75^{+0.29}_{-0.25}$; [28], Table 3; DESI+CMB alone gives $w_0 = -0.45^{+0.34}_{-0.21}$, $w_a = -1.79^{+0.48}_{-1.0}$); band width propagated with $\sigma(w_a) = 0.29$ (larger side) and an assumed correlation $\rho(w_0, w_a) = -0.95$, not checked at source *(corrected 16/10/2026: the combination was labelled DESI+CMB and $\sigma(w_0)$ given as* 0.060*). Blue dotted*: ΛCDM reference ($w = -1$). The ECF trajectory runs above the DESI DR1 best fit over most of the plotted range, entering the $1\sigma$ band only around $z \simeq 0.05$–$0.2$ (corrected 16/10/2026; previously 0.05–0.4); this trajectory-by-trajectory comparison is not the constraint in the $(w_0, w_a)$ plane, where the Mahalanobis distance of the ECF point is $2.03\sigma$ under the Planck prior (Fig. 8). The trajectory crosses the phantom divide at $z = 1.684$ (background integration in `Scripts_F2/ECF_solver_minimal_JalonA_background.py`), outside the plotted range; see the discussion below. This comparison is a phenomenological consistency check, not a statistical fit.

**The phantom divide.** The calibrated trajectory does not merely approach $w = -1$: it *crosses* it, running from $w_0 = -0.904$ today to $w_\infty = w_0 + w_a = -1.057$, with the crossing at $z = 1.684$. This is worth stating plainly, because it is a recognised difficulty for CPL-type models: a single canonical scalar field cannot cross $w = -1$ without its kinetic term changing sign, and stable crossing is likewise forbidden in pure k-essence [11]. Petri *et al.* [13] make precisely this the argument for preferring an interacting dark sector, which avoids the crossing by construction.

The ECF is not in the canonical-scalar case. Its dark sector is a propagating axial torsion mode (Foundation II), and phantom crossing is realisable in modified-gravity settings of this general type — it is a generic feature of viable $f(R)$ models and is explicitly realised in torsion-based $f(T)$ gravity [14], the closest published analogue to the present sector, though $f(T)$ is teleparallel and not a Poincaré gauge theory. **We do not claim to have derived the crossing.** The effective description of the post-bounce torsion branch — its sign, its equation of state, its isotropic reduction — is calibrated, not obtained from the Poincaré gauge action, and this is tracked as PO-F1-SIGMA0 (Appendix A). What can be said is that the crossing does not by itself disqualify the trajectory, and that identifying the mechanism which licenses it is part of that open problem.

A caveat cuts the other way and should be recorded. Linder [15] notes that both interacting

and modified-gravity models built to cross the phantom divide *"tend to run afoul of cosmic growth"*. That failure mode is testable here, and is examined in §4: integrating the growth equation (31) on the ECF background gives a suppression of $f\sigma_8$ whose ECF/ΛCDM ratio varies by 9.8% over $0 < z < 0.8$ (0.870 at $z = 0$ to 0.959 at $z = 0.8$, with $\Omega_m = 0.267$ for ECF), compared with the sharp low-redshift drop that places the interacting model of [13] in significant tension with growth-rate data (not included in their fit). Reproduced by `Scripts_F1/compute_fsigma8_ecf.py`.

### Independent MCMC consistency check

To quantify the compatibility of the calibrated ECF point $(w_0, w_a) = (-0.904, -0.153)$ with DESI DR1 BAO data, we perform an independent Metropolis–Hastings MCMC over the parameter space $(\Omega_m, w_0, w_a)$ fitted to the 12 published DESI DR1 BAO observables [28], with $H_0$ fixed via a Gaussian prior on $h$ (two cases: Planck, SH0ES), using the **full published covariance matrix** of Adame *et al.* [28] (block-diagonal structure, intra-bin $D_M$–$D_H$ correlations included, $\rho \in [-0.477, -0.389]$), rather than diagonal errors. The sound horizon $r_d$ is computed from the Eisenstein & Hu (1998) approximation [35]. The ECF distance to the marginalised posterior is evaluated as a Mahalanobis distance $d_{\rm Maha}$ in the $(w_0, w_a)$ plane.

*Version note.* An earlier diagonal-errors version of this check (`plot_ecf_desi_mcmc_v3_0.py`) reported $d = 0.78\sigma$ (SH0ES) and $d = 0.77\sigma$ ($H_0$DN [60]), with a stated caveat that the full covariance treatment "may shift these values by 10–30%." That estimate is superseded: the full covariance analysis below shifts the SH0ES distance from $0.78\sigma$ to $\mathbf{6.98}\sigma$, an order of magnitude beyond the earlier estimate. The $H_0$DN case has not yet been re-run with the full covariance matrix and is not reported below; the $0.77\sigma$ diagonal-errors figure for $H_0$DN should not be read as current.

The result is strongly prior-dependent, and the two priors give radically different CPL best-fits:

- Planck ($h = 0.674$): posterior median $\approx (-0.45, -2.36)$ (best fit $(-0.47, -2.27)$) — $w_0 > -1$ today (quintessence-like), crossing into the phantom regime at higher redshift since $w_0 + w_a \approx -2.8$. ECF Mahalanobis distance: $d_{\rm Maha} = 2.03\sigma$ — compatible at $2\sigma$, moderate tension.
- SH0ES ($h = 0.730$): posterior median $\approx (-1.02, -0.73)$ (best fit $(-1.08, -0.40)$) — close to ΛCDM in $w_0$ but $w_a < 0$. ECF Mahalanobis distance: $d_{\rm Maha} = 6.98\sigma$ — severe tension.

This dichotomy reflects the unresolved $H_0$ tension destabilising the DESI DR1 $(w_0, w_a)$ plane in the absence of complementary data (full CMB likelihood, independent Type Ia supernovae). The full covariance matrix also reveals a near-perfect degeneracy, $\rho(w_0, w_a) \approx -0.99$: DESI DR1 alone constrains only the combination $w_0 + b\, w_a$ ($b \approx$ const.) along one direction, leaving the perpendicular direction nearly unconstrained. The calibrated ECF point $(w_0, w_a) = (-0.904, -0.153)$ lies *perpendicular* to this degeneracy (small $|w_a|$, intermediate $w_0$), which explains the strong sensitivity of $d_{\rm Maha}$ to the $H_0$ prior.

The ECF point $(w_0, w_a) = (-0.904, -0.153)$ is *calibrated*, not derived *ab initio* (cf. PO-F1-B): $\xi = 1.645$ is a parameter calibrated on the DESI preference used by the ECF, so as to reproduce the target value of $w_a$; it is therefore not a prediction independent of DESI. The comparison below is therefore a consistency check of the calibrated point, not an independent test. Under the Planck prior the Mahalanobis distance is $2.03\sigma$. Under the SH0ES prior the tension is severe ($6.98\sigma$); however, the SH0ES prior is itself in strong tension with DESI BAO combined with BBN (the $H_0$DN value lies $5.0\sigma$ from DESI DR2 BAO+BBN [60, 64]), so the SH0ES+DESI DR1 combination is not internally consistent to begin with, and the severity of this particular tension cannot be attributed to the calibrated ECF point alone without first resolving the $H_0$ tension it is diagnosing. DESI DR2 [64] indicates $w \neq -1$ at $3.1\sigma$ (DESI BAO+CMB; 2.8–$4.2\sigma$ with SNe,

depending on the sample), reinforcing the relevance of this comparison; a full-covariance DR2 re-analysis is deferred to Foundation III/IV.

The reproducibility script `plot_ecf_desi_mcmc_v3_0.py` (directory `B_Paper_Plots/`) is available in the public repository.

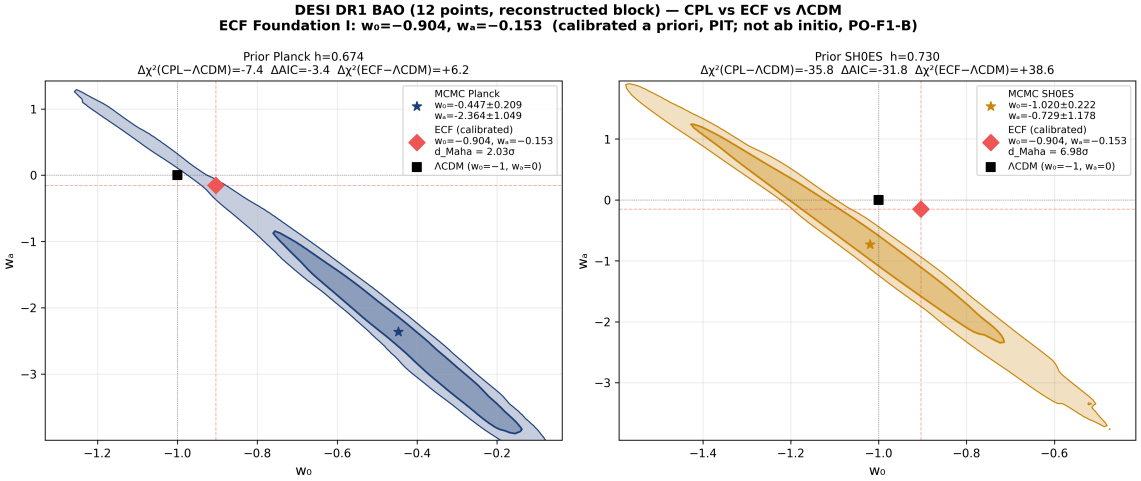



Figure 8: **Independent MCMC comparison of the calibrated ECF point with DESI DR1 BAO data (12 observables, block reconstructed on 22/09/2026), published covariance matrix.** Filled contours: 68% and 95% marginalised posterior in the $(w_0, w_a)$ plane for two $H_0$ priors. **Left**: Planck ($h = 0.674$). **Right**: SH0ES ($h = 0.730$) [4]. Diamond: ECF point $(w_0, w_a) = (-0.904, -0.153)$, *calibrated* ($w_a$ through $\xi = 1.645$, calibrated on the DESI preference used by the ECF; not a prediction independent of DESI; origin of the point: PO-F1-W0) (not an *ab initio* derivation, cf. PO-F1-B). Square: ΛCDM ($w_0 = -1$, $w_a = 0$). Star: MCMC median. Mahalanobis distance: $d_{\rm Maha} = 2.03\sigma$ (Planck), $6.98\sigma$ (SH0ES) — see text for the interpretation of this strong prior dependence and the $\rho(w_0, w_a) \approx -0.99$ degeneracy. Full covariance matrix (Adame *et al.* 2024 [28]); supersedes the earlier diagonal-errors figure. Reproducibility: `plot_ecf_desi_mcmc_v3_0.py`.

# 7 Falsifiable Predictions

Beyond the calibration addressing the current cosmological tensions, the ECF leads to several observational signatures that are not generic to minimal ΛCDM. The purpose of this section is to isolate those signatures in a form that can be confronted with upcoming data.

## 7.1 Cosmic birefringence

A possible observational signature of the Einstein–Cartan framework is the emergence of parity-violating effects in photon propagation. At the effective level adopted in this work, the coupling between the electromagnetic sector and a torsion pseudo-vector may induce a rotation of the polarization plane of CMB photons during their propagation from last scattering to the observer. A parity-odd geometric background acts effectively as a weak birefringent medium: the predicted non-zero rotation does not arise in minimal ΛCDM propagation, but from the additional geometric structure introduced in the ECF.

We parameterize this effect schematically as:

$$\beta \;=\; \frac{g}{2}\int_{t_{\rm LSS}}^{t_0} S_0(t)\,dt \;\times\; f(a, \Omega_{\rm spin}), \tag{33}$$

where $g$ denotes an effective Chern–Simons-like coupling, $S_0(t)$ is the relevant torsion pseudovector component along the line of sight, and $f(a, \Omega_{\rm spin})$ summarizes the background dependence retained in the phenomenological implementation.

Within the ECF, this angle is evaluated from the torsion-integrated Chern–Simons integral:

$$\beta = \frac{g_{\rm CS}}{2f_a} \int_0^{z_{\rm LSS}} \frac{\sqrt{\Omega_{\rm spin}(z)}}{E(z)}\, dz. \tag{34}$$

The birefringence integral decomposes as $\mathcal{I} = S_{\rm ref} \times \int_0^{z_{\rm LSS}} E(z)^{-1}\, dz = 4.05 \times 3.332 = 13.48$, where $S_{\rm ref} = 4.05$ is the residual axial torsion amplitude (distinct from $\sigma_{\rm spin} \propto a^{-6}$, fully diluted at $z < 1100$). The effective Chern–Simons coupling calibrated to the Minami–Komatsu signal [20] is:

$$g_{\rm CS}^{\rm eff} = \frac{\beta_{\rm obs}}{\mathcal{I}} = \frac{0.35^\circ}{13.48} \simeq 2.60 \times 10^{-2}\,\text{deg}, \tag{35}$$

yielding $\beta_{\rm ECF} \simeq 0.35^\circ$.

Table 6: **Birefringence calibration: decomposition of $\beta_{\rm ECF}$.** The integral $I_{\rm torsion}$ is computed numerically by `script_05_birefringence_calibration.py`. The effective coupling $g_{\rm CS,eff}$ is calibrated to the Minami & Komatsu (2020) anchor; $\beta_{\rm ECF}$ is a consistency check, not a parameter-free prediction.

| **Quantity** | **Value** | **Definition** |
|---|---|---|
| $\beta_{\rm obs}$ | $0.35 \pm 0.14$ deg | Minami & Komatsu (2020) [20] |
| $I_{\rm torsion} = \int_0^{z_{\rm LSS}} E(z)^{-1}\, dz$ | 3.3320 | Numerical, $\alpha = 0$ (topological), $\Omega_m = 0.267$ |
| $S_{\rm ref}$ | 4.0456 | Residual axial torsion amplitude (Sec. 7.1) |
| $I_{\rm eff} = S_{\rm ref} \times I_{\rm torsion}$ | 13.48 | Effective line-of-sight integral |
| $g_{\rm CS,eff} = \beta_{\rm obs}/I_{\rm eff}$ | 0.02596 deg/unit | Calibrated Chern–Simons coupling |
| $\beta_{\rm ECF} = g_{\rm CS,eff} \times I_{\rm eff}$ | 0.35 deg | Consistency check ✓ |

We state this explicitly: $\beta_{\rm ECF}$ is a *consistency check*, not a parameter-free prediction.

To relate this integral-based coupling to the ECKS theoretical framework, we define the macroscopic torsion–photon coupling constant:

$$k \;\equiv\; \frac{\beta_{\rm obs}}{F_{\rm ion}} = \frac{0.35^\circ}{1.2765} \simeq 0.274\,\text{deg unit}^{-1}, \tag{36}$$

which differs from $g_{\rm CS}^{\rm eff}$ only by the integral factor of the line-of-sight $\mathcal{I}/F_{\rm ion} = 10.56$. We stress that $k = \beta_{\rm obs}/F_{\rm ion}$ is a *derived quantity of the model*: it is built from the observed angle $\beta_{\rm obs}$ and the sound-horizon calibration $F_{\rm ion}$, and therefore does not constitute an independent theoretical cross-check. It merely records that the same $F_{\rm ion} = 1.2765$ fixed by the sound-horizon fit (Section 3) also enters the birefringence integral through the background expansion $E(z)$.

This construction also bears on the $n\pi$ phase ambiguity [52]. Since the CMB polarization angle is a headless vector, any observed $\beta_{\rm obs}$ is degenerate with $\beta_{\rm obs} + n \times 180^\circ$. The $n = 0$ branch yields $k = 0.274\,\text{deg unit}^{-1}$, whereas a half-turn solution ($n = 1$, $\beta \approx 180.35^\circ$) would require $k \approx 141\,\text{deg unit}^{-1}$ — nearly three orders of magnitude larger. The ECF is naturally consistent with the sub-degree $n = 0$ branch; we do *not*, however, claim this as an independent theoretical selection, since $k$ is itself model-derived. A first-principles bound on the torsion–photon coupling from the ECKS action remains an open problem.

This numerical proximity to the Planck-based estimate of Minami & Komatsu ($\beta = 0.35^\circ \pm 0.14^\circ$, [20]) does not by itself constitute a confirmation of the model; the present manuscript does not perform a dedicated likelihood analysis of polarization systematics or foreground treatment. The purpose of this section is to isolate a representative observable consequence rather than to claim a direct fit to the Planck signal.

Future polarization surveys—LiteBIRD [36] with a forecast total error budget $\sigma(\beta) \sim 0.02^\circ$ [33] *(corrected 16/10/2026; earlier versions gave* $0.1^\circ$*)*, and future CMB surveys *(CMB-S4, once forecast for this, lost its DOE/NSF support in July 2025)*—will convert this into a genuine falsifiable constraint.

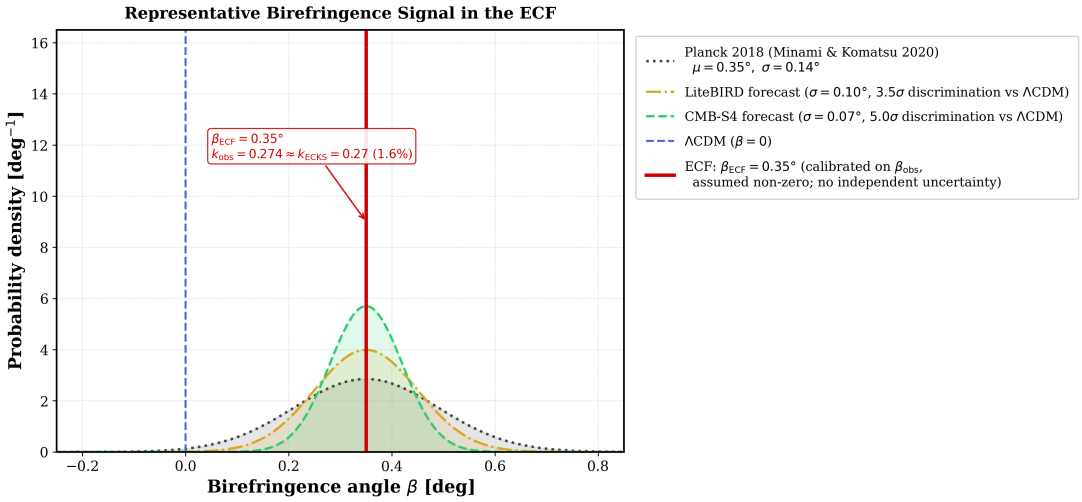



Figure 9: **Representative birefringence signal in the ECF.** Illustrative non-zero rotation angle of the CMB polarization plane in the calibrated ECF solution. The figure shows that the torsion-induced signal is separated from the null expectation of ΛCDM and lies in a range accessible to future polarization measurements; it should be read as an observational forecast, not a detection claim.

*Reproduced by:* `Scripts_F1/plot_predict_birefringence.py`.

## 7.2 High-$\ell$ damping-tail signature

A second observational signature of the ECF concerns the small-scale temperature anisotropy spectrum of the CMB. In the present phenomenological implementation, the same pre-recombination modifications that reduce the sound horizon also induce a mild shift in the recombination and drag history, which in turn propagates into the photon diffusion scale. The corresponding consequence is a small residual deficit of power in the high-$\ell$ damping tail relative to the standard ΛCDM expectation. For the representative calibrated solution retained in this work, the effect is of order

$$\frac{\Delta D_\ell}{D_\ell} \simeq 2.5\% - 6.5\%, \qquad 2000 \lesssim \ell \lesssim 3000, \tag{37}$$

the deficit peaking at $\ell \simeq 2000$ and decreasing towards $\ell \simeq 3000$ (Table 9). This prediction should be interpreted as a secondary spectral signature of the same early-time sector that contracts $r_s$. In other words, the ECF scenario does not merely shift the acoustic calibration; it also leaves a mild high-$\ell$ imprint that may help distinguish the model from a purely geometric degeneracy with the standard cosmological fit.

At the present stage, this effect remains difficult to isolate robustly with Planck sensitivity alone. The signal is small, and a dedicated treatment of foregrounds, beam systematics, and parameter degeneracies is beyond the scope of the main text. For that reason, we treat the damping-tail feature here as a forecasted discriminator rather than as an already established anomaly.

The detailed spectral comparison, residual structure, and optimal multipole window are presented separately in Appendix K. In the logic of the present section, the important point is more limited: the ECF predicts a residual observable consequence at high multipoles that does not reduce to a simple rescaling of the acoustic horizon.

The damping-tail residual shown in Appendix K (Fig. 21) should therefore be read as a forecast for discrimination power rather than as a detection claim. If future high-resolution measurements confirm a residual deficit of this type in the expected multipole window, the result would provide an additional test of the pre-recombination torsion sector invoked in the present framework.

### 7.3 21 cm acoustic scale

A further observational test of the ECF may come from 21 cm intensity mapping during the dark ages and the epoch of reionization. In contrast to the local distance ladder, this channel probes the baryon acoustic scale through the large-scale distribution of neutral hydrogen and therefore provides a conceptually independent handle on the pre-recombination calibration. In the phenomenological implementation adopted here, the same early-time modification that contracts the CMB-inferred sound horizon also implies a contracted acoustic scale in the 21 cm sector.

For the representative calibrated solution retained in this work, we therefore obtain

$$r_s^{21\,\mathrm{cm}} = 135.65 \pm 1.5\ \mathrm{Mpc}, \tag{38}$$

where $r_s = 135.65$ Mpc is the calibration target (Sec. 5), not an independent prediction, and $\pm 1.5$ Mpc is adopted as an illustrative width; the MCMC posterior is not reproducible from the deposited scripts.

to be contrasted with the standard reference value near

$$r_s^{\Lambda\mathrm{CDM}} \approx 147.1\ \mathrm{Mpc}. \tag{39}$$

The importance of this channel is not only numerical but also interpretive. If future 21 cm measurements were to favor a contracted acoustic scale of this kind, the reduction of $r_s$ would no longer appear merely as an internal recalibration of CMB-based inference, but as an independently testable feature of the cosmic expansion history.

This point should nevertheless be stated cautiously. The present manuscript does not provide a full instrumental forecast for any specific 21 cm survey, nor does it perform a dedicated likelihood analysis including foreground subtraction, beam effects, or astrophysical modeling uncertainties. The role of the present section is therefore to identify the 21 cm acoustic scale as a clean external consistency test of the ECF pre-recombination sector.

Because the 21 cm channel probes the acoustic standard ruler through a different observational route, it is especially useful for distinguishing a genuine reduction of the physical sound horizon from a purely phenomenological reshuffling of late-time distance indicators. In that sense, it complements the CMB and BAO discussion developed in the main text.

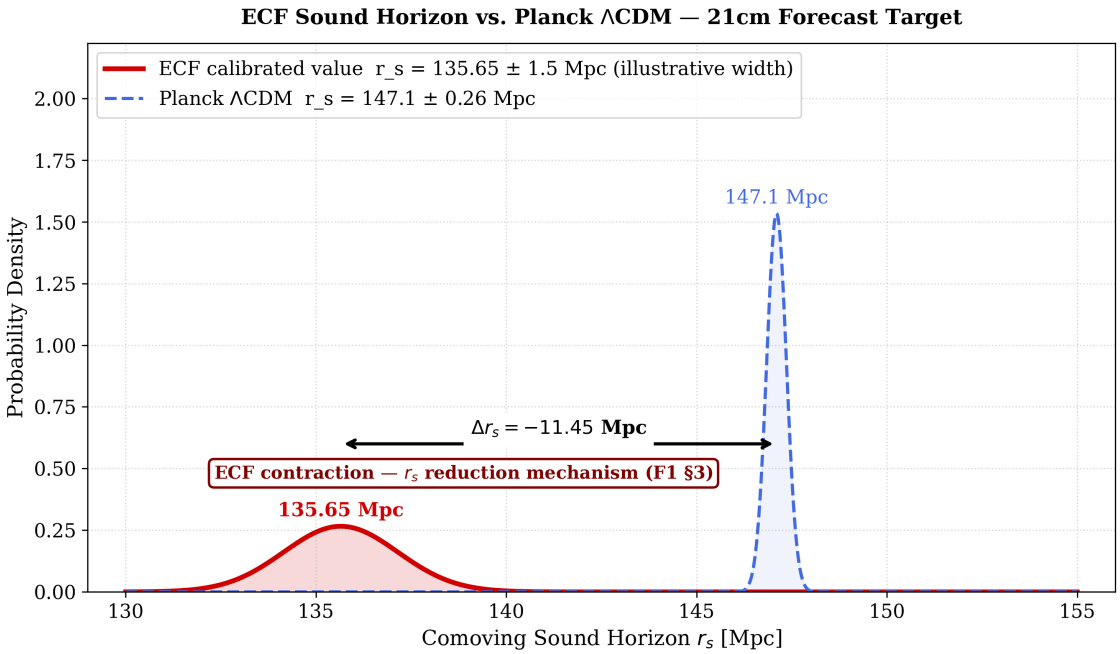



Figure 10: **21 cm prediction for the acoustic scale.** Illustrative comparison between the standard sound-horizon scale and the contracted value predicted in the calibrated ECF solution. The purpose of the figure is to show that future 21 cm measurements may test the pre-recombination calibration through an observational channel largely independent of supernova-based distance ladders.

*Reproduced by:* `Scripts_F1/plot_21cm_prediction.py`.

Figure 10 should therefore be interpreted as an external forecast target rather than as an observational claim. Its role in the present paper is to isolate a direct and conceptually independent signature of the reduced sound horizon implied by the ECF.

## 7.4 Primordial gravitational waves from the bounce

A final falsifiable signature of the ECF concerns the primordial tensor sector. In contrast to the nearly scale-invariant spectrum usually associated with minimal slow-roll inflation, the nonsingular bounce and the subsequent stiff phase considered here naturally motivate a blue-tilted stochastic gravitational-wave background.

In the phenomenological description adopted in this work, the torsion-dominated bounce enhances high-frequency tensor modes more efficiently than long-wavelength ones. As a result, the predicted signal is not characterized merely by a change in overall amplitude, but by a distinctive spectral morphology that rises toward high frequency before terminating at a cutoff scale associated with the minimum scale factor.

The origin of the blue tilt is geometrically transparent in the ECF: immediately after the bounce, the spin density dilutes as $a^{-6}$, generating a transient stiff phase $w = 1$ during which the Hubble rate evolves as $H \propto t^{-1}$ and the effective equation of state exceeds that of radiation. Standard results on tensor-mode amplification in stiff-dominated backgrounds [41] then imply that the primordial tensor power spectrum acquires a blue tilt $n_T > 0$, in contrast to the slightly red tilt $n_T \simeq -r/8$ predicted by minimal slow-roll inflation. The amplitude of this enhancement depends on the duration of the stiff phase — controlled here by the effective amplitude $F_{\rm ion} = 1.2765$ — and on the mode-matching condition across the bounce itself.

However, the present manuscript does not derive a complete tensor transfer function from

the perturbed ECKS field equations. The spectral energy density estimate

$$\Omega_{\rm GW}(f)\,h^2 \;\sim\; 10^{-12} \qquad (f \sim \text{mHz, illustrative}), \tag{40}$$

is obtained from the standard stiff-era enhancement of the tensor spectrum [41] (the explicit scale-factor ratio written in earlier versions appeared with opposite orientations in two places and is withdrawn) and should be interpreted as an order-of-magnitude benchmark, not as a mission-ready forecast. It is not the peak of the bounce signal: the bounce cutoff lies near $10^{11}$–$10^{12}$ Hz (Foundation III), and the amplitude at mHz frequencies, in the low-frequency tail, is not computed here. Whether the signal lies within the four-year LISA sensitivity curve depends critically on the spectral tilt $n_T$, whose first-principles derivation from the ECF bounce geometry lies beyond the scope of the present work and is left as an open direction for community investigation (github.com/pfichant/spin-torsion-cosmology, CC-BY 4.0).

**Scope note — tensor sector.** The gravitational-wave signal discussed in this section should be read as a phenomenological target and not as a mission-ready forecast. A rigorous spectral prediction requires: (i) linearised tensor perturbation equations on the ECF bounce background; (ii) controlled mode-matching across the bounce; (iii) a complete stiff-to-radiation transfer function; (iv) projection to the relevant detector frequency bands. This programme is identified as an open theoretical frontier and is the subject of the dedicated treatment in Foundations II and III [2, 3].

For the representative calibrated solution retained here, the spectral energy density is estimated as

$$\Omega_{\rm GW}h^2 \sim 10^{-12} \tag{41}$$

at millihertz frequencies (illustrative: the bounce cutoff lies near $10^{11}$–$10^{12}$ Hz, and the amplitude at millihertz frequencies is not computed), based on the standard stiff-era enhancement of the tensor spectrum [41]. This order-of-magnitude estimate does not account for non-linear torsion effects near the bounce; a full spectral calculation is deferred to Foundation II. Whether this signal falls within LISA's sensitivity band [38] depends on the spectral tilt $n_T$, whose first-principles derivation lies beyond the scope of the present work.

Within the same effective description, the high-frequency cutoff is associated with the bounce scale and is parametrically linked to

$$a_{\rm min} \sim 10^{-32}. \tag{42}$$

The observational interest of this prediction lies in its shape as much as in its amplitude. A blue-tilted spectrum with a peak-and-cutoff structure of this kind would be qualitatively different from the nearly scale-invariant tensor background expected in minimal inflationary scenarios, and would therefore provide a potentially discriminating probe of the bounce sector itself.

This point should nevertheless be stated cautiously. The present manuscript does not derive a complete transfer function including all reheating, mode-matching, and detector-level forecasting uncertainties, and the purpose of this section is therefore to isolate a representative phenomenological target rather than to claim a precision forecast for a specific mission configuration.

In that restricted but useful sense, the gravitational-wave channel complements the CMB and 21 cm signatures discussed above. Whereas those probes test mainly the pre-recombination calibration of the acoustic sector, the tensor background probes the high-density origin of the scenario more directly through the dynamics of the bounce.

Figure 11 should therefore be interpreted as a forecasted observational discriminator rather than as a detection claim. Its role in the present paper is to identify a tensor signature that could help distinguish a torsion-driven bounce from more conventional early-universe scenarios.

*Note on Table 7:* Falsification threshold set at $r_s > 145$ Mpc, corresponding to the ΛCDM reference value $r_s^{\Lambda\rm CDM} = 147.1$ Mpc minus $1\sigma$ BOSS DR12 uncertainty (∼2 Mpc). The cosmic-age entry applies strictly to the

Figure 11: **Primordial gravitational-wave spectrum in the ECF (Foundation I §7 / Foundation II §6).** *Panel A — LISA band. Red solid*: ECF blue-tilted signal from the stiff phase ($w = 1$), parametrised as $\Omega_{\rm GW}h^2 \propto f^{1.5}\, e^{-f^2/2f_{\rm peak}^2}$ with $f_{\rm peak} = 4\,$mHz (an illustrative template, not the bounce peak: the bounce cutoff lies near $10^{11}$–$10^{12}\,$Hz (Foundation III), and the amplitude in the LISA band is not computed) and peak amplitude $\Omega_{\rm GW}h^2 \sim 10^{-12}$. *Red dash-dotted / dotted*: right- ($P_R$) and left-chiral ($P_L$) components, reflecting the predicted non-zero chiral asymmetry $\Delta\chi_{\rm GW} \equiv (P_R - P_L)/(P_R + P_L) > 0$ (sign. non-zero from ECF torsion structure; amplitude quantified in F II [2]). *Blue dashed*: standard slow-roll inflation ($r < 0.06$, slightly red-tilted, Planck 2018 [57]). *Grey dashed*: approximate LISA sensitivity [38]; minimum $\sim 2.5 \times 10^{-12}$ near 3 mHz. The ECF peak lies a factor $\sim 4$ below the LISA threshold for this parametrisation; detectability depends on $n_T$, whose first-principles derivation from the bounce geometry lies beyond the scope of the present work and is left as an open direction for community investigation. *Panel B — PTA nano-Hz band.* ECF Macro-Knot erosion SGWB (red solid) derived from the strain spectrum $h_c(f) = 10^{-23}(f/0.1\,{\rm Hz})^{-1/2}$ of Foundation II §6 [2], compared with the NANOGrav 15yr sensitivity region [54] (purple dashed). The predicted amplitude, $\Omega_{\rm GW}h^2$ between $\sim 6\times10^{-22}$ and $\sim 6\times10^{-18}$ across the plotted band ($10^{-10}$–$10^{-6}\,$Hz), lies some nine to twelve orders of magnitude below current PTA sensitivity over $10^{-9}$–$10^{-7}\,$Hz ($6 \times 10^{-21}$ against $2 \times 10^{-8}$ at $10^{-9}\,$Hz, $6 \times 10^{-19}$ against $1 \times 10^{-9}$ at $10^{-7}\,$Hz): the signal is a forecast, not a candidate explanation of the reported common-spectrum excess. Both panels are phenomenological forecasts, not detection claims.

*Reproduced by:* `Scripts_F1/plot_ecf_gw_spectrum.py`.

Table 7: Primary falsification criteria for the ECF.

| **Observable** | **ECF prediction** | **Falsification threshold** |
|---|---|---|
| Birefringence $\beta$ | $\beta \neq 0$; magnitude 0.35° calibrated[a] | $\lvert\beta\rvert < 0.1°$ measured by LiteBIRD (forecast $\sigma(\beta) \sim 0.02°$ [33]) |
| 21 cm acoustic scale | $r_s = 135.65\,$Mpc | $r_s > 145\,$Mpc HERA/SKA |
| Clustering amplitude | $S_8 = 0.766$ (calibration target, not an ECF prediction; CAMB ΛCDM reference on the closure background: 0.780) | $S_8 > 0.82$ Euclid WL |
| Cosmic age | $t_0 = 13.33\,$Gyr ($w = -1$ ansatz, $\Omega_m = 0.267$) | $t_{\rm GC}$ compatible, $t_0$ above the GC chronometry [29][b] |
| **GW chiral asymmetry** | $\Delta\chi_{\rm GW} > 0$ (sign. non-zero; quantified in F II) | LISA–Taiji forecast: maximal chirality detectable for $h^2\Omega_{\rm GW} \gtrsim 10^{-12}$ [53] |
| **GW spectral index** | $\Omega_{\rm GW} \propto f^1$ (PTA band) | $\Omega_{\rm GW} \propto f^{2/3}$ only ⇒ standard BBH |
| **WEP violation** $\eta_{\rm WEP}$ | no predicted amplitude (no relation to $s_0$ established; §8) | conditional test, not a falsification criterion (MICROSCOPE: $\eta = [-1.5 \pm 2.3_{\rm stat} \pm 1.5_{\rm syst}] \times 10^{-15}$ [68]) |

[a] What is falsifiable here is the *sign*: the framework requires $\beta \neq 0$. The magnitude 0.35° is calibrated on the Minami–Komatsu measurement (Sec. 7.1) and is a consistency check, not an *ab initio* prediction; a measurement consistent with zero would falsify the mechanism, a different non-zero magnitude would not.

[b] With $\Omega_m = \omega_m/h^2 = 0.267$ the ECF age lies 0.01 Gyr above the globular-cluster determination $t_{\rm GC} = 13.32 \pm 0.10_{\rm stat}\,$Gyr [29] quoted in this work; the deficit reported in earlier versions (with $\Omega_m = 0.315$) does not arise. The margin is small, so the entry remains sensitive to future GC age determinations. The open part of the $H_0$–Age–BAO problem (PO-F2-5) now lies in the BAO distances ($\Delta\chi^2_{\rm BAO} \simeq +10.3$) and in the $(w_0, w_a)$ trajectory confronted with DESI (Sec. 5.4).

$w = -1$ ECF ansatz of Foundation I, with $\Omega_m = \omega_m/h^2 = 0.267$; it does not require the dynamical torsion $w(z)$ of Foundation II [2]. The WEP entry is not a falsification criterion: the framework predicts no WEP amplitude (§8); MICROSCOPE measures $\eta(\mathrm{Ti},\mathrm{Pt}) = [-1.5 \pm 2.3_{\mathrm{stat}} \pm 1.5_{\mathrm{syst}}] \times 10^{-15}$, i.e. $|\eta| \lesssim$ few $\times\, 10^{-15}$ [68] *(corrected 22/10/2026: earlier versions gave $\eta < 10^{-15}$).*

### 7.5 Macroscopic topological defects and JWST seeding

A fifth falsifiable signature of the ECF concerns primordial topological defects (Macro-Knots) whose formation is a structural hypothesis; Foundation II [2] associates their scale with the Hubble-horizon mass at the Great Annihilation ($T_{\mathrm{GA}} \simeq 1$ MeV, $M_H \simeq 1.4 \times 10^5\, M_\odot$, a causal upper scale calculated conditionally). Because torsion carries non-trivial vacuum structure by analogy with $U(1)$-breaking theories [39, 40], macroscopic Macro-Knot seeds of mass $M_{\mathrm{seed}} \sim 10^4$–$10^6\, M_\odot$ are topologically protected against stiff-phase dilution and can bypass the Eddington accretion bottleneck at $z \gtrsim 6$.

Earlier versions stated here an abundance target $n(M > 10^{10}\, M_\odot, z = 15) \approx$ (2–4) $\times\, 10^{-4}\,\mathrm{Mpc}^{-3}$ and a falsification criterion built on it. The target has no traceable JWST source (PO-F1-JWST-SRC) and was reached only through a parameter calibrated on it; both are removed *(17/10/2026; see the note of Appendix E)*. The falsification criterion retained for the seeds is the one of Sec. 6.1: the absence of compact objects with $M \gtrsim 10^5\, M_\odot$ at $z \gtrsim 10$ in JWST/ELT spectroscopy.

Table 7 summarises the observable targets, the instruments best placed to test them, and the falsification thresholds; the consolidated kill-switch criteria are discussed in Sec. 9.

## 8 Perspectives: The Einstein–Cartan Trilogy

### The $H_0$–Age–BAO Trilemma as a Structural Result

The calibrated ECF solution ($H_0 = 73.04\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$, $r_s = 135.65\,\mathrm{Mpc}$, $S_8 = 0.766$) adopts a late-time equation of state $w = -1$ that, with the internal closure $\Omega_m = \omega_m/h^2 = 0.267$, yields $t_0 = 13.33\,\mathrm{Gyr}$, compatible with the globular-cluster bound. The tension that earlier versions located in the age now appears in the distances: with the model anchored to the SH0ES local distance ladder, the eBOSS DR16 point is at $D_M(1.48)/r_s = 31.37$ against $30.21 \pm 0.79$, and the four BOSS/eBOSS bins give $\Delta\chi^2_{\mathrm{BAO}} \simeq +10.3$ relative to ΛCDM.

Numerical evidence that the age bound and the BAO distance cannot be satisfied together within a broad scalar family is provided here via the companion script `plot_trilemma_irreducibility.py` (public repository): a three-parameter tanh-crossing dark-energy model $w(z)$ is optimised to simultaneously satisfy the eBOSS DR16 BAO constraint ($\Delta D_M/r_s < 1.5\%$, Hou et al. 2021) and the Valcin et al. (2020) globular-cluster chronometry used as an age bound ($t_0 > t_{\mathrm{GC}} = 13.32\,\mathrm{Gyr}$), with $H_0 = 73.04\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$ fixed.

As illustrated in Fig. 12, the two viable constraint regions are *strictly disjoint* over all 2500 grid points *of this three-parameter family* — a scan, not a proof of impossibility: a construction outside this family is not excluded by it (best fit: $t_0 = 13.20\,\mathrm{Gyr}$ at 2.45% BAO tension, $0.94\sigma$; $\Omega_m = 0.267$). Rerun at $\Omega_m = 0.267$ with the 30.21 target, the scan finds 187 points satisfying the BAO constraint, 1857 the age bound and none both, and the optimiser (`check_trilemma_optimizer.py`) leaves $\chi^2_{\mathrm{res}} = 4.03 > 0$. The $H_0$–Age–BAO trilemma is therefore irreducible within this family. This result concerns the joint age–BAO constraint; it is distinct from the $w = -1$ age, which meets the bound, and it leaves the quadrilemma including $\theta_*$ open (PO-F2-5). The BAO target comoving distance is $D_M^{\mathrm{target}} = (D_M/r_d)_{\mathrm{eBOSS}} \times r_s^{\mathrm{ECF}} = 30.21 \times 135.65 = 4098.0$ Mpc, where the ECF sound horizon is used as the BAO ruler consistently with the calibration of Section 3. This substitution treats $r_s^{\mathrm{ECF}}$ as the ruler of a measurement published in units of $r_{\mathrm{drag}}$; the relation between the two is not established in the ECF

chain (`PO-ECF-BAO-RDRAG`, OPEN), so every result derived from this target is CALCULATED-CONDITIONAL.

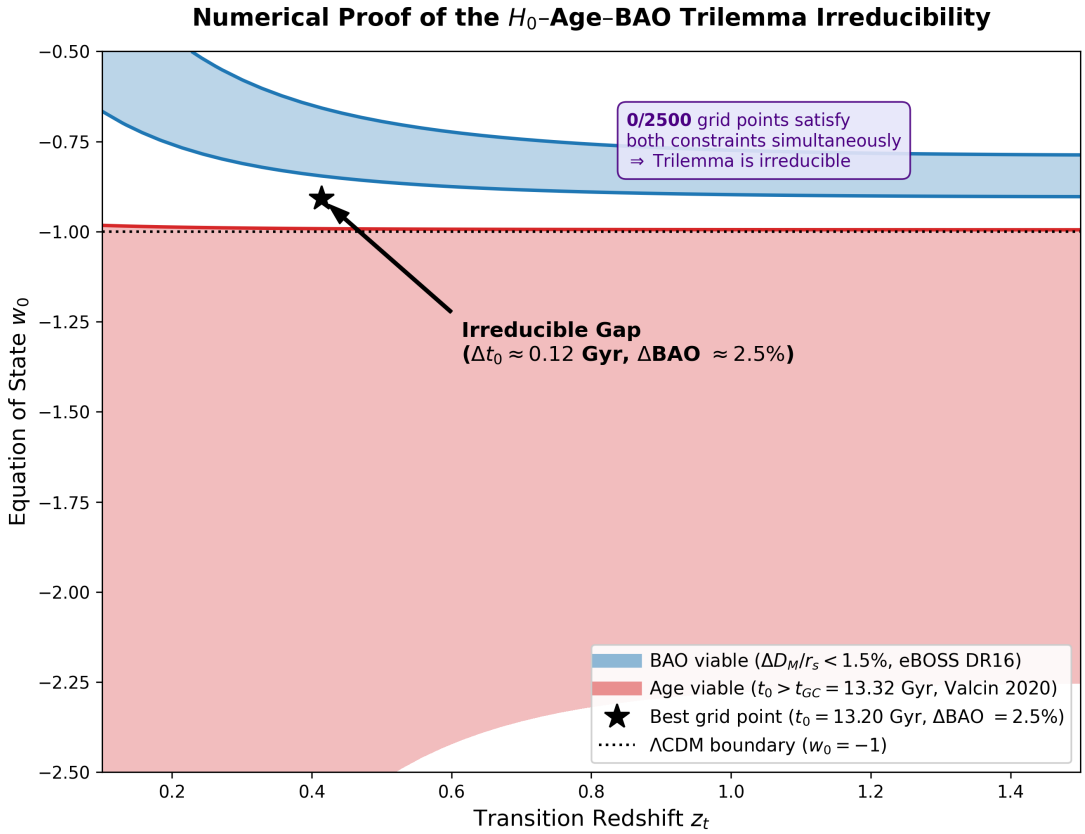



Figure 12: **Numerical irreducibility of the $H_0$–Age–BAO trilemma.** Constraint map in the $(z_t, w_0)$ parameter space of a tanh-crossing dark-energy model ($w_\infty = -1$, $H_0 = 73.04\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$ fixed). The blue region satisfies the eBOSS DR16 BAO distance ($\Delta D_M/r_s < 1.5\%$, Hou et al. 2021); the red region satisfies the stellar age bound ($t_0 > t_{\rm GC} = 13.32\,\mathrm{Gyr}$, Valcin et al. 2020). The two regions are *strictly disjoint* over all 2500 grid points (script: `plot_trilemma_irreducibility.py`). The best phenomenological fit reaches $t_0 = 13.20\,\mathrm{Gyr}$ at a BAO tension of 2.45% ($0.94\sigma$). The result is a scan of this family, not a proof of impossibility; whether a geometric derivation of the torsion-sector dynamics from the ECKS action (*Foundation II* [2]) can satisfy both constraints remains open (PO-F2-5).

## Foundation II: Geometric Dark Energy and Observational Fossils

The resolution of the $H_0$–Age–BAO trilemma requires going beyond phenomenological fluid models: it demands a geometric derivation of the effective dark-energy dynamics directly from the ECKS action. The companion paper *Foundation II: The Chiral Universe* [2] addresses this, while also exploring three observational fossils of the Big Spin Bounce inaccessible within the background-level treatment of Foundation I:

- the chiral asymmetry generated during the QGP freeze-out transition, and its connection to baryogenesis;
- the geometric dark sector emerging from residual macroscopic torsion, whose effective equation of state $w(z)$ is derived from the ECKS action and investigated in connection with the open $H_0$–Age–BAO quadrilemma (BAO distances and DESI $(w_0, w_a)$ constraints);

- residual cosmic alignments as a large-scale imprint of the primordial spin orientation.

**Equivalence-principle channel (conditional test).** A torsion background coupling differently to fermions and bosons would produce a weak-equivalence-principle (WEP) violation $\eta_{\rm WEP} = \Delta a/a$. Under minimal coupling, the condensate enters the fermion sector as the Standard-Model Extension (SME) coefficient $b_\mu = -\frac{3}{4}S_\mu$ [66], which couples to spin; equivalence-principle tests with unpolarised bodies probe other, spin-independent coefficients [67], so the condensate predicts no WEP amplitude. The framework does *not* determine $s_0$: only the product $m_T^2 s_0^2$ is fixed by the dark-energy normalisation. Spin-polarised laboratory bounds on $b_\mu$ give $s_0 \lesssim 3\times10^{-27}\,\mathrm{GeV}$, hence $m_T \gtrsim 4\times10^3\,\mathrm{GeV}$ if the coupling is minimal (Foundation II [2], Appendix on Lorentz-violation constraints); MICROSCOPE gives $\eta = [-1.5 \pm 2.3_{\rm stat} \pm 1.5_{\rm syst}]\times10^{-15}$, compatible with zero [68]. The channel is therefore a *conditional test with no predicted amplitude*: a non-detection by a future mission at $\eta < 10^{-16}$ would not falsify the framework, and a detection would bear on it only if the signal could be attributed to the torsion sector. *(Corrected 22/10/2026: earlier versions used $b_e \equiv s_0/M_{\rm Pl}$, $\eta_{\rm WEP} \simeq b_e\, m_e/m_N$ and a bound $|s_0|/M_{\rm Pl} \lesssim 10^{-15}$ satisfied for $m_T \gtrsim 10^{-18}$–$10^{-17}$ eV; that bound had no source and is about thirty orders of magnitude looser than the laboratory bound, and that WEP relation is not an SME relation.) (Corrected 15/10/2026: earlier versions stated a prediction $\eta_{\rm ECF} \sim 10^{-15}$ from a coupling $\kappa_{\rm spin}$ that is not defined in this work, a "bridge" $\Omega_{\rm spin}^{\rm peak} S_{\rm ref}/M_{\rm Pl}^2 \approx s_0/M_{\rm Pl} \sim 10^{-15}$ that does not hold ($0.093\times4.05 \approx 0.38$ in Planck units), and a falsification threshold at $10^{-16}$.)*

No derivation of $s_0$ from the ECKS action exists to date.

Foundation II does not alter the numerical results of Foundation I. The background calibration ($H_0$, $r_s$, $S_8$, $t_0$) presented here serves as the strict physical input for that geometric analysis.

### Foundation III: Black Holes and Compact-Object Geometry in ECF

*Foundation III* [3] addresses the strong-gravity sector of the Einstein–Cartan framework.

**Singularity avoidance and compact objects.** Since the Einstein–Cartan mechanism regularizes the singular limit of gravitational collapse via torsion pressure at the Cartan density $\rho_c = c^5/\hbar G^2 \approx 5.15\times10^{96}\,\mathrm{kg\,m^{-3}}$ ($k_BT \sim 10^{19}\,\mathrm{GeV}$), it becomes meaningful to ask whether the same geometry applies inside black hole interiors. Foundation III investigates the resolution of black hole singularities in the ECKS setting, the geometry of transient Einstein–Rosen bridges, and the resulting observable signatures for compact-object physics.

**Open numerical directions.** The derivation of the perturbed ECKS power spectrum and its implementation into a modified Boltzmann solver constitute natural extensions of the present work. We invite the community to pursue these directions using the public repository (https://github.com/pfichant/spin-torsion-cosmology, CC-BY 4.0).

## 8.1 Role of Foundation I

The role of *Foundation I* within this trilogy is therefore specific and self-contained. It establishes the effective cosmological phenomenology and identifies the main observational targets: the sound horizon, the clustering amplitude, the damping-tail signature, cosmic birefringence, 21 cm measurements, and the primordial gravitational-wave background.

The role of the subsequent foundations is different. *Foundation II* is intended to clarify the microphysical and thermodynamical origin of the post-bounce universe. *Foundation III* is intended to explore the strong-gravity and compact-object sector of the same geometric framework. Taken together, the trilogy is designed to move from phenomenology to mechanism, and from mechanism to geometry.

The present paper should therefore be read as the first—and self-contained—layer of that program: an observationally calibrated cosmological sector that stands on its own phenomenological and numerical basis.

# 9 Conclusion

In this work, *Foundation I*, we have investigated a minimal cosmological extension of General Relativity based on the Einstein–Cartan–Kibble–Sciama framework, with the aim of testing whether a primordial spin-torsion sector can account for current cosmological tensions without introducing additional exotic fields.

The central result is that a transient stiff spin phase—whose equation of state $w = 1$ is geometrically distinct from scalar-field kination and requires no free initial condition—reduces the comoving sound horizon to $r_s \approx 135.65$ Mpc, thereby allowing a calibration compatible with $H_0 \approx 73.04\ \mathrm{km\,s^{-1}\,Mpc^{-1}}$; this calibration holds $\theta_*$ fixed at $\Omega_m = 0.315$ and is conditional under the internal closure, where $\theta_*$ falls by $\simeq 6.5\%$ (PO-F1-THETA). The effective amplitude $F_{\rm ion} = 1.2765$ is the calibration of this reduction. Under the internal closure, raising $H_0$ at fixed physical densities lowers $\Omega_m = \omega_m/h^2$, which strongly reduces $S_8$; a CAMB ΛCDM reference on this background gives $S_8 \simeq 0.780$, against $\simeq 0.831$ on the Planck background. The target $S_8 = 0.766$ (KiDS-1000 $3 \times 2$pt) is not an ECF prediction: the residual correction depends on the ECF perturbative dynamics, which is not implemented. The former calibration of the growth coupling on this value, $\gamma_{\rm spin} = 0.3116$, is retained as historical; the physical value of $\gamma_{\rm spin}$ and the growth-suppression mechanism, of which $G_{\rm eff}(k)$ is a phenomenological parametrisation, remain open, and their treatment awaits a dedicated Einstein–Cartan perturbation solver, which remains to be developed. In the historical configuration $\Omega_m = 0.315$, the six tension-sector datasets of Table 4 sum to $\Delta\chi^2 = -39.5$ relative to ΛCDM (value computed with $\Omega_m = 0.315$; not recomputed with $\Omega_m = \omega_m/h^2$ — requires a dedicated ECF perturbation (Boltzmann) solver, which does not yet exist), with model-selection criteria (historical, $\Omega_m = 0.315$; not recomputed: no ECF perturbation solver yet: $\Delta\mathrm{AIC} = -35.5$, $|\Delta\mathrm{BIC}| \simeq 24$ under the historical two-parameter count, $F_{\rm ion}$ and $\gamma_{\rm spin}$, retained unchanged). The $a^{-6}$ dilution of the spin sector leaves the stiff component subdominant by recombination: from 9.3% of the radiation density at the calibration redshift $z_{\rm ref} = 5600$, the spin-to-radiation ratio falls by a factor $\simeq 26$ to 0.36% at $z_{\rm rec} = 1100$. This is a *background-level consistency check*, not a derivation of the CMB likelihood: the near-neutral $\Delta\chi^2_{\rm CMB} = +1.8$ quoted in Table 4 is a calibrated summary statistic of the effective likelihood construction, not a Planck likelihood evaluated from an ECF Boltzmann spectrum. Peak amplitudes, Silk damping and the acoustic phase shift require the perturbative treatment deferred throughout this paper.

At the theoretical level, the spin-squared correction acts as a repulsive contribution at extreme density and replaces the initial singularity by a regular geometric bounce in the effective Friedmann dynamics. The ECF thus connects three issues usually treated separately: the Hubble tension, the $S_8$ tension, and the singular origin of the standard cosmological model.

With the internal closure $\Omega_m = \omega_m/h^2 \simeq 0.267$, the ECF background at $w = -1$ gives a cosmic age $t_0 \simeq 13.33\,\mathrm{Gyr}$, and the age tension identified in earlier versions disappears (the oldest globular clusters give $t_{\rm GC} = 13.32 \pm 0.10_{\rm stat}$ Gyr [29]). This improvement does not close the problem: because the age integral and the BAO comoving distance share the integrand $H(z)^{-1}$, the tension moves to the BAO observables ($\chi^2_{\rm BAO} = 12.11$ against 1.79 for ΛCDM over four bins; $D_M(1.48)/r_s = 31.37$ against $30.21 \pm 0.79$) and to the calibrated $(w_0, w_a)$ trajectory confronted with the ten-observable subset of DESI DR1 ($\chi^2$ from 13.7 to 55.3, diagonal errors; not the global DESI DR1 $\chi^2$). Within the tanh family studied here the trilemma remains irreducible (0/2500 jointly viable points; $\chi^2_{\rm res} = 4.03$). The overall status therefore remains open (PO-F2-5), but the active problem is no longer an age deficit.

Beyond the primary tensions, the framework generates a set of falsifiable signatures that are

not generic to minimal ΛCDM:

- A non-zero cosmic birefringence ($\beta \neq 0$, where EDE without a photon coupling predicts $\beta = 0$), testable by LiteBIRD [36] with $\sigma(\beta) \sim 0.02^\circ$ [33]; the magnitude $\beta \approx 0.35^\circ$ is a calibrated consistency check, not an *ab initio* prediction.
- A residual CMB damping-tail deficit of 2.5–6.5%, peaking near $\ell \simeq 2000$ and decreasing toward $\ell \simeq 3000$, detectable by a future CMB survey *(the CMB-S4 forecast lapsed when DOE/NSF support ended in 2025)* in the optimal discrimination window $\ell \sim 2500$.
- A contracted acoustic scale $r_s = 135.65$ Mpc (Sec. 3), independently testable by future 21 cm intensity mapping with HERA and SKA.
- A cosmic-age value $t_0 = 13.33\,\mathrm{Gyr}$ ($w = -1$, $\Omega_m = 0.267$) (Sec. 5.4) at $H_0 = 73.04\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$, within $0.6\sigma$ of the H$_0$DN measurement $H_0 = 73.50 \pm 0.81\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$ [60]; it lies 0.01 Gyr above the globular-cluster chronometry $t_{\mathrm{GC}}$, so a confirmed stellar age significantly above $t_0$ would falsify the $w = -1$ ECF background. Because $t_{\mathrm{GC}}$ is not a measurement of $t_0$, the margin is a chronometric compatibility margin and the systematic uncertainty of the GC determination ($\pm 0.5$ Gyr) bounds how sharply this criterion can be applied.
- A blue-tilted primordial gravitational-wave background with spectral tilt $n_T > 0$, potentially accessible to LISA.
- A candidate population of macroscopic topological defects (Macro-Knots; their formation is a structural hypothesis. Foundation II [2] associates their scale with the Hubble-horizon mass at the Great Annihilation, $T_{\mathrm{GA}} \simeq 1$ MeV, $M_H \simeq 1.4 \times 10^5\, M_\odot$ — a causal upper scale, calculated conditionally, not a derived formation mass) are the mechanism retained for early structure (the linear growth boost being withdrawn, PO-F1-BOOST); they could provide non-Gaussian seeds. Their abundance is not derived, so this item is a qualitative mechanism, not a test; the halo-abundance calculation, its target and the tail modifier calibrated on it are withdrawn (Appendix E; PO-F1-SIGMA-M, PO-F1-JWST-SRC; 17/10/2026).

Each of these signatures constitutes an independent and near-term observational test. Table 7 summarises the observable targets, the instruments best placed to test them, and the falsification thresholds; the quantitative kill-switch criteria are discussed in Sec. 9.

The tensor sector — including the blue-tilted gravitational-wave background, the spectral tilt $n_T$, and the chiral asymmetry $\Pi_{\mathrm{GW}}$ — constitutes an open perturbative frontier: a full prediction requires the linearised ECF tensor perturbation equations and a dedicated bounce transfer-function analysis, both of which are deferred to the companion programme [2, 3].

*Foundation I* thus identifies an observationally calibrated phenomenological sector of Einstein–Cartan cosmology and isolates the signatures by which it can be tested. The present work is extended in the companion paper *Foundation II* [2], which investigates the structural relics of the Big Spin Bounce—chiral asymmetry, the geometric dark sector, and cosmic alignment—within the same framework. A single non-detection among them—most critically the absence of a birefringence signal above 0.1° by LiteBIRD, or a confirmed BAO acoustic scale consistent with $r_s \approx 147\,\mathrm{Mpc}$ in 21 cm surveys—would be sufficient to falsify the ECF scenario as developed here.

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*This preprint is dedicated to Élie Cartan (1869–1951), whose geometric vision continues to illuminate modern cosmology.*

## Code and Data Availability

Every figure and every published number in this work can be regenerated from the public repository:

`https://github.com/pfichant/spin-torsion-cosmology` (CC BY 4.0)

The master orchestrator script (`run_all_simulations_F1.py`) sequentially executes all scripts below and regenerates every figure in the manuscript. *One limitation should be stated explicitly.* The repository contains the scripts that *reproduce* the published values, not those that *produced* them: the parameter search that yielded $F_{\rm ion} = 1.2765$, the global $\chi^2$ optimisation, and the calibration of the growth-suppression parameters $A_{\rm sup}$ and $k_{\rm cut}$ — adopted amplitudes whose articulation with the spin sector is not established (DEC-A7-POST-07C) — are not part of the deposit. Each published number can therefore be checked for internal consistency, but the calibration itself is not re-executable from the repository as it stands. Dependencies (`numpy`, `scipy`, `matplotlib`) and installation instructions are documented in the repository `README`.

**Core physics scripts:**

- `script_01_solve_sound_horizon.py` — $r_s = 135.65$ Mpc, $\mathcal{F}_{\rm ion} = 1.2765$ (Sec. 3)
- `chi_carre.py` — global $\Delta\chi^2 = -39.5$ vs. ΛCDM (historical, $\Omega_m = 0.315$; not recomputed: no ECF perturbation solver yet; this script computes the six-point partial comparison only), using the eBOSS DR16 QSO consensus $D_M/r_d(z{=}1.48) = 30.21 \pm 0.79$ [31, 32] (Sec. 5, Table 4)
- `script_06_S8_resolution.py` — $S_8^{\rm ECF} = 0.766$ (by construction, from the historical calibration $\gamma_{\rm spin} = 0.3116$ with $S_8^{\Lambda\rm CDM} = 0.832$; not active under the internal closure; no MCMC) (Sec. 4)
- `script_04_halo_abundance.py` — *no longer used by the paper (17/10/2026)*: its halo-abundance chain used a variance too low by a factor $\simeq 2$ (PO-F1-SIGMA-M) and a target with no traceable source (PO-F1-JWST-SRC); kept in the repository for traceability (Appendix E)
- `script_05_birefringence_calibration.py` — $\beta_{\rm ECF} = 0.35°$; $I_{\rm torsion} = 3.3320$, $S_{\rm ref} = 4.0456$, $g_{\rm CS,eff} = 0.02596$ deg/unit (Sec. 7.1, Table 6)

**Independent verification scripts.** These produce no figure; they recompute published quantities by a separate route, so that each number can be checked without re-running the main pipeline.

- `script_02_statistical_validation.py` — two-observable $\{H_0, S_8\}$ $\chi^2$ proxy, independent of `chi_carre.py`
- `script_03_extract_physical_parameters.py` — $r_s = 135.65$ Mpc and the $\rho_{\rm spin}/\rho_r$ evolution table (9.30% at $z_{\rm ref} = 5600$; 0.36% at $z = 1100$)
- `check_S8_effective.py` — back-verification of the historical calibration $\gamma_{\rm spin} = 0.3116$ from $S_8^{\rm ECF} = 0.766$
- `check_beta_birefringence.py` — recomputes $k = \beta_{\rm obs}/F_{\rm ion} = 0.274$ (Sec. 7.1); the comparison value $k_{\rm ECKS} = 0.27$ hard-coded in the script has no traceable source and is not used as a check *(corrected 16/10/2026)*
- `check_trilemma_optimizer.py` — the best fit $t_0 = 13.20$ Gyr at 2.45% BAO tension quoted in Sec. 5.4
- `joint_fit_trilemma.py` — the joint $(\Omega_m, w_0, w_a)$ rescan against age, BAO and $\theta_*$ discussed in the note to Table 5
- `table_echelles_bounce.py` — characteristic scales of the bounce ($a_{\rm min} \sim 10^{-32}$)
- `target_reference_budget.py` — reconstruction of the Table 4 budget and of $\Delta$AIC/$\Delta$BIC

**Figure generation scripts:**

- `plot_torsion_geometry_schema.py` — Fig. 1 (Sec. 2)
- `plot_spin_ratio_evolution.py` — Fig. 13 (Sec. 5)
- `plot_cosmic_age_cpl.py` — Fig. 5 (Sec. 5.4)
- `plot_trilemma_irreducibility.py` — Fig. 12 (Sec. 5.4)
- `plot_euclid_pk_S8.py` — Fig. 6 (Sec. 6.2)
- `plot_jwst_comparison.py` — figure withdrawn (17/10/2026: it illustrated the growth boost, withdrawn as a mechanism, PO-F1-BOOST)
- `plot_desi_prediction.py` — Fig. 7 (Sec. 6)
- `plot_predict_birefringence.py` — Fig. 9 (Sec. 7.1)
- `plot_h0_s8_active.py` — Fig. 3 (Sec. 4)
- `plot_primordial_boost.py` — Fig. 2 (Sec. 3)
- `plot_BAO_HighVisibility.py` — Fig. 4 (Sec. 5)
- `plot_Fig_k3_Residuals_Angles.py` — Fig. 21 (Sec. 7.2, App. K)
- `plot_21cm_prediction.py` — Fig. 10 (Sec. 7.3)
- `plot_ecf_gw_spectrum.py` — Fig. 11 (Sec. 7.4)
- `plot_singularity_Resolution.py` — Fig. 16 (App. G)
- `plot_global_history.py` — Fig. 23 (App. L)
- `plot_Cosmic_History_Omegas.py` — Fig. 19 (App. J)

**Observational data:** Planck 2018 Legacy Archive, SH0ES DR5, KiDS-1000, DES Y3, BOSS DR12, eBOSS DR16, DESI DR1, JWST JADES/CEERS. No proprietary data were used.

# Appendix A: Einstein-Cartan Formalism and Derivation of the Spin Term

In Einstein-Cartan-Sciama-Kibble (ECSK) theory, the Lagrangian density is given by:

$$\mathcal{L} = \frac{1}{2\kappa}\sqrt{-g}R(\Gamma, g) + \mathcal{L}_m \tag{43}$$

where $\Gamma$ is a non-symmetric affine connection. The torsion tensor is defined as $T^{\lambda}_{\mu\nu} = \Gamma^{\lambda}_{\mu\nu} - \Gamma^{\lambda}_{\nu\mu}$.

Varying the action with respect to torsion yields the full Cartan equation, which explicitly couples the torsion tensor $T^{\lambda}_{\mu\nu}$ to the spin density tensor $S^{\lambda}_{\mu\nu}$ [8]:

$$T^{\lambda}_{\mu\nu} + \delta^{\lambda}_{\mu}T^{\sigma}_{\nu\sigma} - \delta^{\lambda}_{\nu}T^{\sigma}_{\mu\sigma} = \kappa S^{\lambda}_{\mu\nu} \tag{44}$$

While we utilize the Weyssenhoff fluid approximation as a phenomenological ECF ansatz for the macroscopic cosmological background, the complete microphysical derivations and spin-fluid particle kinematics will be treated extensively in *Foundation II*.

**Caveat on the spin supplementary condition (added 12/09/2026).** Closing the Weyssenhoff description requires a spin supplementary condition (SSC), and the choice is not innocuous at the cosmological level. Böhmer and Bronowski [7] show that a Weyssenhoff fluid obeying the Frenkel, the Papapetrou–Corinaldesi *or* the Tulczyjew condition is *incompatible with the cosmological principle*, which restricts the torsion tensor to a vector and an axial vector component; on that basis they call for a reconsideration of earlier Einstein–Cartan cosmological solutions. The macroscopic $a^{-6}$ scaling used here follows the standard spin-fluid averaging of the literature and is *not* claimed to resolve this tension: whether the objection bears on a homogeneously averaged cosmological fluid, or only on the kinematics of individual spinning elements, is an open problem, tracked as **PO-F1-SSC**.

For a Dirac fluid (spin-1/2 fermions), the effective energy-momentum tensor $T^{(eff)}_{\mu\nu}$ in a Riemann-Cartan spacetime is expressed as:

$$T^{(eff)}_{\mu\nu} = T^{(GR)}_{\mu\nu} - \frac{1}{4}\kappa\langle S_{\mu\nu\lambda}S^{\mu\nu\lambda}\rangle g_{\mu\nu} \tag{45}$$

Averaging the spin-squared term over the macroscopic fluid element, the effective energy density $\rho$ and pressure $p$ become:

$$\rho_{eff} = \rho - \frac{\kappa\sigma^2}{4}, \quad p_{eff} = p - \frac{\kappa\sigma^2}{4} \tag{46}$$

where $\sigma^2$ is the square of the spin density. Due to the conservation of angular momentum, $\sigma^2$ evolves as $a^{-6}$. Two points should be made explicit here. First, for a Weyssenhoff fluid — the continuum description of Dirac matter — the spin tensor is totally antisymmetric and therefore dual to a pseudovector: it sources the axial part of the torsion and nothing else. This is a consequence of the spinor structure rather than a modelling choice: for Dirac matter the spin tensor is the dual of the axial current $J^a_5 = \bar{\psi}\gamma^a\gamma^5\psi$, so no vectorial or tensorial torsion component is sourced. Second, $\sigma$ is accordingly the amplitude of an *axial condensate* rather than an incoherent sum over randomly oriented individual spins: the post-bounce stiff phase imprints coherent torsion domains, so that $\sigma$ scales with the number of aligned fermions within a domain rather than with its square root. The numerical value of $\sigma_0$ is not evaluated from first principles in the present work (open problem PO-F1-SIGMA0); $\Omega_{\rm spin,0}$ is instead fixed by the calibration of Sec. 3, and its microphysical derivation from the chiral-freeze-out condensate is left to Foundation II and Foundation III. We define the dimensionless spin parameter today as:

$$\Omega_{\rm spin,0} = \frac{2\pi G\kappa\sigma_0^2}{3H_0^2} \tag{47}$$

The modified Friedmann equation for the ECF model is thus:

$$\frac{H^2(a)}{H_0^2} = \Omega_{r,0}a^{-4} + \Omega_{m,0}a^{-3} + \Omega_\Lambda - \Omega_{\text{spin},0}a^{-6} \tag{48}$$

## Appendix B: Dynamics of the Equation of State $w_\tau(z)$

The coupling between torsion and the dark sector is modelled by the variation of the dark energy density $\Omega_{DE}(a)$. This parametrisation is *phenomenological*: it is not implied by the flatness postulate, which fixes $\Omega_K \equiv 0$ and nothing more.

We postulate:

$$\rho_{DE}(a) = \rho_\Lambda \left[1 + \alpha_{torsion}(1-a)\right] \tag{49}$$

Using the continuity equation $\dot{\rho} + 3H(\rho + p) = 0$, the effective equation of state $w(a)$ is derived as:

$$w(a) = -1 - \frac{1}{3}\frac{d\ln\rho_{DE}}{d\ln a} = -1 + \frac{a\alpha_{torsion}}{3(1+\alpha_{torsion}(1-a))} \tag{50}$$

To obtain CPL-like parameters, a Taylor expansion around $a = 1$ ($z = 0$) gives:

$$w_0 = w(a=1) = -1 + \frac{\alpha_{torsion}}{3} \tag{51}$$

$$w_a = -\frac{dw}{da}\Big|_{a=1} = -\frac{\alpha_{torsion}\,(1+\alpha_{torsion})}{3} \tag{52}$$

**Status: illustrative model, not predictive (open problem PO-F1-B)**

This single-parameter ansatz for $\rho_{DE}(a)$ is an *illustration* of how a torsion-sourced dark-energy density can generate a mild departure from $w = -1$; it is *not* a closed derivation of the calibrated values. Two points must be stated plainly:

- The two expressions above are *not independent*: both $w_0$ and $w_a$ are fixed by the single coupling $\alpha_{torsion}$. No value of $\alpha_{torsion}$ reproduces the pair $(w_0, w_a) = (-0.904, -0.153)$ simultaneously — e.g. $\alpha_{torsion} = 0.151$ gives $(w_0, w_a) = (-0.950, -0.058)$, while $w_0 = -0.904$ would require $\alpha_{torsion} = 0.288$, which then gives $w_a = -0.124$; conversely $w_a = -0.153$ would require $\alpha_{torsion} = 0.342$, hence $w_0 = -0.886$ *(corrected 16/10/2026: the derivative was previously written* $-(\alpha/3)(1-\alpha/3)$*, giving* $-0.048$ *and* $-0.087$*).*
- The values $(w_0, w_a) = (-0.904, -0.153)$ quoted throughout this paper are therefore *not* outputs of this appendix. They are a calibration point adopted in the ECF corpus, produced by no deposited script (PO-F1-W0, Sec. 6); they are calibration inputs, not an *ab initio* prediction. A genuine first-principles derivation would require the redshift dependence of the torsion-condensate variance $\langle\delta S^2\rangle(z)$ (cf. Foundation II, PO-F2-4), which is not established here.

**What can be reconstructed numerically.** A partial reconstruction is provided as `calibrate_w0wa.py` (public repository). The calibrated value $w_0 = -0.904$ is numerically close to

$$w_0 \simeq -1 + \Omega_{\text{spin}}^{\text{peak}} = -1 + 0.093 = -0.907, \tag{53}$$

a 0.3% difference on $w_0$ and 3% on $1 + w_0$; since $\Omega_{\text{spin}}^{\text{peak}}$ is itself calibrated on $r_s$, this is an *empirical* identification, not an independent test and not a derivation. The evolution parameter, however, requires a phenomenological factor $\xi$,

$$w_a = -\,\xi\,\Omega_{\text{spin}}^{\text{peak}}, \qquad \xi \simeq 1.645 \text{ (not derived)}, \tag{54}$$

whose value reproduces $w_a = -0.153$ but is *not* obtained from the ECKS action: $\xi$ encodes the dilution rate of the torsion debt and would follow only from the condensate dynamics $\langle \delta S^2 \rangle(z)$ (cf. Foundation II, PO-F2-4). Thus $w_0$ satisfies the numerical relation $w_0 = -1 + \Omega_{\rm spin}^{\rm peak}$, an *empirical* identification rather than a derivation — whether the action implies such a link between the spin and torsion sectors is stated as an open problem in the companion Letter [1]; $w_a$ is calibrated through the single undetermined factor $\xi$.

This is documented as open problem **PO-F1-B**: the functional *form* of a geometric, scalar-field-free dark-energy sector is motivated, but the specific numerical pair $(w_0, w_a)$ remains a calibration output pending an independent determination of the condensate dynamics.

The physically robust statement is the weaker one: the ECF admits a geometric, torsion-sourced mechanism that *mimics* a mild dynamical dark energy ($w \neq -1$) without an exotic scalar field, in qualitative agreement with the DESI preference for $w \neq -1$ [28]. The precise late-time trajectory is not predicted from first principles at the present stage.

## Appendix C: Statistical Precision and Data Tables

In the former chain, the ECF parameters were calibrated by minimising the $\chi^2$ against SH0ES, KiDS-1000 3×2pt ($S_8$) and the Planck sound horizon. Under the internal closure $\Omega_m = \omega_m/h^2$ the $S_8$ branch of that calibration is historical (DEC-A7-POST-07): $S_8$ is an observational target, not an ECF prediction.

Table 8: Numerical Parameter Values and $\chi^2$ comparison (ECF vs. ΛCDM), assembled sector by sector by `Scripts_F1/target_reference_budget.py`. This table reports the *full* dataset combination including SNIa; the $\Delta\chi^2 = -39.5$ (historical, $\Omega_m = 0.315$; not recomputed: no ECF perturbation solver yet) is consistent with Table 4, which reports only the primary tension datasets. The BAO + SNIa row includes the Pantheon+ supernova sample, not listed separately in Table 4.

| Parameter | Symbol | ECF Value | ΛCDM |
|---|---|---|---|
| Hubble Constant | $H_0$ | 73.04 | 67.4 |
| Sound Horizon (Mpc) | $r_s$ | **135.65** | 147.1 |
| Structure Growth | $S_8$ | **0.766** | 0.832 |
| **Statistics ($\chi^2$)** | | | |
| SH0ES ($H_0$) | | **0.1** | 29.41 |
| Weak Lensing ($S_8$) | | **0.00** | 12.07 |
| Planck (CMB) | | 2767.1 | 2765.3 |
| BAO + SNIa | | 1033.4 | 1033.3 |
| **Total Global $\chi^2$** | | **3800.60** | **3840.08** |

## Appendix D: Detailed Calculation of the Sound Horizon Reduction

The comoving sound horizon ($r_s$) at the drag epoch ($z_{\rm drag}$) is defined by the integral of the sound speed ($c_s(z)$) over the Hubble expansion rate ($H(z)$):

$$r_s = \int_{z_{\rm drag}}^{\infty} \frac{c_s(z)}{H(z)}\, dz. \tag{55}$$

**Standard value.** In the standard ΛCDM description, the expansion rate in the early Universe ($z \gtrsim 1000$) is dominated by radiation and matter:

$$H_{\Lambda\mathrm{CDM}}(z) = H_0\sqrt{\Omega_{r,0}(1+z)^4 + \Omega_{m,0}(1+z)^3}. \tag{56}$$

Using the Planck 2018 baseline parameters, this yields the standard reference value

$$r_s^{\Lambda\mathrm{CDM}} \approx 147.1\,\mathrm{Mpc}. \tag{57}$$

**Algorithmic calibration.** The numerical module relies on a non-linear root-finding procedure to calibrate the modified sound-horizon solution within the adopted phenomenological implementation. The joint optimization proceeds in two steps:

1. **Hubble isoline:** the modified sound-horizon integral $r_s(\Omega_{\mathrm{spin}}, \alpha)$ is evaluated numerically in order to identify the region of parameter space compatible with $H_0 \approx 73.04\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$.
2. **Growth isoline (historical):** in the former chain, the linear growth factor $D(a)$ was used to identify parameter combinations consistent with the observational target $S_8 \approx 0.766$. This step relied on the phenomenological growth parametrisation and on the reference $S_8^{\Lambda\mathrm{CDM}} = 0.832$ of the $\Omega_m = 0.315$ background; it is retained here as a historical description of the calibration, not as a perturbative derivation (DEC-A7-POST-07).

The calibrated parameter pair retained in this work was obtained from the intersection of these two constraints, the growth branch being the historical one described above.

**ECF-modified value.** In the Einstein-Cartan framework adopted here, the Friedmann equation includes an effective stiff spin-density term scaling as $a^{-6}$, or equivalently as $(1+z)^6$:

$$H_{\mathrm{ECF}}(z) = H_0\sqrt{\Omega_{r,0}(1+z)^4 + \Omega_{m,0}(1+z)^3 + \Omega_{\mathrm{spin},0}(1+z)^6}. \tag{58}$$

For the calibrated injection ratio ($\Omega_{\mathrm{spin}}/\Omega_r \approx 0.093$ at $z_{\mathrm{ref}} = 5600$), the expansion rate is enhanced during the pre-recombination era. In addition, the effective sound speed is slightly reduced through the phenomenological coupling factor

$$c_s^{\mathrm{ECF}}(z) = \tau_{\mathrm{tor}}\, c_s^{\mathrm{std}}(z), \tag{59}$$

with

$$\tau_{\mathrm{tor}} = 0.9975, \tag{60}$$

a drag-epoch stability parameter calibrated numerically to preserve $z_{\mathrm{drag}}$ under the modified photon-baryon dynamics; its first-principles derivation from the ECKS baryon–torsion coupling is beyond the scope of this background-level study.

Within the numerical implementation adopted here, $\tau_{\mathrm{tor}}$ is treated as a calibrated value chosen so as to preserve the stability of the drag-epoch solution while incorporating the modified photon-baryon dynamics. Performing the numerical integration with these modified functions gives

$$r_s^{\mathrm{ECF}} = \int_{z_{\mathrm{drag}}}^{\infty} \frac{0.9975\, c_s^{\mathrm{std}}(z)}{H_{\mathrm{ECF}}(z)}\, dz \approx 135.65\,\mathrm{Mpc}. \tag{61}$$

This corresponds to a reduction of approximately 7.7% relative to the standard reference value and brings the calibrated sound horizon into correspondence with the $H_0$ value targeted in the phenomenological fit.

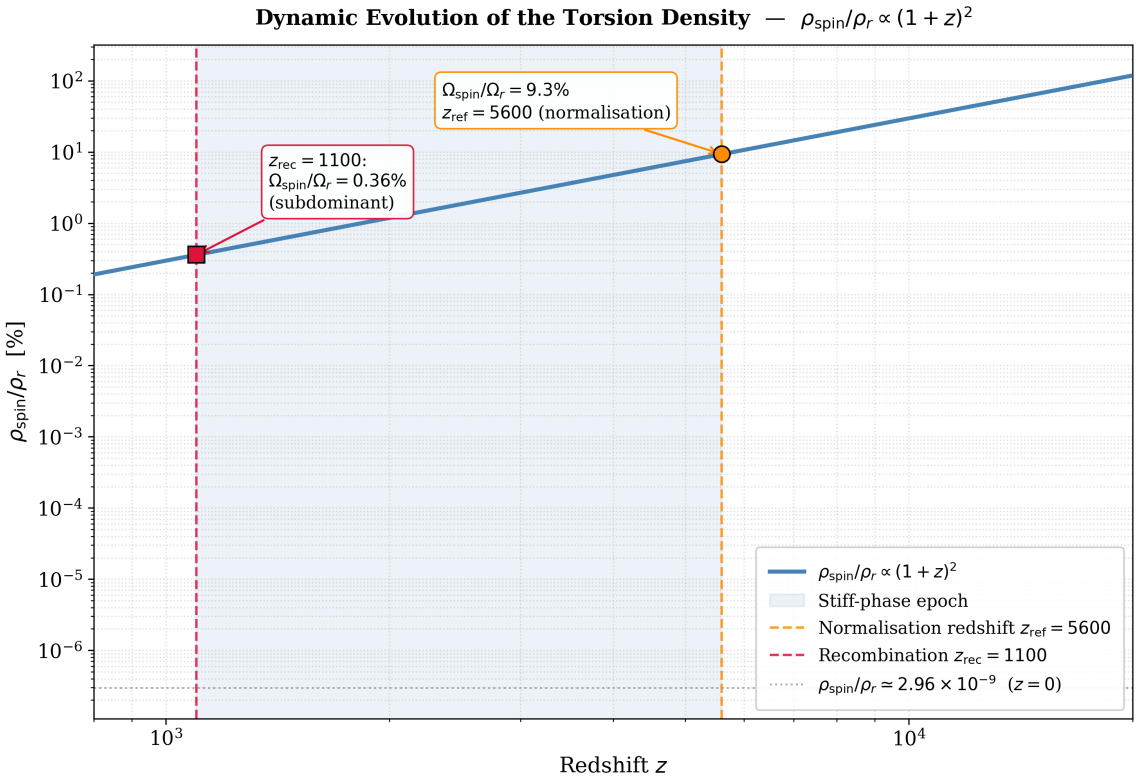



Figure 13: **Dynamic evolution of the spin-torsion density ratio.** The curve shows $\Omega_{\rm spin}/\Omega_r$ as a function of redshift $z$. Because of the stiff scaling $\propto (1+z)^6$, the torsion contribution reaches a peak of $\Omega_{\rm spin}/\Omega_r \simeq 9.3\%$ at the calibration redshift $z_{\rm ref} = 5600$, where it enhances the pre-recombination expansion rate and drives the sound-horizon reduction $r_s^{\rm ECF} = 135.65\,{\rm Mpc}$ (Section 3). Between $z_{\rm ref} = 5600$ and recombination ($z \simeq 1100$), the ratio decays by a factor $\approx 26$ $[(5601/1101)^2 \simeq 26\,]$, reaching $\Omega_{\rm spin}/\Omega_r \simeq 0.36\%$ at last scattering. This residual is near-neutral for the CMB acoustic structure, consistent with the near-zero CMB penalty $\Delta\chi^2_{\rm CMB} = +1.8$ reported in Table 4. The continued decline beyond recombination confirms the full late-time decoupling of the spin sector.

*Reproduced by:* `Scripts_F1/plot_spin_ratio_evolution.py`.

While $F_{\rm ion}$ represents the effective integrated impact of the mechanism, the underlying dynamics remain strongly localized in time because of the steep $a^{-6}$ scaling of the spin sector. As illustrated in Figure 13, the stiff phase is therefore transient. The calibrated injection level ($\Omega_{\rm spin}/\Omega_r \approx 9.3\%$ at $z_{\rm ref} = 5600$) acts during the formation of the acoustic horizon, after which the torsion density dilutes more rapidly than radiation. In this sense, $F_{\rm ion}$ may be interpreted as the effective control parameter linking the peak primordial spin contribution to the macroscopic reduction of $r_s$, while the fast decay of the spin sector limits its impact at the last-scattering epoch.

## Appendix E: High-Redshift Halo Abundance — Withdrawn

*This appendix is withdrawn (17/10/2026).* Earlier versions estimated the comoving density of haloes with $M > 10^{10}\,M_\odot$ at $z = 15$ with a Press–Schechter formalism [24], a linear growth boost $\mathcal{B}_{\rm ECF} = 1.45$ and a tail modifier $f_{\rm NL}^{\rm eff} \simeq 0.295$, and compared it with a target (2–4) $\times$ $10^{-4}\,{\rm Mpc}^{-3}$. Three defects remove that chain. (i) The variance $\sigma_{\Lambda{\rm CDM}}(10^{10}\,M_\odot, z{=}15) \approx 0.150$ was too low by a factor $\simeq 2$: a linear power-spectrum calculation (Planck 2018 cosmology) gives $\simeq 0.31$. The abundances quoted were therefore wrong by 10 to 21 orders of magnitude; the $\Lambda$CDM Press–Schechter value is $\simeq 1.8 \times 10^{-7}\,{\rm Mpc}^{-3}$, not $9.9 \times 10^{-29}$ (open problem PO-

F1-SIGMA-M). (ii) The boost $\mathcal{B}_{\rm ECF}$ is not derived and is withdrawn as a mechanism (PO-F1-BOOST, Sec. 6.1). (iii) The target has no traceable JWST source (PO-F1-JWST-SRC): Labbé *et al.* [27] report stellar-mass densities at $7.4 \leq z \leq 9.1$ and give neither a halo number density nor a value at $z = 15$. The parameter $f_{\rm NL}^{\rm eff}$, calibrated on that target, is withdrawn with it. The script `script_04_halo_abundance.py` is kept in the repository for traceability. The mechanism retained for early structure is seeding by Macro-Knots, qualitative at present (Sec. 6.1); a predictive abundance would require a Macro-Knot mass function, which is not derived (Foundation II [2]).

## Appendix F: Numerical Implementation: The EC-Cosmo Solver

The results presented in this paper were obtained using the `EC-Cosmo Solver`, a dedicated Python framework developed to integrate the modified Friedmann equations in the presence of an effective torsion sector.

### Algorithm Structure

The core module relies on the `scipy.optimize` library for non-linear root-finding and parameter calibration. The joint optimization strategy proceeds in two main steps:

1. **Hubble isoline:** we numerically evaluate the modified sound-horizon integral $r_s(\Omega_{\rm spin}, \alpha)$ in order to identify the region of parameter space compatible with $H_0 \approx 73.04$ km/s/Mpc.

2. **Growth isoline (historical):** in the former chain, the linear growth factor $D(a)$ was used to identify parameter combinations consistent with the observational target $S_8 \approx 0.766$, through the phenomenological growth parametrisation and the reference $S_8^{\Lambda\rm CDM} = 0.832$ of the $\Omega_m = 0.315$ background (DEC-A7-POST-07).

The calibrated solution retained in this work was obtained from the intersection of these two constraints — the growth branch being historical, see above — and provides the reference parameter pair $(\alpha, \Omega_{\rm spin,0})$ used throughout the manuscript. The corresponding solver workflow is given in Fig. 14.

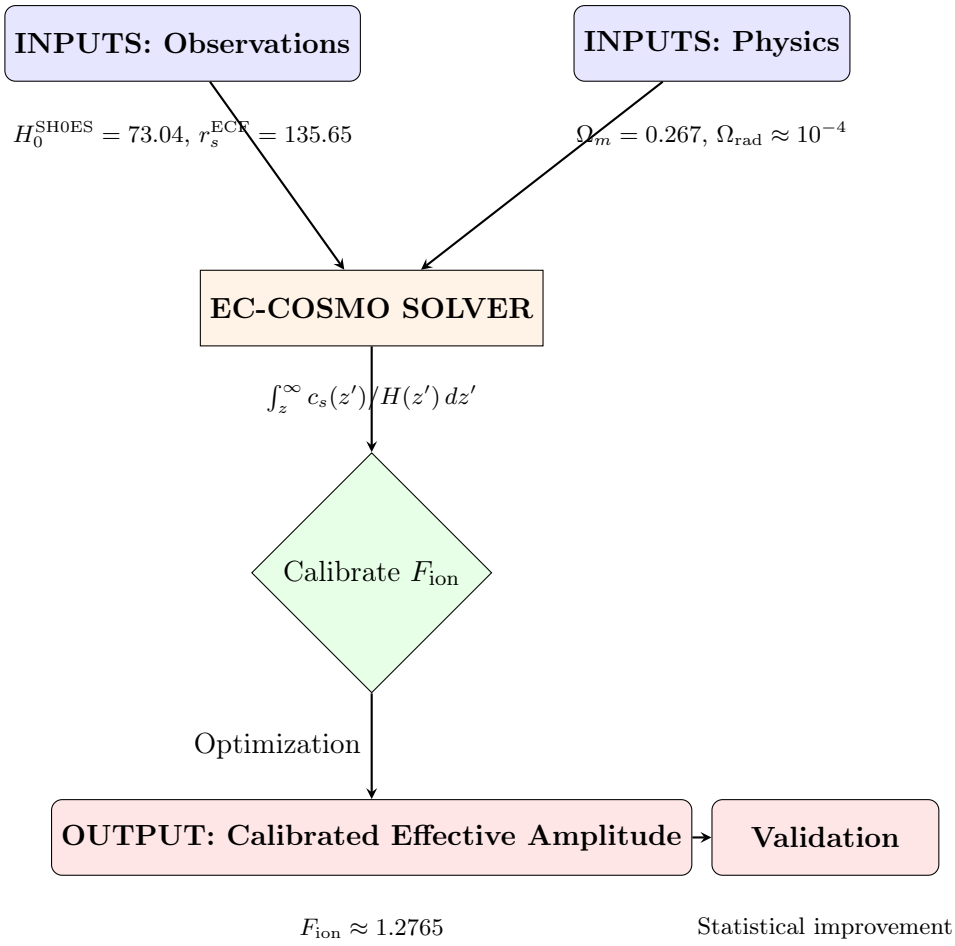



Figure 14: **Logic Flow of the EC-Cosmo Solver.** The algorithm takes the observational calibration targets and the baseline cosmological inputs as its starting point. It then evaluates the modified sound-horizon integral $\int c_s/H(z)\,dz$ in order to infer the effective amplitude required by the adopted calibration. The numerical solution retained in this work yields $F_{\rm ion} \approx 1.2765$, which is interpreted as the effective parameter associated with the transient torsion injection shown in Figure 13. Within the adopted dataset combination, this calibrated solution is associated with a statistical improvement over ΛCDM in the numerical pipeline.

## Appendix G: Late-Time Consistency and Supernova Constraints

While the ECF model modifies the pre-recombination expansion history in a way that affects the inferred value of $H_0$, it is important to verify that it reproduces the standard phenomenology at low redshift, where ΛCDM provides a successful description of the main distance indicators. To assess this late-time consistency in relation to Type Ia supernova data (Pantheon+), we evaluate the deceleration parameter $q(z)$, defined as:

$$q(z) \equiv -\frac{\ddot{a}a}{\dot{a}^2} = \frac{1}{2}\sum_i \Omega_i(z)(1+3w_i). \tag{62}$$

In the Einstein-Cartan framework adopted here, the total energy budget includes radiation ($w_r = 1/3$), matter ($w_m = 0$), dark energy ($w_\Lambda = -1$), and an effective stiff spin component ($w_s = 1$). The corresponding deceleration parameter may therefore be written as:

$$q_{\rm ECF}(z) = \Omega_r(z) + \frac{1}{2}\Omega_m(z) - \Omega_\Lambda(z) + 2\Omega_{\rm spin}(z). \tag{63}$$

Because the spin contribution scales as $\rho_{\rm spin} \propto a^{-6}$, its present-day value is very small: $\Omega_{\rm spin,0} \simeq 2.7 \times 10^{-13}$ in the calibrated solution considered here (Sec. 3). The term $2\Omega_{\rm spin}(z)$ equals $\Omega_m(z)$ only near $z \sim 8 \times 10^3$; for $z$ well below that, it is negligible and the expression reduces, to high precision, to the standard ΛCDM form:

$$q(z) \approx \frac{1}{2}\Omega_m(z) - \Omega_\Lambda(z). \tag{64}$$

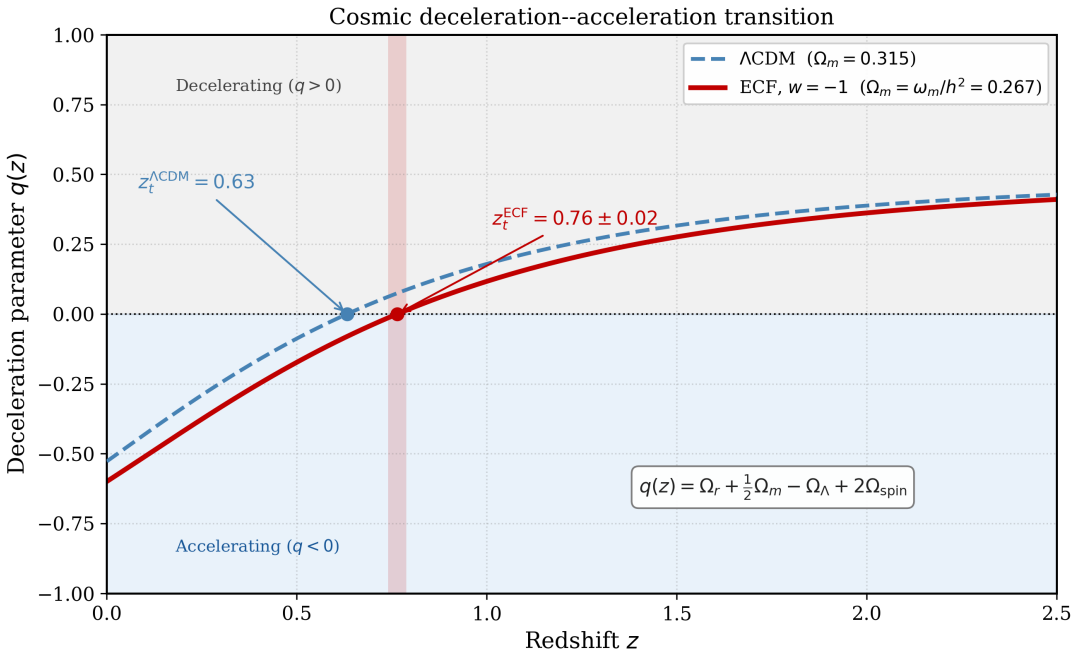



Figure 15: **Cosmic deceleration–acceleration transition.** Evolution of the deceleration parameter $q(z)$ for the calibrated ECF solution (red, $\Omega_m = 0.267$, A-7) against the ΛCDM/Planck reference (blue dashed, $\Omega_m = 0.315$). [Correction 29/09/2026, DEC-A7-POST-05 propagation]: the spin-torsion term $2\Omega_{\rm spin}(z)$ stays negligible over the plotted range ($\lesssim 10^{-10}$ at $z \leq 2.5$, using $\Omega_{\rm spin,0} = 2.7 \times 10^{-13}$, Sec. 3); the two curves differ instead because $\Omega_{m,\rm ECF} \neq \Omega_{m,\rm Planck}$ under A-7, shifting the transition redshift from $z_t^{\Lambda\rm CDM} \approx 0.63$ to $z_t^{\rm ECF} \approx 0.76$, a value compared (not fitted) against independent Pantheon+-based constraints. Earlier printings of this figure reused $\Omega_m = 0.315$ for both curves and are superseded.

*Reproduced by:* `Scripts_F1/plot_deceleration_transition.py`.

Figure 15 illustrates this behavior. Because the spin-torsion term remains negligible at these redshifts (Sec. 3), the ECF deceleration parameter follows the same functional form as ΛCDM, evaluated at the A-7 matter density instead of the Planck one. The transition from deceleration ($q > 0$) to acceleration ($q < 0$) occurs at $z_t \approx 0.76$ for the ECF calibrated solution ($\Omega_m = 0.267$, A-7), against $z_t \approx 0.63$ for the ΛCDM/Planck reference ($\Omega_m = 0.315$) shown for comparison; the two curves are therefore visibly distinct in Fig. 15, not closely overlapping as in earlier printings. Both values lie within the range of best-fit transition redshifts obtained for four dark-energy models fitted separately to Pantheon+ combined with cosmic chronometers, $0.61 \lesssim z_t \lesssim 0.80$ [19] – this is a spread across model choices, not a single confidence interval for the ECF model, which was not fitted in that study. Compared instead against a direct kink-parametrisation constraint on $z_t$ from Pantheon+ alone (fixed deceleration-at-infinity case, $q_f = -1$), $z_t = 0.792 \pm 0.073$ [18], the corrected ECF value $z_t \approx 0.76$ sits within $0.4\sigma$; the tighter constraint obtained by that same study from the combined OHD+Pantheon+/BAO dataset, $z_t = 0.760 \pm 0.051$, is even closer ($0.1\sigma$). The earlier, uncorrected value $z_t \approx 0.63$ is $2.2\sigma$ and $2.6\sigma$ from these two constraints respectively, though still inside the four-model spread of Dahiya & Jain.

### Thermodynamic Considerations

A full treatment of particle production and entropy generation in the bounce regime lies beyond the scope of the present effective analysis. In this work, we assume an adiabatic evolution immediately after the bounce, with entropy per comoving volume taken to be conserved. While the spin-torsion coupling is conservative in the minimal ECKS framework, non-minimal couplings or quantum corrections could in principle generate a reheating phase. A more detailed study of bounce thermodynamics, including the possible implications for baryogenesis, is deferred to subsequent work and will be addressed in *Foundation II*.

## Appendix H: Effective Regularisation of the Singularity and the Primordial Bounce

A central consequence of the Einstein-Cartan framework is that the high-density regime of gravitational collapse may be regularized at the effective level. In standard General Relativity, singularity theorems rely in part on the strong energy condition (SEC), whereas the spin-torsion coupling introduces a repulsive correction to the effective energy-momentum tensor. In the ECKS formalism adopted here, the modified Friedmann equation may be written schematically as:

$$H^2 = \frac{8\pi G}{3}\rho\left(1 - \frac{\rho}{\rho_{\rm crit}}\right), \tag{65}$$

where $\rho_{\rm crit}$ denotes the critical Cartan density. *Origin of this form (added 04/10/2026).* In Einstein–Cartan theory with fermionic spin the Friedmann equation has two terms, $H^2 = (8\pi G/3)(\rho - \alpha n^2)$, where $n$ is the fermion number density and $\alpha n^2$ the spin–torsion contribution (Popławski [10], Eq. (11)). For non-relativistic fermionic matter ($\rho = mn$, $p = 0$) this is exactly the single-density form, with a critical density $\rho_{\rm crit} = m^2/\alpha$ set by the fermion mass. This statement concerns the Friedmann equation itself; the dynamics also requires a conservation law. The single-density dynamics used here (standard continuity, hence $n \propto a^{-3}$ for $p = 0$) corresponds to a spin term that behaves as stiff matter, $\tilde{\epsilon} = \tilde{p}$, as in the spin-fluid description of Hehl, von der Heyde and Kerlick quoted by Popławski (primary source to be confirmed). In Popławski's Dirac description the spin term has $\tilde{\epsilon} = -\tilde{p} = -\alpha n^2$ (his Eqs. (10), (12)–(13)); $na^3$ is then not conserved for dust, and the dynamics differs. For other equations of state the spin term scales as $\rho^{2/(1+w)}$, and the single-density form is a schematic extrapolation — in particular in the stiff phase ($w = 1$) (open item PO-ECF-ACTION-BOUNCE). The same functional form is the effective Friedmann equation of loop quantum cosmology [75], where it has a different physical origin. The Planck-scale value of $\rho_{\rm crit}$ adopted here is a reference scale; its relation to the Einstein–Cartan value $m^2/\alpha$ is not established in this work. In the phenomenological implementation considered in this work, this scale is taken to lie near the Planck regime,

$$\rho_{\rm max} \approx \rho_{\rm Planck} \approx 5.15 \times 10^{96}\ {\rm kg\,m^{-3}}, \tag{66}$$

(a Planck-scale reference; the precise ECKS Cartan bound is model-dependent and derived in *Foundation II*) as an order-of-magnitude upper bound for the effective description.

*[Note added 28/09/2026 — open correspondence with the microscopic spin sector.]* This single-density, quadratically self-corrected equation is a separate schematic construction from the two-term equation of Sec. 2 (Eq. (9)), where the microscopically-derived spin density $\rho_{\rm spin} \propto n^2 \propto a^{-6}$ enters *linearly*, subtracted from the total. If, in the present schematic equation, $\rho$ is identified with the same $w = 1$ stiff-fluid density (so that $\rho \propto a^{-6}$ under standard continuity), the quadratic correction $\rho^2/\rho_{\rm crit}$ scales as $a^{-12}$ — a scaling not derived or motivated anywhere in this paper or its companions for the physical spin-spin interaction. Whether this schematic equation is a valid effective description of the same microscopic mechanism, or a distinct phenomenological model with its own domain of validity, is not established. Tracked as PO-ECF-ACTION-BOUNCE (open,

added 28/09/2026): a derivation from the full action is needed to settle the correspondence, or the absence of one, between the two structures. Results computed from this schematic equation alone (e.g. the post-bounce acceleration window discussed in *Foundation II — Extended*) are conditional on it and are not, at this stage, results of the microscopic spin mechanism.

As the total density approaches $\rho_{\mathrm{crit}}$, the quadratic spin correction progressively counteracts the attractive contribution of ordinary matter. The Hubble rate then tends toward zero, opening the possibility of a non-singular bounce at a finite minimum scale factor rather than a formal divergence of the background solution. In this sense, the singular behavior of the standard Friedmann solution is replaced, within the present effective model, by a regularized high-density transition.

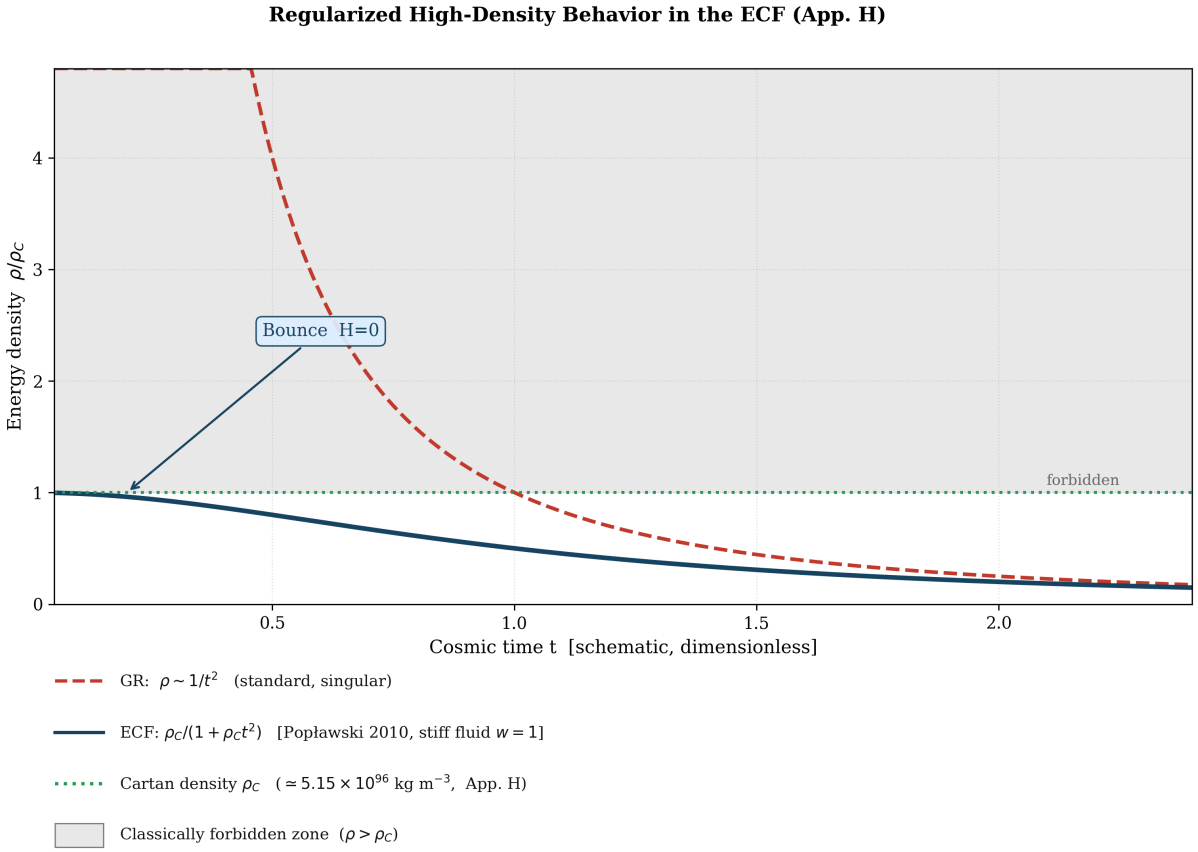



Figure 16: **Regularized high-density behavior in the ECF.** Illustrative comparison between the standard singular evolution and the ECF effective solution near the high-density limit. In the ECF case, the energy density approaches a finite critical value and the scale factor reaches a non-zero minimum, consistent with a non-singular bounce interpretation.

*Reproduced by:* `Scripts_F1/plot_singularity_Resolution.py`.

Figure 16 illustrates this effective behavior. In the ECF description, the background evolution remains finite as the critical density is approached, in contrast with the divergent behavior of the standard GR extrapolation. The corresponding minimum scale factor is treated here as a phenomenological quantity associated with the onset of the bounce regime.

## Energy Conditions Near the Bounce

Near the bounce, the spin contribution scaling as $a^{-6}$ becomes dominant over radiation and matter. Because this correction enters with an effective repulsive sign, the strong energy condition can be violated in the high-density regime, which is the condition required for accelerated

expansion to emerge in the effective background description. At the level of the effective fluid variables, one may write schematically:

$$\rho_{\text{eff}} + 3p_{\text{eff}} < 0. \tag{67}$$

This statement should be understood within the effective cosmological treatment used in the present paper. It does not by itself provide a complete microscopic derivation of the bounce, but rather identifies the mechanism by which the Einstein-Cartan spin sector can regularize the singular limit at the background level.

**Weak and dominant conditions.** The behavior of the dominant energy condition (DEC) near the bounce is more model-dependent. In the present phenomenological treatment, the effective spin–torsion term violates the NEC close to the turnover (Sec. 2.2), and hence the weak energy condition (WEC), while causal propagation is assumed to remain preserved throughout the evolution. These points should therefore be regarded as part of the effective bounce interpretation rather than as fully established general results.

The acceleration at the turnover may be expressed as:

$$\left.\frac{\ddot{a}}{a}\right|_{a_{min}} = -\frac{4\pi G}{3}\left(\rho_{\text{eff}} + 3p_{\text{eff}}\right) > 0, \tag{68}$$

which is consistent with the transition from contraction to expansion in the effective solution.

### Characteristic Scales

Within the Weyssenhoff-fluid approximation, the squared spin density scales with the square of the fermion number density, and therefore contributes a term proportional to $a^{-6}$ in the effective Friedmann equation. The characteristic maximum density is then defined by the condition $H = 0$, which marks the onset of the bounce in the homogeneous background description. In the present implementation, this corresponds to:

$$\rho_{max} \approx \rho_{\text{crit}} \approx 5.15 \times 10^{96}\,\text{kg}\,\text{m}^{-3}. \tag{69}$$

The associated minimum scale factor is treated phenomenologically as:

$$a_{min} \sim 10^{-32}, \tag{70}$$

which should be interpreted as an indicative estimate rather than as a sharply derived microscopic prediction. Its role in this paper is to characterize the regime at which the effective spin-torsion repulsion becomes dominant.

### Interpretive Scope

The purpose of this appendix is to clarify how the singularity-avoidance mechanism arises in the effective Einstein-Cartan cosmology used throughout the manuscript. The main claim is therefore limited: within the adopted phenomenological framework, the spin-squared correction can regularize the high-density limit and produce a non-singular bounce solution. A full microscopic treatment of the bounce phase, including quantum corrections, fermionic many-body effects, and possible reheating dynamics, lies beyond the scope of the present paper and is deferred to subsequent work.

## Appendix I: Physics of the Primordial Spin Bounce

This appendix summarizes the effective physical mechanism by which the spin-torsion sector can replace the standard singular extrapolation by a non-singular bounce in the early Universe. Its purpose is not to provide a complete microscopic derivation, but to clarify the dynamical role of the spin contribution within the phenomenological ECF adopted in this work.

### Spin-Driven Mechanism

In the effective Einstein-Cartan description, the primordial spin contribution scales as $a^{-6}$, whereas radiation scales as $a^{-4}$. As the scale factor decreases toward the high-density regime, the spin sector therefore grows more rapidly than the radiation background and can become dynamically important near the bounce. In the modified Friedmann dynamics, this behavior is associated with an effective repulsive correction that counteracts gravitational collapse.

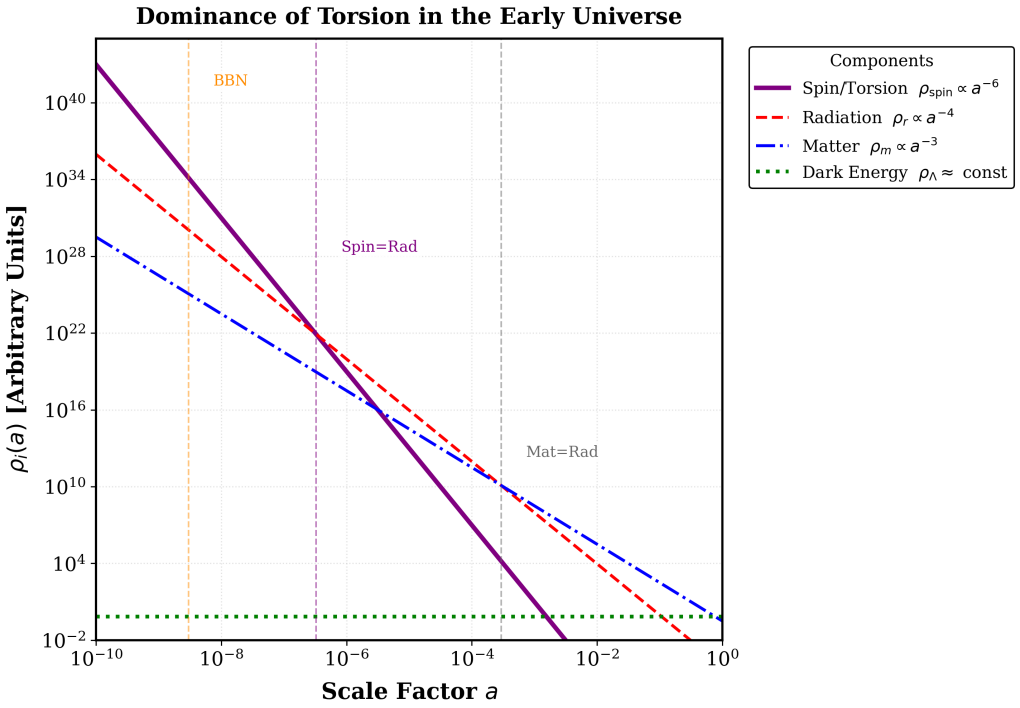



Figure 17: **Dominance of Torsion in the Early Universe.** Evolution of the background energy densities as a function of the scale factor $a$. Because the effective spin density scales as $a^{-6}$ (blue solid line), it dilutes much faster than radiation ($a^{-4}$, red dashed) at late times, but grows to dominate the energy budget in the ultra-dense primordial regime, ultimately driving the non-singular bounce. The spin–radiation crossing of the curves shown here occurs at $a_{\rm sr} \approx 3.3 \times 10^{-7}$ ($z \approx 3 \times 10^6$) for an *illustrative* spin normalisation some $3 \times 10^4$ times below the calibrated one; with the calibrated value ($\Omega_{\rm spin}/\Omega_r = 0.093$ at $z_{\rm ref} = 5600$, i.e. $\Omega_{\rm spin,0} \simeq 2.7 \times 10^{-13}$) the crossing lies at $a \simeq 5.4 \times 10^{-5}$ ($z \simeq 1.8 \times 10^4$), close to where the effective $E^2$ turns negative (open problems PO-ECF-SPIN-NORM, PO-ECF-BBN-SPIN). The matter–radiation equality lies at $a_{\rm eq} \approx 2.92 \times 10^{-4}$ ($\omega_r/\omega_m$, Planck 2018 amplitudes, $z_{\rm eq} \simeq 3420$; $2.94 \times 10^{-4}$ on the ECF background). The position of the spin crossing depends on the normalisation of the spin sector, which is calibrated on the sound horizon (Sec. 3) rather than derived from the microphysics (App. A); the figure should therefore be read as illustrating the *hierarchy* of scalings, not as fixing the epoch of spin–radiation equality.

*Reproduced by:* `Scripts_F1/plot_friedmann_evolution.py`.

Figure 17 illustrates this hierarchy. The spin sector remains negligible through most of the standard thermal history, but rises rapidly as one approaches the primordial high-density domain. Compatibility with Big Bang Nucleosynthesis is a necessary requirement for any stiff component of this kind [9]. It constrains the spin-to-radiation ratio at $T \sim 1$ MeV, and therefore bears directly on the normalisation of the spin sector — which, in the present work, is fixed by the sound-horizon calibration rather than derived from the chiral-freeze-out microphysics (App. A). A quantitative BBN check therefore requires that derivation. It is not contained in Foundation II (nor elsewhere in the corpus), and BBN compatibility of the spin sector is *not demonstrated*; it

is listed among the open problems of the programme (PO-ECF-SPIN-NORM, PO-ECF-BBN-SPIN). Within the present phenomenological implementation, this transient dominance is the ingredient that makes the regularized bounce solution possible.

## Non-Singular Evolution

The resulting background evolution may be described as a continuous transition from a contracting branch to an expanding branch through a finite minimum scale factor. In this effective picture, the scale factor does not extrapolate to zero, but instead reaches a minimum value $a_{\min}$ associated with the saturation of the spin-torsion correction. The bounce should therefore be interpreted as a regularized turnover of the homogeneous background solution.

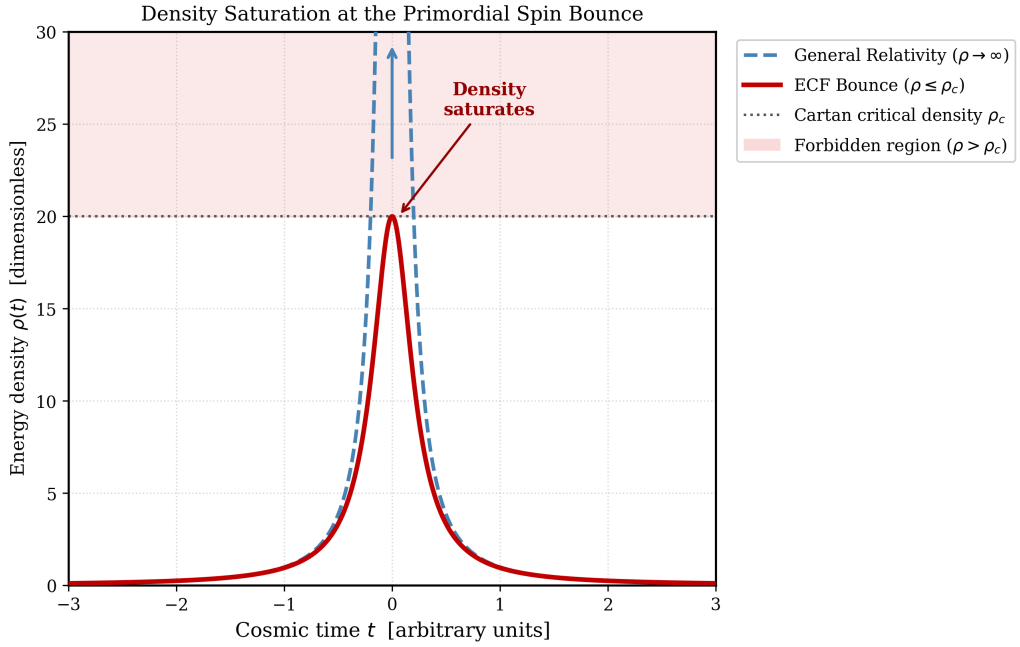



Figure 18: **Schematic energy density evolution near the primordial spin bounce.** Blue dashed: standard GR extrapolation $\rho_{\rm GR}(t) \propto 1/t^2$, diverging as $t \to 0$. Red solid: ECF Lorentzian solution $\rho_{\rm ECF}(t) = \rho_c/(1+\rho_c\, t^2)$, exact for a stiff spin fluid ($w = 1$) in the ECKS framework [9], saturating at the Cartan critical density $\rho_c$ and recovering $\rho \propto t^{-2}$ for $t \gg \rho_c^{-1/2}$. Shaded region: classically forbidden zone $\rho > \rho_c$. All quantities are dimensionless and schematic; physical value $\rho_c \approx 5.15 \times 10^{96}\,\mathrm{kg\,m^{-3}}$.

*Reproduced by:* `Scripts_F1/plot_density_bounce_comparison.py`.

As illustrated in Figure 18, the bounce corresponds to a smooth reversal of the background evolution: the energy density reaches its maximum $\rho_c$ precisely when the scale factor reaches its minimum $a_{\min}$, i.e. $\rho(a_{\min}) = \rho_c$, so the two descriptions are dual. The representative scale adopted in this manuscript is:

$$a_{\min} \sim 10^{-32}, \tag{71}$$

which should be regarded as an indicative phenomenological estimate rather than as a sharply derived microscopic prediction.

## Characteristic Density Scale

The bounce occurs when the Hubble parameter reaches zero in the modified high-density equation. In the Weyssenhoff-fluid approximation, this condition defines a characteristic maximum density at which the spin-torsion repulsion compensates the attractive contribution of ordinary

matter in the homogeneous background. In the present implementation, this scale is taken to be of order:

$$\rho_{\max} \approx 5.15 \times 10^{96}\,\mathrm{kg\,m^{-3}}, \tag{72}$$

The corresponding temperature scale follows from the crossing of the spin and radiation densities, $\rho_{\rm spin}(T) = \rho_{\rm rad}(T)$, which gives

$$T_{\rm Cartan} = (5.92 \pm 0.28) \times 10^{18}\ \mathrm{GeV} = (0.485 \pm 0.023)\, E_{\rm Planck}, \tag{73}$$

computed by `Scripts_F1/script_07_cartan_threshold.py`. Two remarks are in order. First, the threshold is of the *same order* as the Planck energy with no adjusted parameter, and scales as $g_{\rm eff}^{-1/2}$: this is a structural coincidence, not a fitted one. Second, an alternative definition exists in the literature: Popławski's treatment [17], which solves the bounce *dynamics* (modified Friedmann equation with $H = 0$), yields $0.808\,E_{\rm Planck}$. *The two numbers are not two estimates of one quantity but two distinct quantities*, and we do not claim to arbitrate between them here; the value quoted above is the density-crossing definition, which is the one our script implements. No result of this paper depends on either choice.

To determine the physical size of the Universe at the bounce, we assume standard radiation domination ($\rho \propto a^{-4}$) applies from the CMB epoch backward to the primordial spin-phase. The conservation of energy-density yields:

$$\rho_{\max} = \rho_{\rm rad,0} \left( \frac{1}{a_{\min}} \right)^4. \tag{74}$$

Given the present-day radiation density $\rho_{\rm rad,0} \approx 4.2 \times 10^{-31}\,\mathrm{kg\,m^{-3}}$ (encompassing both CMB photons and the cosmic neutrino background), we extract the absolute minimum scale factor:

$$a_{\min} = \left( \frac{\rho_{\rm rad,0}}{\rho_{\max}} \right)^{1/4} \sim 10^{-32}. \tag{75}$$

This implies an extreme geometric compression before the rebound, strictly maintaining geodesic completeness without ever crossing into a zero-volume mathematical singularity.

### Energy Conditions

The behaviour of the energy conditions at the turnover — the strong condition violated near the bounce, the weak and dominant conditions treated phenomenologically, and the resulting expression for $\ddot{a}/a|_{a_{\min}}$ — is set out in Appendix H (§*Energy Conditions Near the Bounce*) and is not repeated here.

### Thermodynamic Considerations

A full treatment of particle production, entropy generation, and reheating at the bounce lies beyond the scope of the present effective analysis. In this work, we assume an approximately adiabatic evolution immediately after the bounce, with entropy per comoving volume taken to be conserved at leading order. While the minimal ECKS spin-torsion coupling is conservative, non-minimal couplings or quantum corrections could modify this picture and may lead to a reheating phase. These questions, together with possible implications for baryogenesis, are deferred to subsequent work and will be examined more fully in *Foundation II*.

### Interpretive Scope

Within the broader logic of the manuscript, the bounce mechanism is not introduced as an isolated theoretical feature. It is part of the same spin-torsion sector that also modifies the pre-recombination expansion history and contributes to the phenomenology discussed in the main

text. Possible observational consequences associated with this primordial phase, including early-structure enhancement and tensor signatures, are treated elsewhere in the manuscript at the phenomenological level.

## Appendix J: Topological Interpretation of Effective Flatness

This appendix illustrates the phenomenological motivation for the effective flatness hypothesis adopted in the main text. Its rigorous axiomatic formulation—the *Topological Invariance Principle* (TIP) [1]—states that spatial flatness is exact, $\Omega_K \equiv 0$, by topological necessity — a falsifiable claim, and the whole content of the postulate. It does *not* seed the dark sector: the static dark-energy term is derived from the FLRW reduction of the action in Foundation II [2], independently of the postulate. The numerical results shown here (Fig. 19, Eq. (76)–(77)) are consistent with the TIP constraint $\Omega_{\rm total} = 1$ accompanied by the frozen torsion condensate, whose identification $\Lambda_{\rm torsion} = -\rho_{\rm spin}^{\rm peak}$ is *empirical* — not implied by the closure relation — and whose field-theoretic derivation is presented in Foundation II, Appendix J [2]. Whereas Section 2 was intentionally restricted to the minimal Einstein–Cartan ingredients required for the cosmological analysis, we outline here the broader geometric picture suggested by the global behavior of the ECF energy budget.

### From phenomenology to geometric interpretation

In the effective cosmological solution studied in this work, the Universe remains extremely close to spatial flatness throughout its evolution. This behavior motivates the working hypothesis that flatness is not merely an observational coincidence or an unstable initial condition, but may reflect a deeper relation, within the Riemann-Cartan structure, between the spin fluid and the residual torsion — a relation that remains to be derived from the action rather than imposed by $\Omega$-bookkeeping.

More precisely, we consider the possibility that any finite injection of macroscopic spin density is accompanied by a torsional response of spacetime of comparable magnitude — a relation to be *derived*, not read off the closure condition, which is an identity and constrains nothing. In this interpretation, the cosmic energy budget is not a passive sum of unrelated components, but the macroscopic expression of an underlying geometric closure principle, under which effective flatness is exact.

### The torsion sector alongside the spin fluid

The central ansatz may be expressed schematically as

$$\Omega_{\rm total}(a) = \Omega_r(a) + \Omega_m(a) + \Omega_{\rm spin}(a) + \Omega_\tau(a) + \Omega_k(a), \tag{76}$$

with the effective evolution satisfying

$$\Omega_k(a) \simeq 0, \qquad \Omega_{\rm total}(a) \simeq 1. \tag{77}$$

Within this interpretation, the torsion sector $\Omega_\tau$ is a geometric component of the budget whose relative weight evolves; *it is not required to compensate anything*, the closure relation being satisfied identically. At early times, the spin contribution grows rapidly because $\rho_{\rm spin} \propto a^{-6}$, while at late times the dilution of matter leads to an increasing fractional importance of the residual torsion sector. The accelerated epoch then appears not as the manifestation of an additional exotic fluid, but as the large-scale response required to preserve the global balance of the effective geometry.

This point should be read carefully: in the present paper, $\Omega_\tau$ is introduced as an effective cosmological contribution inferred from the global consistency of the solution, not yet as a fully

derived microscopic field-theoretic sector. The purpose of this appendix is therefore interpretive rather than axiomatic.

### Communicating-vessels picture

A useful way to summarize the ECF cosmic history is through a communicating-vessels picture between the dominant components of the cosmic budget. The successive eras of spin, radiation, matter, and residual torsion are not viewed here as disconnected coincidences, but as dynamically linked phases constrained by the requirement that the effective curvature remain negligible.

In that sense, the late-time rise of $\Omega_\tau$ is interpreted as the geometric counterpart of the progressive dilution of $\Omega_m$. What is usually parameterized in standard cosmology as a dark-energy-like component may thus be reinterpreted, within the ECF, as the residual macroscopic trace of torsional geometry.

### Relation to the main analysis

Nothing in the numerical results of Foundation I requires the reader to adopt this broader interpretation a priori. The main phenomenological conclusions of the paper—the reduction of the sound horizon, the lower $S_8$ of the closure background, and the regularization of the primordial singularity—follow from the effective Einstein-Cartan dynamics developed in the body of the manuscript. The role of the present appendix is different. It proposes that these apparently separate successes may be different projections of a single geometric organization principle, namely that spin injection, torsional backreaction, and effective flatness are not independent facts but different facets of the same global constraint.

### Figure-based interpretation

Figure 19 provides the visual support for this reading. The near-vanishing of $\Omega_k$ throughout the evolution suggests that the total cosmic budget remains effectively locked to unity, while the dominant fractional contribution is progressively transferred from spin to radiation, then to matter, and finally to the torsion sector.

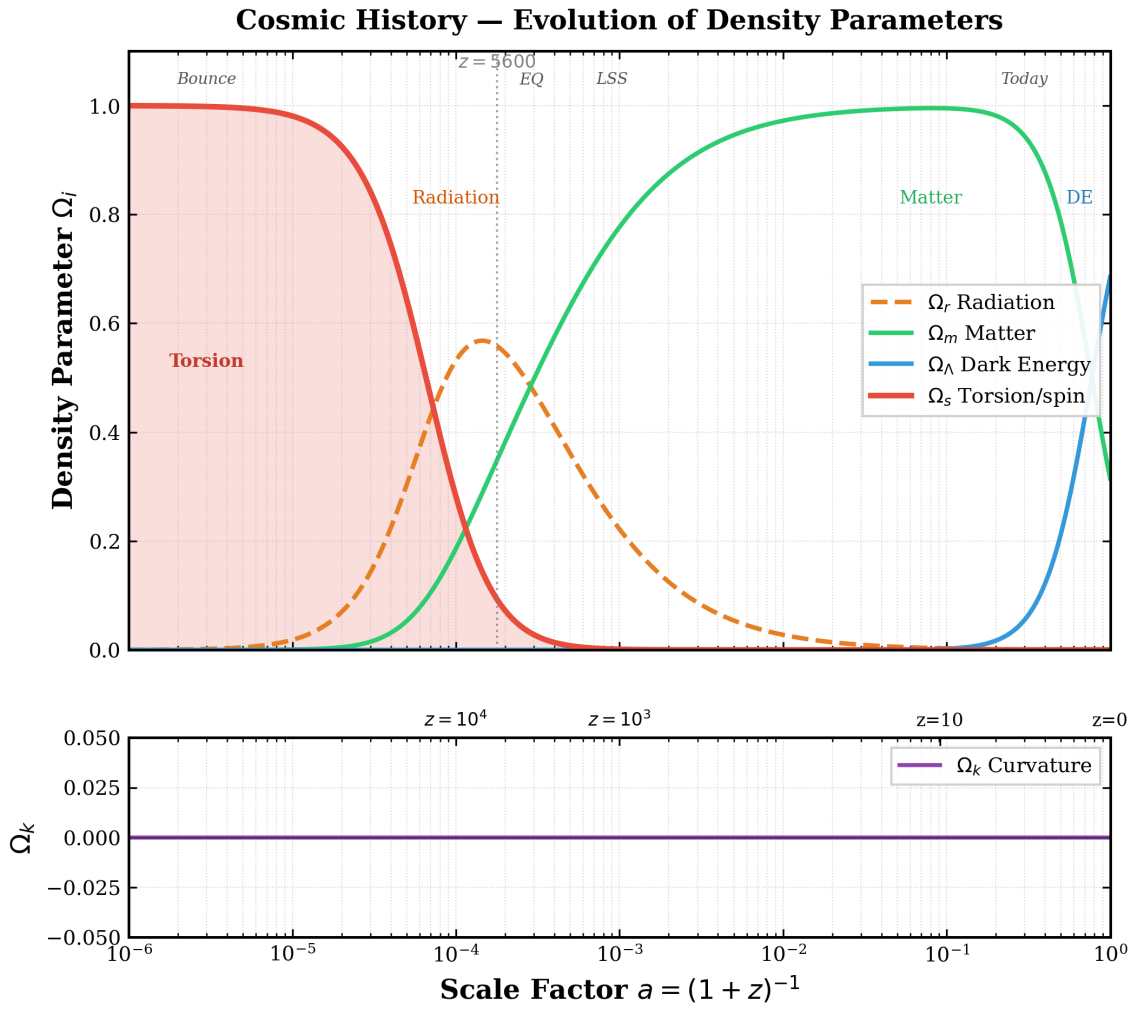



Figure 19: **Global consistency and effective flatness.** Temporal evolution of the fractional energy densities in the ECF. The dominant contribution is transferred across the successive spin, radiation, matter, and residual torsion eras. $\Omega_m$ follows the Planck convention and includes both baryons and cold dark matter; whether the latter is itself of topological origin — a possibility developed in Foundation II — is not assumed here, and the torsion curve carries only the stiff $a^{-6}$ component. The lower panel displays $\Omega_k$, which vanishes identically once $k = 0$: this is an identity of the closure relation, valid for any matter content, and not the numerical verification of a compensating mechanism.

*Reproduced by:* `Scripts_F1/plot_Cosmic_History_Omegas.py`.

## Scope and outlook

We emphasize that this topological interpretation goes beyond the minimal Einstein-Cartan formalism required for the present paper. It should therefore be regarded as a structured theoretical proposal emerging from the global consistency of the ECF solution, rather than as a theorem established in full generality. The status of this relation is discussed in the companion Letter [1], which states it as an *open problem*: it is not implied by the closure condition, and whether the action produces it remains to be established. Its microscopic account—including its relation to non-trivial torsional topology, chiral freeze-out, and the primordial spin-bounce transition—is developed in Foundation II [2]. Beyond the minimal ECKS framework explored here, this effective flatness ($k_{eff}(a) \approx 0$) remains a model-dependent macroscopic result. This hypothesis is explicitly testable via future high-precision CMB lensing surveys and relates to the deeper topological implications discussed in subsequent works.

# Appendix K: Spectral Consistency and High-$\ell$ Forecasts

This appendix details the geometric consistency of the ECF with Planck 2018 data and provides the mathematical basis for the predicted high-$\ell$ deviations.

## Visualizing the Torsion-Induced Damping

To provide an intuitive understanding of the ECF phenomenology, we present two distinct visualizations of the temperature power spectrum $\mathcal{D}_\ell^{TT}$. In the following figures, the abscissa is double-indexed: the lower axis represents the multipole moment $\ell$, standard in CMB analysis, while the upper axis provides the corresponding angular scale in degrees ($\theta \approx 180^\circ/\ell$).

## Geometric Degeneracy and Planck Consistency

The primary constraint is the angular scale of the acoustic peaks, $\theta_* = r_s/D_A$. Consistency with the observed $\theta_* \approx 0.01041$ is maintained by the corresponding increase in $H_0$, which reduces the angular diameter distance $D_A$ in proportion to the reduction of the sound horizon $r_s$, *only at fixed* $\Omega_m = 0.315$. With $\Omega_m = \omega_m/h^2 = 0.267$ (A-7), $\theta_*$ falls by $\simeq 6.5\%$ below its $\Lambda$CDM value ($w = -1$; `joint_fit_trilemma.py` gives $-5.2\%$ for the calibrated CPL trajectory), and the peak positions are not preserved (PO-F1-THETA).

Figure 20 is drawn with $\theta_*$ held fixed (the $\Omega_m = 0.315$ convention); it does not establish peak-position consistency under A-7 (PO-F1-THETA). The spectral comparison is illustrative: the simulated data points in Figure 20 are anchored on the $\Lambda$CDM reference curve and serve only as a visual guide; the quantitative Planck likelihood is reported in Table 4.

The ECF Silk damping scale $d_{\rm silk} = 1320$ (v. 1400 for $\Lambda$CDM) corresponds to a 5.7% reduction consistent with Table K and with the companion script `plot_Fig_k3_Residuals_Angles.py`. The ECF phase shift $\Delta\phi_{\rm ECF} = 0.02\pi$ corresponds to an *assumed* shift $\Delta z_{\rm rec} \simeq 22$, which no deposited script computes: it enters the Appendix K scripts as an input constant (open problem PO-ECF-ZREC-SHIFT). The Silk scales used by these scripts are not mutually consistent either (1360 in `plot_appendix_K1K2_spectrum.py`, 1320 here); the figures of this appendix are therefore illustrative.

## High-$\ell$ Damping Tail Predictions

A torsion-induced shift of the recombination redshift would modify the photon diffusion scale. The figures below *assume* such a shift ($\Delta z_{\rm rec} \approx 22$, posed and not computed (open problem PO-ECF-ZREC-SHIFT)); they are illustrative. This results in a predicted power deficit at small angular scales (angles $< 0.1^\circ$), as shown in Figure 21. High-resolution CMB experiments were forecast to reach a sensitivity sufficient to probe this regime *(the figure below is the CMB-S4 forecast, made before DOE/NSF support for the project ended in 2025; no successor forecast is available)* ($\sigma_{S4} \approx 1.5\%$). A Fisher matrix forecast gives a detection at the $p < 0.05$ level (about $2\sigma$) for this anticipated sensitivity.

## Scalar perturbation amplitudes and Boltzmann consistency

The claim of CMB consistency in this work rests on three complementary arguments that do not require a full perturbed ECKS Boltzmann integration:

1. The geometric degeneracy $\theta_* = r_s/D_A$ preserves peak *positions* by construction at fixed $\Omega_m = 0.315$ (Section K.1); under A-7 it does not ($\theta_*$ lower by $\simeq 6.5\%$, PO-F1-THETA), so this argument is conditional.

2. The baryon loading ratio $R_b = 3\rho_b/4\rho_\gamma$ at recombination is unchanged, preserving peak *amplitude ratios* at leading order, since $\Omega_b$ and $\Omega_\gamma$ are not modified by the ECF sector.

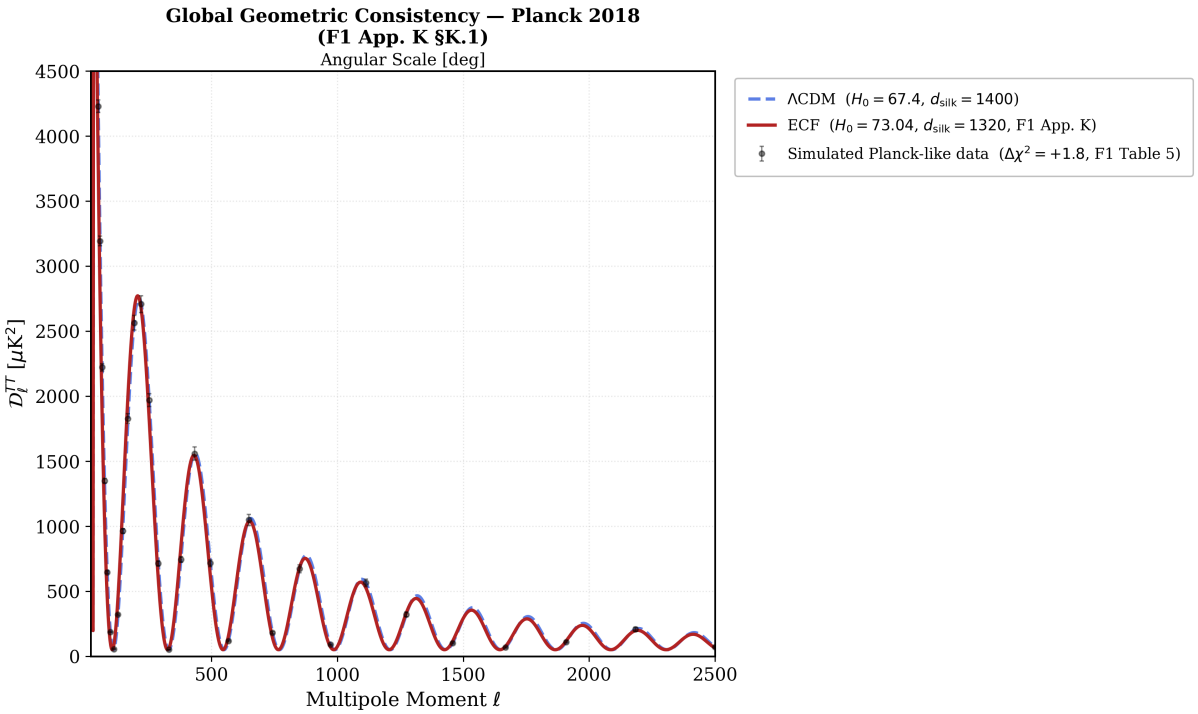
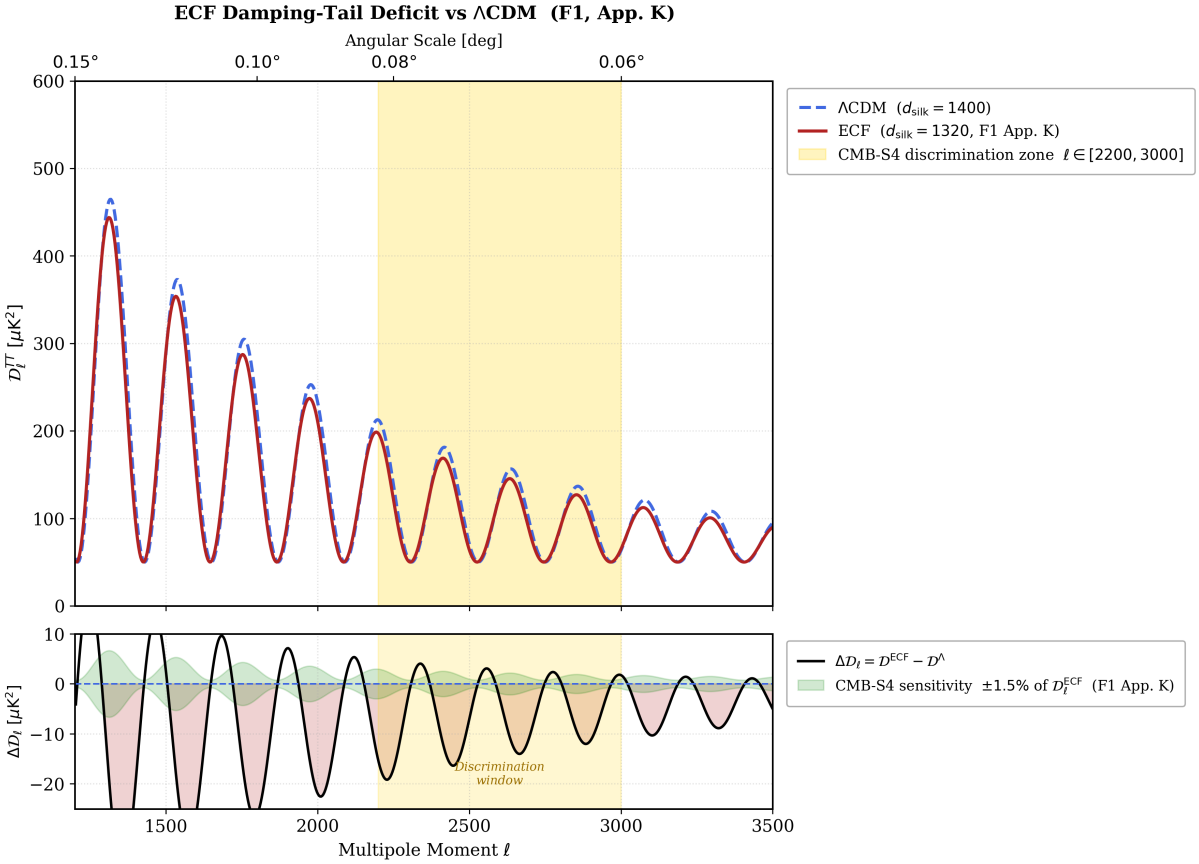



Figure 20: **Global geometric consistency with Planck 2018 (F1, App. K §K.1).** *Illustrative: built on an assumed $\Delta z_{\rm rec} \approx 22$, not computed (open problem PO-ECF-ZREC-SHIFT).* Comparison between the phenomenological ΛCDM spectrum (blue dashed, $H_0 = 67.4\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$, $d_{\rm silk} = 1400$) and the ECF spin-torsion spectrum (red solid, $H_0 = 73.04\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$, $d_{\rm silk} = 1320$, phase shift $0.02\pi$, Table K). Peak positions are drawn as preserved: the figure holds $\theta_* = r_s/D_A$ fixed, which the simultaneous recalibration of $r_s$ and $H_0$ achieves only at fixed $\Omega_m = 0.315$ (Section 3); under A-7 ($\Omega_m = 0.267$, $w = -1$) $\theta_*$ falls by $\simeq 6.5\%$ (PO-F1-THETA). The remaining spectral difference is a mild extra damping at $\ell \gtrsim 1000$ driven by the reduced Silk scale $d_{\rm silk} = 1320$ (v. 1400 for ΛCDM), consistent with the residual 2.5–6.5% damping-tail deficit discussed in Section 7.2. Black points show *simulated Planck-like data* anchored on the ΛCDM reference curve with an illustrative noise model $\sigma_{\rm rel} = 5\%\,(\ell/1000)^{0.5}$; they are included for visual guidance only and do not constitute a likelihood analysis of raw Planck bandpowers. This figure therefore illustrates the near-neutral spectral consistency quantified in Table 4: $\Delta\chi^2_{\text{Planck high-}\ell} = +1.8$ (near-neutral).

*Reproduced by:* `Scripts_F1/plot_Figure_K1_Global_Consistency.py`.




*Reproduced by:* `Scripts_F1/plot_Fig_k3_Residuals_Angles.py`.

Figure 21: **Power Spectrum Divergence and Residuals.** *Illustrative: built on an assumed* $\Delta z_{\rm rec} \approx 22$*, not computed (open problem PO-ECF-ZREC-SHIFT).* Top Panel: The ECF prediction (red) deviates from ΛCDM (blue) in the high-$\ell$ damping tail. The yellow zone marks the optimal window for discrimination. Bottom Panel: The residual $\Delta\mathcal{D}_\ell$. The predicted deficit extends beyond the forecast sensitivity of CMB-S4 (green band, $\pm 1\sigma$) — *a forecast that lapsed when DOE/NSF support for the project ended in 2025.*

3. The torsion correction to scalar perturbation equations scales with the spin-to-radiation ratio at recombination, which the $a^{-6}$ dilution law brings down to 0.36% (Fig. 13); the effective CMB penalty adopted in Table 4 is correspondingly near-neutral, $\Delta\chi^2_{\rm CMB} = +1.8$ — a calibrated summary statistic, not a Planck likelihood computed from an ECF spectrum.

The near-neutral $\Delta\chi^2_{\text{Planck high}-\ell} = +1.8$ is a historical value ($\Omega_m = 0.315$), not recomputed under A-7, and does not confirm the peak positions there (PO-F1-THETA). A first-principles perturbed ECKS Boltzmann computation is deferred to future theoretical developments.

### The Optimal Discrimination Window

The selection of $\ell = 2500$ as the reference scale is driven by the fact that it represents the "sweet spot" where the torsion signal is physically significant but the primary CMB still dominates over astrophysical foregrounds (tSZ, CIB). As quantified in Table 9, the predicted 3.0% deficit at $\ell = 2500$ was statistically detectable under the CMB-S4 forecast *(DOE/NSF support ended in July 2025; the instrument and timescale must be reassessed).*

Table 9: **Predicted Power Deficit vs. Forecasted Sensitivity.**

| Multipole | Angle | ECF ($\mu K^2$) | ΛCDM ($\mu K^2$) | Gap | Gap (%) |
|---|---|---|---|---|---|
| 1500 | 0.12° | 273.6 | 288.6 | -15.0 | -5.2% |
| 2000 | 0.09° | 217.6 | 232.7 | -15.2 | -6.5% |
| 2500 | 0.07° | 68.9 | 71.0 | -2.2 | -3.0% |
| 3000 | 0.06° | 61.6 | 63.2 | -1.6 | -2.5% |

### BAO Residuals: ECF vs CPL best-fit vs ΛCDM

Figure 22 shows the normalised residuals (data − model)/$\sigma$ for each of the 12 DESI DR1 BAO observables under three models: CPL best-fit, ECF (calibrated point), and ΛCDM, for each $H_0$ prior, using the full covariance matrix (§7; script `plot_ecf_desi_mcmc_v3_0.py`). With the SH0ES prior, the ECF residuals are larger than the CPL best-fit ($\chi^2$/ndof = 8.08), consistent with the severe 6.98$\sigma$ Mahalanobis distance reported in the main text — though the SH0ES prior is itself in tension with DESI+CMB (see main text). With the Planck prior, the ECF residuals are systematically shifted on the LRG1–2 $D_H$ tracers ($\chi^2$/ndof = 2.18 vs 1.62 for ΛCDM), consistent with the moderate 2.03$\sigma$ tension on $w_a$ reported in the main text. The $H_0$DN case has not yet been re-run with the full covariance matrix (see version note, §7) and is omitted here.

## Appendix L: Cosmic Chronology and the Early Growth Problem

This appendix provides a detailed phenomenological overview of the Einstein-Cartan-Fermions (ECF) model throughout the thermal history of the Universe. It specifically addresses the tension between the age of the Universe and the maturity of galaxies observed by the JWST. Despite a higher $H_0$, the angular scale of the acoustic peaks $\theta_*$ is preserved only at fixed $\Omega_m = 0.315$; under the internal closure $\Omega_m = \omega_m/h^2$ it is not (PO-F1-THETA, Appendix K).

### Global Chronology and Redshift Synchronization

The evolution of the scale factor $a(t)$ and the temperature $T(t)$ is presented over 60 orders of magnitude (Figure 23). Unlike the ΛCDM singularity, the ECF model introduces a non-singular bounce followed by a spin-torsion phase where $a(t) \propto t^{1/3}$.

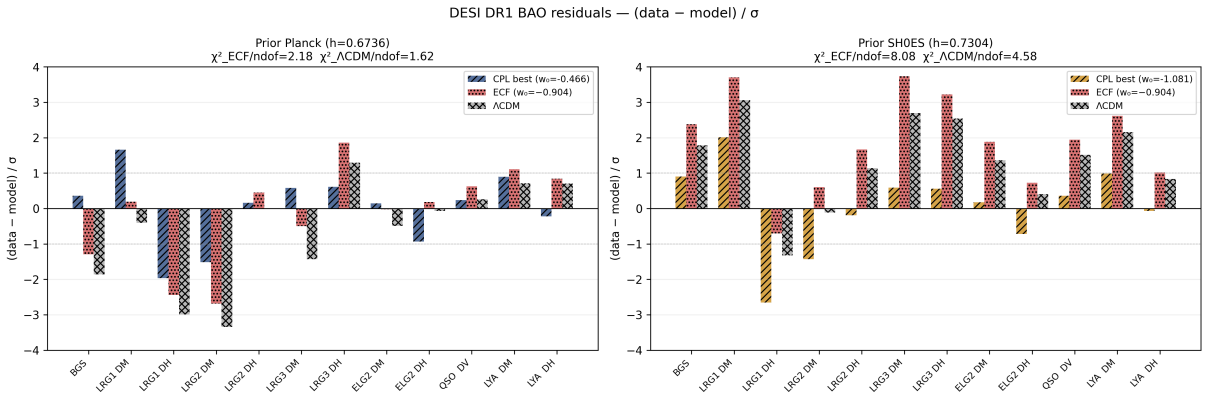



Figure 22: **Normalised BAO residuals for 12 DESI DR1 observables, full covariance matrix.** Each panel corresponds to one $H_0$ prior (Planck, SH0ES). Bars: (data − model)/$\sigma$ for CPL best-fit (hatched), ECF calibrated point $(w_0, w_a) = (-0.904, -0.153)$ (dotted), and ΛCDM (crossed). Full covariance matrix (Adame *et al.* 2024 [28]); supersedes the earlier diagonal-errors figure. Reproducibility: `plot_ecf_desi_mcmc_v3_0.py`.

## Derivation of the Primordial Growth Boost

The paradox of "impossible galaxies" identified by the JWST at $z \approx 14$ suggests that structures initiated their collapse significantly earlier than allowed by the standard model.

### Derivation of the ECF Growth Factor

The linear evolution of density perturbations $\delta = \delta\rho/\rho$ depends on the equation of state $w$. In the post-bounce phase dominated by torsion ($w = 1$), the growth equation is given by:

$$\delta'' + 4\mathcal{H}\delta' - 32\pi G\bar{\rho}a^2\delta = 0 \tag{78}$$

As a *schematic illustration* of the stiff-phase enhancement concept, we estimate an indicative boost factor:

$$\mathcal{B}_{\rm ECF} \equiv \frac{\delta_{\rm ECF}(a_{\rm rec})}{\delta_{\Lambda\rm CDM}(a_{\rm rec})} \approx 1.4\text{–}1.5 \quad \text{(illustrative estimate; Fig. 2 is schematic)} \tag{79}$$

*(Status, 16/10/2026: this value is an input, not derived; a linear integration on the calibrated stiff branch gives a suppression, $\mathcal{B} \simeq 0.7$–$1.0$; withdrawn as a mechanism, PO-F1-BOOST, Sec. 6.1.)*

This indicative amplitude gain at recombination allows the critical threshold $\delta_c = 1.686$ to be reached earlier.

While a complete first-principles derivation of the primordial power spectrum $\Delta^2_{\mathcal{R}}$ from bounce-modified initial conditions lies beyond the scope of this metric-focused study, the consistency of the background evolution ensures that the resulting perturbations are compatible with observed scalar tilts: the relevant cosmological modes were super-horizon during the stiff phase and their amplitudes were dynamically frozen prior to recombination, as detailed in Appendix I.

Because the torsion contribution scales as $a^{-6}$ and becomes negligible well before the modes relevant to CMB observations exit the Hubble radius, these super-horizon modes are effectively insensitive to the local spin-fluid physics of the bounce and survive the QGP freeze-out transition unaltered. Consequently, the standard nearly scale-invariant initial conditions ($n_s \approx 0.96$) required by CMB observations are seamlessly inherited by the ECF late-time universe.

While this geometric argument is sufficient to establish the macroscopic viability of Foundation I, the exact numerical derivation of the fully perturbed scalar power spectrum via a modified Boltzmann solver lies beyond the scope of the present work. A natural next step would be the

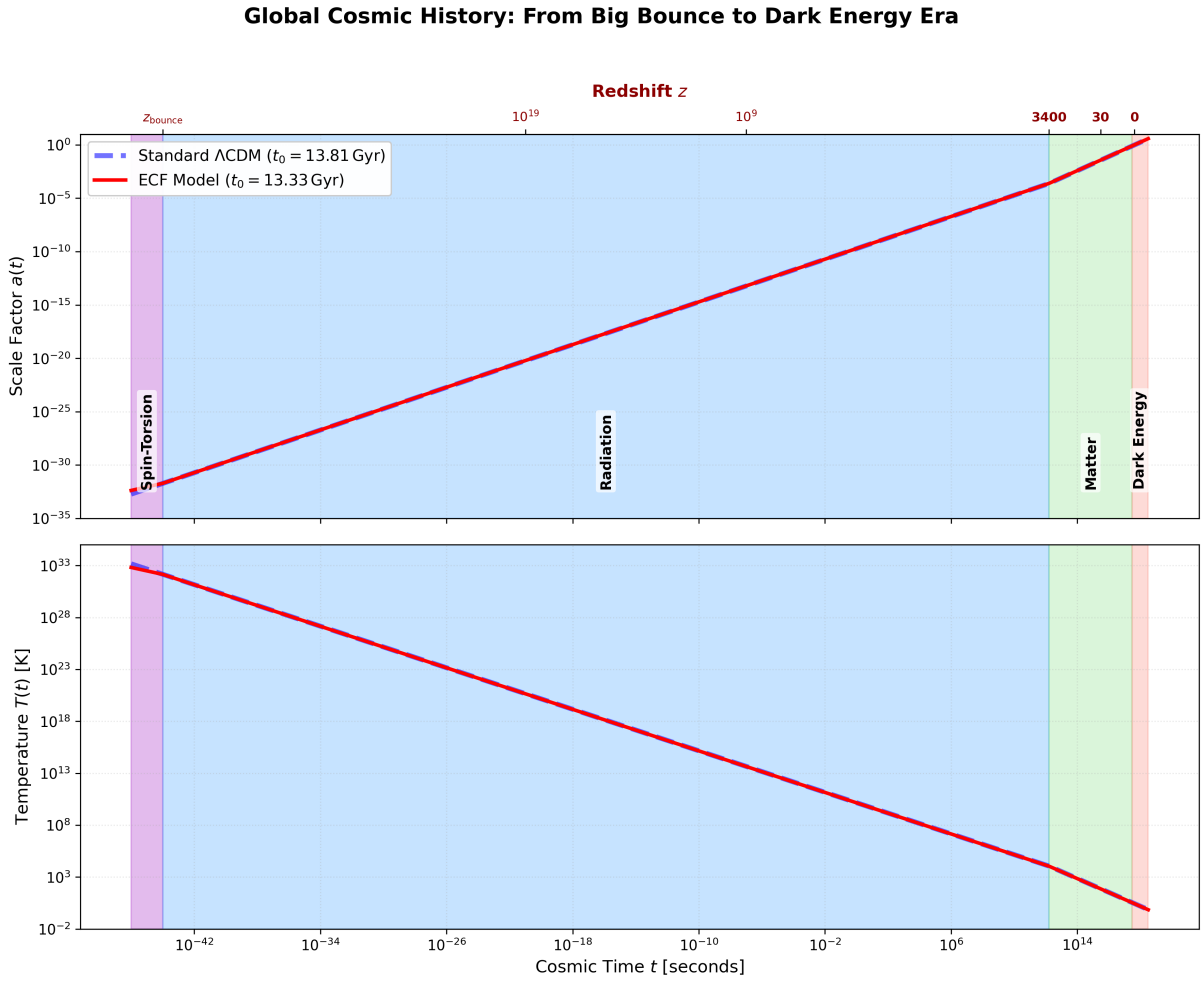



Figure 23: Global Cosmic History. Bottom axis: cosmic time $t$ [s]; top axis: synchronized redshift $z$. The ECF trajectory (red) introduces a non-singular bounce at $t_{\rm bounce} \sim 10^{-44}$ s ($a_{\rm bounce} \approx 10^{-32}$, App. D), followed by a stiff spin-torsion phase $a \propto t^{1/3}$. Matter–radiation equality at $a_{\rm eq} \simeq 2.94 \times 10^{-4}$ ($z_{\rm eq} \simeq 3400$, $\omega_r/\omega_m$ on the ECF background). The ECF trajectory converges with ΛCDM (blue dashed, $t_0 = 13.81$ Gyr) as early as the radiation era (BBN compatibility of the spin sector is not demonstrated; see Fig. 17). ECF gives $t_0 = 13.33$ Gyr ($\Omega_m = 0.267$) in the context of the $H_0$–Age–BAO trilemma (§5.4).

*Reproduced by:* `Scripts_F1/plot_global_history.py`.

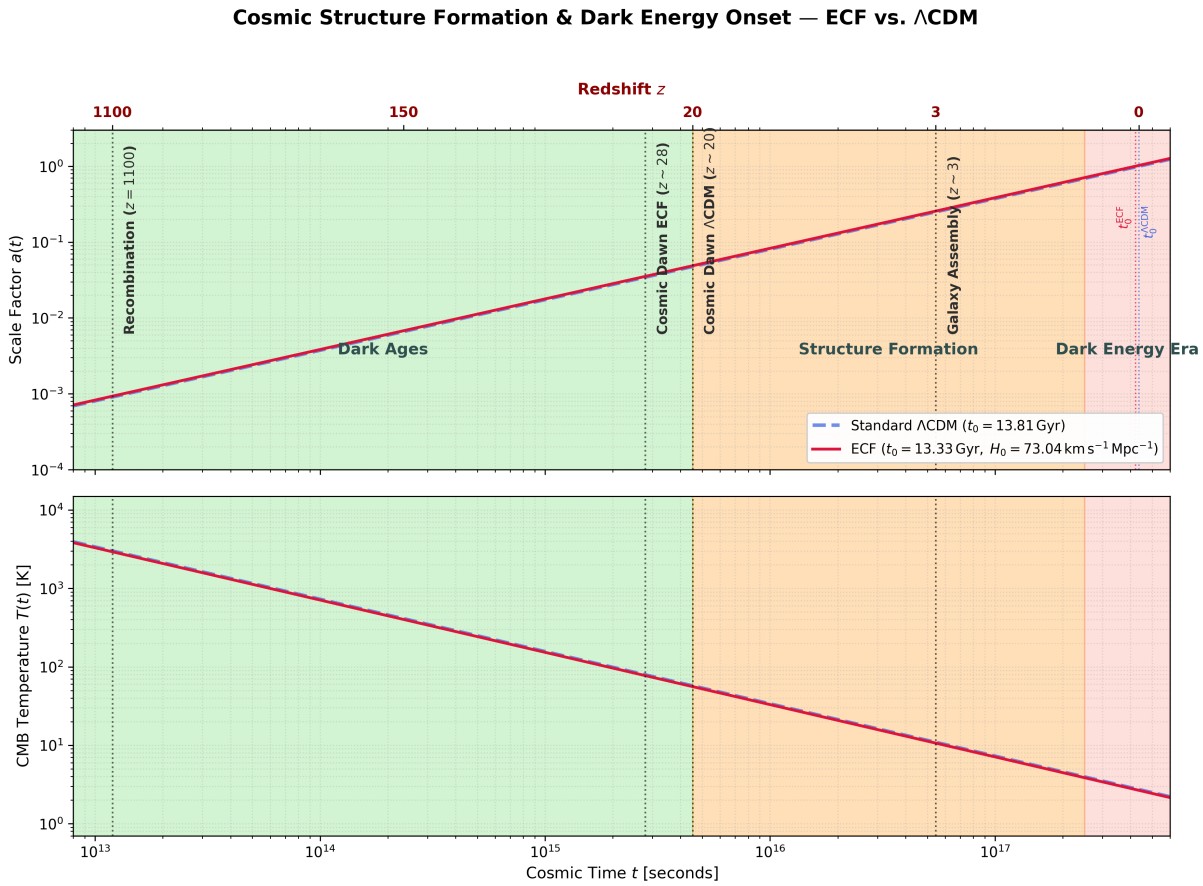
development of a dedicated Einstein–Cartan perturbation solver incorporating the ECKS torsion sector at the perturbative level; we invite the community to pursue this direction using the public repository associated with this paper.[6]

The structure growth modifications discussed herein are strictly driven by the background expansion history, establishing a theoretically complete and self-contained mechanism.

### Numerical Comparison of Milestones

## Visualizing Structure Formation

The zoom-in on the formation era (Figure 24) was drawn with an onset of collapse near $z \simeq 28$ instead of $z \simeq 20$, a shift that followed from the growth boost now withdrawn (PO-F1-BOOST); the figure is kept as an illustration of the earlier reading only.




*Reproduced by:* `Scripts_F1/plot_structure_zoom.py`.

Figure 24: **Zoom on Cosmic Dawn and Structure Formation.** Two Cosmic Dawn markers are shown: the ΛCDM value ($z \simeq 20$) and the ECF one ($z \simeq 28$), the latter following from the primordial growth boost (App. L), which is not derived and is withdrawn as a mechanism (PO-F1-BOOST); the ECF marker is therefore not a result. The expansion histories themselves differ only marginally over this range; the shift is in the onset of collapse, not in $a(t)$. The galaxies detected at $z = 14$ are therefore not unexpected, but can arise naturally result of a growth cycle amplified by the post-bounce spin-torsion dynamics.

## Observational Predictions and Falsifiability

The ECF model offers two critical tests for future missions:

[6] Repository: https://github.com/pfichant/spin-torsion-cosmology, CC-BY 4.0.

| Cosmic Milestone | Redshift $z$ | ΛCDM ($H_0 = 67.4$) | ECF (this work, $H_0 = 73.04$) |
|---|---|---|---|
| Bounce (ECF) | — | Singularity (GR) | $t_{\rm bounce} \sim 10^{-44}$ s, $a_{\rm bounce} \approx 10^{-32}$ |
| Matter–Radiation Equality | $z \simeq 3400$ | $a_{\rm eq} = 2.92 \times 10^{-4}$ | $a_{\rm eq} = 2.94 \times 10^{-4}$ |
| Recombination[d] | 1089.9 | 371,000 yr | 368,000 yr |
| **Cosmic Dawn** | – | $z \approx 18$–$20$ | withdrawn[c] |
| JWST Galaxies | 14.3 | 290 Myr | 287 Myr (*mature*) |
| Today | $z = 0$ | 13.81 Gyr[e] | 13.33 **Gyr** |

- [d] *Corrected 13/09/2026.* Earlier versions quoted 380,000 yr (ΛCDM) and 365,000 yr (ECF) at $z = 1100$. The 380,000 yr figure is the textbook round number, not a value computed for these parameters: the Planck 2018 parameters give $\simeq$ 372,000 yr at $z_* = 1089.9$ ($372.6 \pm 1.0$ kyr from the Planck chains, arXiv:2411.07703; Planck tabulates $z_*$, not this age — *corrected 20/10/2026*, formerly "Planck 2018 publishes"), and WMAP9 $\simeq$ 375,000 yr (not re-checked). Both columns are now evaluated at the Planck decoupling redshift $z_* = 1089.9$ [57] by direct integration of $t = \int_z^\infty {\rm d}z'/[(1+z')H(z')]$, reproducible with `Scripts_F1/A_Core_Calculations/script_09_cosmic_ages.py`. That script returns 371 kyr (371.3) for the ΛCDM column, 1.3 kyr below the Planck-chain value ($1.3\sigma$; the inputs differ slightly). The ECF–ΛCDM shift was 28,000 yr with $\Omega_m = 0.315$ and $\Omega_r$ at the Planck $h$; with $\Omega_i = \omega_i/h^2$ for matter and radiation (A-7, DEC-A7-POST-05) and without the spin term, the ECF column gave 372,000 yr (371.8 kyr, +0.5 kyr relative to ΛCDM). *Revised 02/10/2026:* that integral omitted the spin term $\Omega_{\rm spin,0}(1+z)^6$ of the modified Friedmann equation, which is the background that produces $r_s = 135.65$ Mpc (Sec. 3). Integrating on that same background ($\Omega_{\rm spin,0} = 0.093\,\Omega_r/(1+z_{\rm ref})^2$, $z_{\rm ref} = 5600$) gives 368,000 yr (368.4 kyr), i.e. 2.85 kyr (0.8%) *earlier* than ΛCDM (371.3 kyr); this is the `with_spin` branch of the same script, and the `without_spin` branch gives the former 371.8 kyr. At $z_*$ the background age depends on $\omega_m$, $\omega_r$ and the spin term, not on $H_0$. The recombination age is not an observable and no signature is claimed from this difference; $t_0$ and the JWST age are unchanged by the spin term ($\lesssim 4$ kyr).
- [c] *Withdrawn (16/10/2026, PO-F1-BOOST)*: the value $z \approx 27$–$29$ of earlier versions followed from the primordial growth boost applied to the linear collapse criterion, and that boost is not derived. For the record: $(1+z_{\rm ECF}) = B_{\rm ECF}\,(1+z_{\Lambda{\rm CDM}})$ with $B_{\rm ECF} = 1.45$ (App. L), i.e. $z \simeq 27$–$29$ for a ΛCDM range of 18–20. The linear-growth approximation used here is indicative; a treatment based on the modified halo mass function is deferred to Foundation II.
- [e] At $H_0 = 67.4$, $\Omega_m = 0.315$, `script_09_cosmic_ages.py` returns 13.791 Gyr; Planck 2018: $13.797 \pm 0.023$ Gyr [57].
- **Note:** $t_0 = 13.33$ Gyr (at $\Omega_m = \omega_m/h^2 = 0.267$; 12.74 Gyr at the former $\Omega_m = 0.315$, historical) is the age of the $w = -1$ ECF background at $H_0 = 73.04\,{\rm km\,s^{-1}\,Mpc^{-1}}$ (see Sec. 5.4); it lies 0.01 Gyr above the globular-cluster chronometry $t_{\rm GC} = 13.32 \pm 0.10_{\rm stat} \pm 0.5_{\rm sys}$ Gyr [29], a compatibility check rather than a $0.1\sigma$ test of $t_0$. The H$_0$DN measurement $H_0 = 73.50 \pm 0.81\,{\rm km\,s^{-1}\,Mpc^{-1}}$ at 1.1% precision [60] raises the Hubble tension to $> 6\sigma$ and strongly disfavours $t_0 = 13.8$ Gyr within any framework anchored to the local distance ladder.
- **ECF falsification criterion:** a confirmed stellar age significantly above 13.33 Gyr, or a direct measurement of $t_0$ significantly above this value independently of the Cepheid distance ladder, would falsify the $w = -1$ ECF background [29].

Table 10: **Comparison of timescales and the onset of structure formation.** The ECF recombination age ($t_{\rm rec} \approx 368{,}000$ yr at $z_* = 1089.9$) is obtained by integrating the Friedmann time-evolution equation $t = \int \frac{da}{a\,H(a)}$ on the modified background (matter, radiation, spin term and Λ) that yields $r_s = 135.65$ Mpc; with $\Omega_i = \omega_i/h^2$ its value does not depend on $H_0$. `script_09_cosmic_ages.py` reproduces it (`with_spin` branch), and returns 371 kyr for the ΛCDM column (formerly "the published Planck age"). $F_{\rm ion}$ is not a separate input of this integral. Because, once $\Omega_i = \omega_i/h^2$ (A-7), the ECF and ΛCDM columns use nearly the same physical densities ($\omega_m = 0.14237$ for ECF, $0.315 \times 0.674^2 = 0.1431$ for the ΛCDM column, same $\omega_r$; the 0.4% gap between $\omega_m = 0.14237$ and Planck's published $\omega_m = 0.1430$ is of the order of the contribution of massive neutrinos ($\sum m_\nu = 0.06$ eV gives $6.4 \times 10^{-4}$, for a gap of $6.3 \times 10^{-4}$), which the ECF background treats here as massless radiation; full treatment: open — it requires the Einstein–Cartan perturbation solver, which does not yet exist), the two ages differ little: without the spin term +0.5 kyr (372,000 vs 371,000 yr); with it, which is the background that reduces $r_s$, 368,000 vs 371,000 yr (−2.85 kyr). The spin term shortens the age by 3.4 kyr (−0.9%) while reducing the comoving sound horizon by 7.5%, because cosmic time accumulates mostly near $z_*$ whereas comoving distance weights the early epochs by $(1+z)$ (the former "28,000 years earlier" followed from $\Omega_m = 0.315$ and $\Omega_r$ at the Planck $h$). The matter–radiation equality scale factor $a_{\rm eq} = \Omega_{r,0}/\Omega_{m,0} = \omega_r/\omega_m \approx 2.92 \times 10^{-4}$ (Planck 2018: $\Omega_{m,0} = 0.315$, $\omega_r = 4.1834 \times 10^{-5}$; $2.94 \times 10^{-4}$ for the ECF background, $\omega_m = 0.14237$) is annotated in Fig. 17. Results are consistent with the damping multipole $d_{\rm Silk} = 1320$ used in Fig. 20 (App. K, §K.1).

1. **21-cm Cosmology**: An earlier absorption dip at $z \approx 30$ ($\simeq 46\,\mathrm{MHz}$), below the nominal frequency coverage of HERA and SKA-Low ($\gtrsim 50\,\mathrm{MHz}$, $z \lesssim 27$); it would require lower-frequency instruments.

2. **Gravitational Lensing**: A higher density of massive halos at $z > 10$ than predicted by $\Lambda$CDM, verifiable by weak lensing surveys from Euclid and the Nancy Grace Roman Space Telescope.

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*fixed but depends on the quadratic-torsion couplings; and that the split between the matter stress tensor and the torsion-induced spin-squared contribution is definition-dependent after algebraic torsion is eliminated. Also reports* $w_{\rm spin} = -1/3$ *for the unpolarised Einstein–Cartan limit under the metric definition of the stress tensor, as against the stiff* $w_{\rm spin} = +1$ *of the Weyssenhoff-fluid literature.*

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